

Power weighted Dyck moments

(OEIS A290569, A216966, A218221, A227887, A338634 and A291333)

Part I: harmonic perturbations, absolute amplitudes and polynomial weights

Part II: quartic Dyck moments and noncrossing cumulants

Part III: first relative corrections in three regimes

Part IV: the diagonal $p = n$

Four research reports of one external research session
(bundle Reports 201, 199, 202 and 204), bound into one ProveIt report

All Parts: 4 October 2026 merged: 7 October 2026

Abstract

Let $Z_n(\lambda)$ be the total weight of Dyck paths of semilength n in which each descent from height h carries the weight λ_h ; for $\lambda_h = h^k$ these are the columns of OEIS A290569. Part I fixes $p > 0$ and proves $Z_n(h^p) \sim C_p d_p^n (n!)^p n^{-1/2}$ with a relative $O_p((\log n)^{-2})$ remainder, the formula posted by Kotěšovec in A216966 in 2020, using the Freud leading-coefficient theorem of Kriecherbauer and McLaughlin as restated by Claeys, Krasovsky and Minakov; it proves exact amplitude transfer for harmonic and parity-harmonic perturbations, and absolute amplitudes for every positive polynomial weight, with two-ceiling inverses. Part II treats $p = 4$: the first correction $-(1 + \pi)/(16n)$ for the moments and for their free cumulants A338634, an expansion to every fixed order in shifted moments, a matching-quadruple interpretation, and a proof of the parity criterion that A338634 records as a conjecture. Part III proves first relative corrections under summable logarithmic changes in three regimes, a cumulative-tail transfer, a second route to Part II's correction, and corrections for zero-harmonic quartic and quadratic weights; its general- p correction C_p is conditional on an unverified Freud envelope and is recorded as an open question. Part IV takes $p = \tau n + \sigma$ and proves a full expansion for $Z_n(\tau n + \sigma)/(n!)^{\tau n + \sigma}$, uniform on compact sets, with the diagonal A291333 as $\tau = 1$, a total-variation correction and integer-safe inverses. The write adds the OEIS entries and b-files, a numerical test of Part III's conditional coefficient at $p = 3$, and a comparison of every inverse with the repository's transseries volume. The report is unrefereed, and nothing in it is formalized.

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Guide to this report

The four Parts are four manuscripts of the session bundle of Reports 1–243 (arrival commit 60f54ea06), placed in this directory by commit d451ef3d8 (batch 111).

Part	Bundle report	Delivered title	Labels
I	201 (base)	Harmonic perturbations of power weighted Dyck moments: Absolute amplitudes and polynomial weight asymptotics	<code>pwd:hm:</code>
II	199	Quartic Dyck moments and noncrossing cumulants: Leading amplitude and first correction for OEIS A338634	<code>pwd:qt:</code>
III	202	First relative corrections for power weighted Dyck moments: Three regimes and cumulative perturbations	<code>pwd:rc:</code>
IV	204	Diagonal power weighted Dyck paths: All orders asymptotics and integer safe inversion for OEIS A291333	<code>pwd:dg:</code>

All four are dated 4 October 2026.

One object. Every Part studies the weighted Dyck moments $Z_n(\lambda) = \sum_D \prod_h \lambda_h^{N_h(D)}$, N_h the number of descents from height h , with descent weights of size about h^p ; for $\lambda_h = h^k$ they are the entries $A(n, k)$ of the array A290569, given by the Stieltjes fraction with coefficients $1, 2^k, 3^k, \dots$. The Parts differ in the regime of p :

	Part I (201)	Part II (199)	Part III (202)	Part IV (204)
Regime	every fixed $p > 0$	$p = 4$	every fixed p , relative	$p = \tau n + \sigma$
Main result	$Z_n(h^p) \sim C_p d_p^n (n!)^p n^{-1/2}$, rate $(\log n)^{-2}$; harmonic, parity and polynomial amplitudes	$m_n/F(n) = 1 - \frac{1+\pi}{16n} + o(n^{-1})$, same for cumulants	first corrections, three regimes; cumulative transfer	full expansion of $Z_n/(n!)^{\tau n + \sigma}$, uniform
OEIS	A216966, A218221, A227887, A290569	A338634, A227887	A227887, A144853, A261000, A338634	A291333, A290569
External input	Freud coefficient theorem (KM 1999 via CKM 2023)	Berg–Valent quartic moments	Berg–Valent; secant identity	none beyond Stirling
Inverse	factorial core; piecewise-linear staircase	factorial core; direct bracket	piecewise-linear staircase	factorial core after $X = t^2$; direct bracket

How the Parts relate. Report 201 is the base: it has the greatest generality (every $p > 0$, harmonic and polynomial perturbations), and Report 202 builds on it. Report 199 is the case $p = 4$, with the cumulant transform A338634 and its parity; its correction $-(1 + \pi)/(16n)$ is re-derived in Part III (Section 40.2) as an application of the relative theorem, a second route through the same Berg–Valent normalization. Report 202 proves first relative corrections; its conditional general- p coefficient C_p would contain Part II’s value at $p = 4$. Report 204 is the diagonal $p = n$ of the same array, by a different method (an exact staircase-partition identity and Boltzmann partition moments), and uses nothing from the other Parts.

Arrangement and duplication. The base of the merge is Report 201: its `Report201.tex` was staged as this `article.tex`. The order is Report 201 (Part I), Report 199 (Part II), Report 202 (Part III, which cites both), Report 204 (Part IV). Three foundations occur more than once. (i) Part I’s uniform endpoint-occupation estimate and excursion deletion (Sections 2–3) appear in Part II as a $p = 4$ version with explicit constants (Section 24) and in Part III’s Appendix A, which reproduces Part I’s argument and adds the discrete log-convexity used in Section 35. (ii) Part I’s arch concentration (Section 6) appears in Part II (Section 26, for the quartic singular potential) and in Part III’s Appendix B, adapted to the scaled family and completed by the free-energy evaluation. (iii) The Berg–Valent quartic normalization is used in Sections 12, 23 and 40.2. These are kept, each with a dated note naming Part I’s counterpart: the later Parts cite them by their own labels, and each version adds something (explicit constants, the convexity, the scaled family, the free energy). The batch-111 placement proposed printing Report 202’s appendices only once; because Appendix A adds the convexity and Appendix B the scaled family and the free energy, both of which Part III uses, the write kept them in full with notes rather than drop delivered content.

Numbering. Sections are numbered continuously; statements and equations within sections, as delivered. Part I keeps all its numbers. Report 199’s Section k is Section $k + 18$ here, Report 202’s Section k is Section $k + 33$ (its appendices keep the letters A and B), and Report 204’s Section k is Section $k + 43$; statements and equations move with their sections. The labels carry `pwd:hm:`, `pwd:qt:`, `pwd:rc:` and `pwd:dg:`; labels added by the write carry the Part prefix, or `pwd:` for this Guide. The bibliography is merged at the end: Part II’s `oeis` is `oeisA338634`, Part IV’s `AD` is Part I’s `avdoshkin` (the same paper), and the entries cited by several Parts are printed once with each Part’s notes.

[write] **Provenance, added 7 October 2026, batch 111.** This report was built from four manuscripts, all arrived in 60f54ea06 and placed in d451ef3d8. *Part I:* Report 201, archive `Report201.zip` (3,546,920 bytes, SHA-256 571a7cad97c6...fedb07da95c77a6, 21 files in `Report201/`); text `Report201.tex` (1382 lines, 35 pages). *Part II:* Report 199, archive `Report199.zip` (2,872,519 bytes, SHA-256 3d8a699b123f...a7bc6ca27ead5d9, 12 files in `Report199/`); text `Report199.tex` (773 lines, 22 pages). *Part III:* Report 202, archive `Report202.zip` (726,805 bytes, SHA-256 2b83e3588e25...2fde19fb05c8187, 18 files in `Report202/`); text `Report202.tex` (1088 lines, 27 pages). *Part IV:* Report 204, archive `Report204.zip` (555,616 bytes, SHA-256 c6b5724246f1...3c60d087a8d972a, 16 files, no wrapper directory); text `Report204.tex` (652 lines, 16 pages). Author lines and PDF author fields read “Report 201”, “Report 199”, “Report 202”; Report 204’s PDF author field is empty and its author line reads “Report 204”. None names a person, tool or addressee; none carries a “prepared for private review” line, an e-mail address or personal data, and none records a ProveIt commit. The programs are shipped under `code/` and data, receipts and recorded outputs under `data/`, with the prefixes 201-harmonic-, 199-quartic-, 202-relcorr- and 204-diagonal-; the reproduction READMEs of Reports 201 and 202 are shipped at the top level, and the source manifests of Reports 199, 201 and 202, which also record an engine banner and an epoch, under `data/`. Report 201’s delivered `README.md` was staged as the report README and is replaced by it. Not shipped, and retrievable from the arrival commit: the four PDFs; the texts of Reports 199, 202 and 204 (Parts II–IV of this file); their delivered READMEs; the pure checksum manifests (the package manifests of Reports 199, 201 and 202, and Report 204’s `MANIFEST.sha256`), all verified at the write; and two large regenerable data files, Report 199’s `data/exact_values.json` (2,257,761 bytes) and Report 201’s `repro/generated/moments.json` (2,605,540 bytes), which the intake regenerated byte for byte (Report 199: `code/exact.py -output`; Report 201: `repro/check.py -out`). The write prefixed the 150, 81, 105 and 95 delivered labels and updated the 126, 65, 92 and 66 references to them (91, 52, 74, 53 \eqref; 35, 13, 18, 13 \ref); it labelled the two unlabelled sections of Report 204 and added this Guide, the Part headers, the remarks and the dated notes marked [write]. The added remarks are the last statements of their sections and the added displays are unnumbered, so every delivered number is unchanged up to the stated section shifts (label tables of builds of the four delivered texts and of this one compared).

The sources, as the write read them (7 October 2026). OEIS A290569, A216966, A218221, A227887, A338634

and A291333 in the internal format, and the b-files of A216966, A218221, A227887, A338634 and A291333; the transseries volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex` (`p0:def:core`, `p0:prop:factorial-core`, `p0:thm:core-reversion`, `p0:def:three-inverses`, `p0:thm:staircase`, `p0:thm:backward-error`). Not read by the write: the cited literature, in particular neither Kriecherbauer–McLaughlin nor Claeys–Krasovsky–Minakov (the Freud input of Part I), Berg–Valent, nor Mastroianni–Milovanović; the Parts’ accounts of what they inspected are reported, not confirmed.

What was checked at intake. The batch-111 dossier (7 October 2026) read the four manuscripts, found no error in them, checked A338634’s parity for $n \leq 130$, regenerated the two large data files byte for byte, and reran the delivered suites on copies: their outputs reproduce, the only failures being Windows artifacts (line-ending bytes in hashed outputs, a recorded Python version string, a console code page), none mathematical. *What the write checked.* Every proof line by line, with no error found, including by hand the Freud telescoping and prefactor algebra ((5.14)), the regularized harmonic integral ((7.5)), Part II’s integral $C_* = -\pi/16$ ((27.6)) and its Stirling bookkeeping, the parity proof over $\mathbb{F}_2$, Part III’s $J_2 = 0$, $J_4 = -\pi/4$ and the conditional C_p at $p = 2, 4$, and Part IV’s coefficients c_1 and the factorial-core substitution. With its own code (Python 3.14.4, mpmath 1.3.0) it computed $Z_n(h^3)$ to $n = 400$, $Z_n(h^4)$ to 500, the weights of A218221 to 198 and the comparison moments of Part II to 500 (all b-file terms of A216966, A218221 and A227887 agree), A338634 to $n = 130$ by Lagrange inversion (all 131 b-file terms agree; the odd terms for $1 \leq n \leq 130$ are exactly at $n = 1, 2, 4, \dots, 128$), and A291333 to $n = 30$ (all b-file terms agree); it reproduced every entry of Part II’s diagnostic table and Part IV’s $Q(1)$, c_1 , total-variation constant and $E_0(10)$, $E_0(20)$, $E_0(40)$; it confirmed that d_p , C_p of Part I equal Kotěšovec’s $d(s)$, $c(s)$ at $p = 3, 4$ and that (1.14) gives A218221’s posted constants to twenty digits. Finite and floating computations are observations, not certificates.

Relation to the repository. Before batch 111 no file of the repository named A290569, A216966, A218221, A227887, A338634 or A291333. No statement of this report is formalized in Lean or Rocq, and its place in the collection confers no formal status. Part I’s source screen (Section 17) mentions “the nearby peak-weight Newton-edge problem” in the repository, and Part IV’s (Section 53) nearby weighted-Dyck reports with peak-weight models; the write identifies the former as `a292692-weighted-dyck-newton-diagonal` (Dyck paths with peak weights $k + a/b$, a different model), in the same directory `SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/oeis-sequence-asymptotics/`. It shares no result with this report, so no reciprocal note is proposed. The inverses are compared with the transseries volume in Remarks 15.3, 29.2, 41.2 and 51.4.

[write] **What is not claimed, collected from the four Parts.** *Part I.* No effective onset, no uniformity in p , no all-orders expansion, no first-proof claim: the absolute formula is Kotěšovec’s (A216966, 2020), and its proof rests on the Freud leading-coefficient estimate of Kriecherbauer and McLaughlin (1999, Theorem 1.5) used only as restated by Claeys, Krasovsky and Minakov (2023, Eq. (2.3)); the original 1999 theorem text was not inspected, and that trust boundary is retained throughout. The harmonic and polynomial equivalents have only relative $o(1)$ errors; the pure-power logarithmic rate does not transfer to them; the parity theorem uses the two-step structure only; no transfer for arbitrary ℓ^1 or periodic perturbations; the shape result is fixed-neighbourhood concentration only; the inverses are not certified finite-threshold methods and an $o(1/(t \log t))$ radius cannot follow from a leading equivalent; the formal saddle geometry is Avdoshkin–Dymarsky’s; exact checks and diagnostics are not proofs. *Part II.* No effective onset, no $O(n^{-2})$ raw-moment remainder, no all-orders inverse-power expansion of the raw moments, no unconditional single-ceiling inverse, no absolute priority; the leading equivalent is Kotěšovec’s, the comparison model Berg and Valent’s; the shifted-moment expansion is of Bender–Richmond type, not a new principle; the source review was bounded (Chen–Ismail, Mlotkowski, Disanto–Munarini not fully read). *Part III.* No first-proof claim, no effective onset, no all-orders claim, no new unconditional general-power absolute coefficient: the coefficient C_p is conditional on an unverified Freud envelope (Mastroianni–Milovanović, Theorem 2.4.10, a secondary lead only); a zero displayed coefficient gives only a little- o statement; the cumulative theorem needs its envelope hypothesis; the classical elliptic formulas are not claimed new. *Part IV.* No convergence, divergence, Borel summability, optimal truncation, exponential scale or transseries; no statement for $\tau \rightarrow 0$ or $\tau \rightarrow \infty$; the Gevrey bound is an upper bound; the printed decimals are not interval certificates; the inverse constants are existential; the leading equivalent is Kotěšovec’s; no global priority. *The write adds:* its

checks are floating, finite or hand computations; the numerical agreement of C_3 is evidence, not a proof; the transseries remarks claim no novelty for any inversion.

Reading conventions (letters the Parts use differently, with the tempting false readings; no symbol of any source was renamed; symbols of the transseries volume carry the subscript “vol” in the transseries remarks).

Symbol	Meanings, by Part
p, k	I, III: the fixed power; IV: $p = \tau n + \sigma$ grows with n ; k the column of A290569, a deletion length (I), a block size (II)
m_n, b_n, a_n	II: $m_n = Z_n(h^4)$ (A227887), b_n the Berg–Valent comparison, a_n the cumulants (A338634); IV: $a_n = Z_n(n)$ (A291333); I: $m_n = Z_n(h^p)$ in Section 15
$I_p, H_p, r_p, d_p, \mathcal{C}_p$	I, III: $\int_0^1 (1 - u^p)^{-1/2} du$, its inverse, $2^{2/p}/I_p$, $4/I_p^p$, the amplitude; II writes $I = I_4$, $r = r_4$, $H = 1/I$
K_p, K_0, K	I: $\log H_p + 2 \log 2/p$; II, III: $K_0 = \Gamma(1/4)^2/(4\sqrt{\pi})$; II: a block bound K in Theorem 19.2
$C_p, \mathcal{C}_p, C$	III: the conditional first coefficient (42.1); I: the amplitude $\mathcal{C}_p$; generic constants C, C_1, C_f throughout
$F, F_p, F(n)$	II: $F(x)$ the quartic template; III: $F_p(n)$ the general template, and $F, \tilde{F}$ piecewise-linear interpolations in Section 41; I: F the interpolation in Section 15
$Q, Q_j, Q(\tau)$	I, III: a polynomial weight $Q(x)$; II: $Q_j(n) = [z^j]M^{-2n}$; IV: $Q(\tau) = \prod (1 - e^{-\tau j})^{-1}$
$J, J_p, J_{p,q}$	III: the regularized integrals; II: $J = \int_0^1 u^2(1 - u^4)^{-1/2} du$; also truncation indices
$A, A_0, A(x)$	I: $A = 4Ce^p/c$; III: A_0 the same constant, and A the cumulative coefficient; II: $A(x)$ the cumulant series; IV: A_r and A in the inverse
B, B_r, B_0	I, III: the bound B_0 ; IV: $B_r(\lambda)$ partition coefficients and B the Gevrey constant; II: B in the inverse (29.1)
D, D_k, d	II: $D = 4 \log(rt)$; IV: $D_k(\lambda)$ statistics; d : II’s $(13 - 3\pi)/48$, IV’s $1/12 + \log Q(1)$, I’s degree d
c_r, ℓ_r	IV: coefficients and logarithmic coefficients; I: c_j shifts of a polynomial; II, III: c the coefficient $(1 + \pi)/16$ or generic
t, L	I, II, IV: t the Lambert-scale centre, $L = \log Y$; I, II: t also the interpolation parameter $\lambda(t)$; III: $L = \lfloor \rho n \rfloor$ a cutoff
$N(Y), N_h$	the integer threshold; the number of descents from h
$X_{\text{vol}}, \Lambda_{\text{vol}}, h_{\text{vol}}$	the transseries volume’s core variable, core slope and core nonlinearity, never the Parts’ symbols

Part I

Harmonic perturbations of power weighted Dyck moments: absolute amplitudes and polynomial weight asymptotics

Bundle Report 201, archive `Report201.zip`, delivered as `Report201.tex` (35 pages), the base of the merge. Delivered title block “Harmonic perturbations of power weighted Dyck moments / Absolute amplitudes and polynomial weight asymptotics / Report 201 / 4 October 2026”. Printed as delivered, with `[write]` remarks and notes; section, statement and equation numbers are those of the delivery, and its labels carry `pwd:hm:.` Its absolute theorem rests on a cited Freud coefficient theorem read only as restated (note at the end of Section 5). This Part is self-contained otherwise.

Abstract of this Part (as delivered). For every fixed $p > 0$, we derive the full leading amplitude of Dyck moments with descent weights h^p , together with a relative $O_p((\log n)^{-2})$ remainder. The absolute formula was already posted by Kotěšovec on OEIS in 2020. Our route combines a uniform endpoint occupation estimate with the strong asymptotics of Freud orthogonal-polynomial leading coefficients, using the precise all-positive-exponent statement restated by Claeys, Krasovsky, and Minakov. Separately, a direct uniform variational argument gives the exact relative amplitude under logarithmic perturbations $a/h + O(h^{-1-\delta})$. The constant is a renormalized Euler product times $\exp(aK_p)$, where $K_p = -\log I_p + 2 \log 2/p$ and $I_p = \int_0^1 (1-u^p)^{-1/2} du$. Exact two-step conditioning extends this to a parity-harmonic term and supplies an independent cubic derivation from the classical Dixon continued fraction. Gamma products now give the absolute amplitude for every real polynomial of positive degree that is positive at the positive integers, including A216966, A218221, and classical quartic examples. We prove inclusive two-ceiling Lambert- W inverses: a general little- o envelope and a sharper logarithmic envelope for pure powers. Portable exact checks, separate numerical diagnostics, and a reproducible TeX/PDF package accompany the proofs. No effective onset, uniformity in p , all-orders expansion, or first-proof claim is made.

1 Results and their scope

Let $\mathcal{D}_n$ be the Dyck paths of semilength n : nearest-neighbor paths of $2n$ steps, starting and ending at zero, which remain nonnegative. For $h \geq 1$, let $N_h(D)$ count descents from h to $h-1$. For a positive weight sequence $\lambda = (\lambda_h)_{h \geq 1}$ put

$$W_\lambda(D) = \prod_{h \geq 1} \lambda_h^{N_h(D)}, \quad Z_n(\lambda) = \sum_{D \in \mathcal{D}_n} W_\lambda(D), \quad Z_0(\lambda) = 1. \quad (1.1)$$

Thus $\sum_h N_h = n$. Our index is always semilength, not total path length or the index of an exponential generating function. First-return decomposition gives the formal Stieltjes fraction

$$\sum_{n \geq 0} Z_n(\lambda) z^n = \frac{1}{1 - \frac{\lambda_1 z}{1 - \frac{\lambda_2 z}{1 - \frac{\lambda_3 z}{\ddots}}}}. \quad (1.2)$$

For the growing weights studied here this ordinary series need not converge at any nonzero z . All uses of (1.2) are formal. Scaling all descent weights by $\kappa > 0$ multiplies Z_n by κ^n .

Fix $p > 0$ and define

$$I_p = \int_0^1 \frac{du}{\sqrt{1-u^p}} = \frac{\Gamma(1/p)\sqrt{\pi}}{p\Gamma(1/p+1/2)}, \quad H_p = I_p^{-1}, \quad K_p = \log H_p + \frac{2\log 2}{p}, \quad r_p = e^{K_p}. \quad (1.3)$$

The beta integral and duplication formula give

$$r_p = \frac{2^{2/p}}{I_p} = \frac{2p\Gamma(2/p)}{\Gamma(1/p)^2}, \quad r_p^p = \frac{4}{I_p^p}. \quad (1.4)$$

Theorem 1.1 (Absolute pure-power amplitude). *For every fixed real $p > 0$, let*

$$d_p = r_p^p = \frac{4}{I_p^p}, \quad C_p = \sqrt{\frac{p d_p}{2\pi}} = \sqrt{\frac{2p}{\pi}} I_p^{-p/2}. \quad (1.5)$$

Then

$$Z_n(h^p) = C_p d_p^m (n!)^p n^{-1/2} \left(1 + O_p((\log n)^{-2})\right). \quad (1.6)$$

The constants and onset in O_p may depend on the fixed p . No effective values, power-rate error, or uniformity as p varies are asserted.

Theorem 1.2 (Harmonic amplitude transfer). *Suppose, with fixed $a \in \mathbb{R}$, $\delta > 0$, and $C_f < \infty$,*

$$\lambda_h = h^p e^{f_h}, \quad f_h = \frac{a}{h} + r_h, \quad |r_h| \leq C_f h^{-1-\delta}.$$

Then

$$\frac{Z_n(\lambda)}{Z_n(h^p)} \sim n^a e^{aK_p} P_f, \quad P_f = \lim_{N \rightarrow \infty} N^{-a} \prod_{h=1}^N e^{f_h} = \exp \left(a\gamma + \sum_{h \geq 1} r_h \right) > 0. \quad (1.7)$$

No absolute asymptotic for $Z_n(h^p)$ is assumed. The error in this equivalent is relative $o(1)$; no universal power rate is asserted.

Theorem 1.3 (Parity-harmonic amplitude transfer). *The same conclusion (1.7) holds if instead*

$$f_h = \frac{a}{h} + \frac{b(-1)^h}{h} + r_h, \quad |r_h| \leq C_f h^{-1-\delta}, \quad a, b \in \mathbb{R}, \quad (1.8)$$

where now

$$P_f = \exp \left(a\gamma - b \log 2 + \sum_{h \geq 1} r_h \right). \quad (1.9)$$

The alternating term supplies its convergent product constant but no additional power of n or macroscopic shape constant.

The endpoint estimate below is useful independently of either transfer theorem. Write $\mathbb{E}_{n,\lambda}$ for expectation under the probability measure $W_\lambda/Z_n(\lambda)$.

Theorem 1.4 (Uniform endpoint occupation). *Fix $p > 0$ and $0 < c \leq C < \infty$. There are $\rho > 0$, $C_1 < \infty$, and n_0 , depending only on p, c, C , such that every positive sequence with*

$$ch^p \leq \lambda_h \leq Ch^p \quad (h \geq 1) \quad (1.10)$$

satisfies

$$\mathbb{E}_{n,\lambda} |N_h - 1| \leq C_1(h/n)^p \quad (n \geq n_0, \quad 1 \leq h \leq \rho n). \quad (1.11)$$

No monotonicity, smoothness, or asymptotic expansion of the weights is required.

Theorem 1.5 (Quantitative summable stability). *Suppose λ and $\tilde{\lambda}$ both satisfy (1.10) and $|\log(\tilde{\lambda}_h/\lambda_h)| \leq Kh^{-q}$ for fixed $K < \infty$ and $q > 1$. Then*

$$\frac{Z_n(\tilde{\lambda})}{Z_n(\lambda)} = \left(\prod_{h \geq 1} \frac{\tilde{\lambda}_h}{\lambda_h} \right) (1 + O(R_n)), \quad R_n = \begin{cases} n^{1-q}, & 1 < q < p+1, \\ n^{-p} \log n, & q = p+1, \\ n^{-p}, & q > p+1. \end{cases} \quad (1.12)$$

The implicit constant depends only on fixed p, c, C, K, q . The product converges absolutely in logarithms and is positive.

The general absolute baseline and harmonic transfer give exact amplitudes in every polynomial degree.

Theorem 1.6 (Absolute polynomial amplitudes in every degree). *Let $Q(x) = \kappa \prod_{j=1}^d (x + c_j)$ be a real polynomial of degree d , with $\kappa > 0$ and $Q(h) > 0$ at every positive integer. Roots are counted with multiplicity, and put $a = \sum_j c_j \in \mathbb{R}$. Then for every $d \geq 1$,*

$$\frac{Z_n(Q)}{Z_n(h^d)} \sim \kappa^n \frac{(r_d n)^a}{\prod_{j=1}^d \Gamma(1 + c_j)}. \quad (1.13)$$

For every such d the absolute equivalent is

$$Z_n(Q) \sim \frac{C_d r_d^a}{\prod_{j=1}^d \Gamma(1 + c_j)} (\kappa r_d^d)^n (n!)^d n^{a-1/2}. \quad (1.14)$$

Here C_d and r_d are the constants in (1.5) and (1.4) evaluated at $p = d$. The error for this general shifted family is only relative $o(1)$: the pure-power logarithmic rate does not automatically transfer. For $d = 3$ this specializes to

$$Z_n(Q) \sim \frac{\sqrt{6/\pi} I_3^{-3/2} r_3^a}{\prod_{j=1}^3 \Gamma(1 + c_j)} \left(\frac{4\kappa}{I_3^3} \right)^n (n!)^3 n^{a-1/2}. \quad (1.15)$$

For $d = 4$, set $r = r_4 = 8\sqrt{\pi}/\Gamma(1/4)^2$. Then

$$Z_n(Q) \sim \frac{\sqrt{2/\pi} r^{2+a}}{\prod_{j=1}^4 \Gamma(1 + c_j)} (\kappa r^4)^n (n!)^4 n^{a-1/2}. \quad (1.16)$$

The gamma products in these formulas are real and strictly positive, even when some individual linear factors are negative at small integers or occur in nonreal conjugate pairs.

1.1 Prior formulas and the inputs used here

OEIS A216966 already records, in its COMMENTS, Kotěšovec’s general- p absolute formula dated 24 September 2020:

$$Z_n(h^p) \sim c(p)d(p)^n \frac{(n!)^p}{\sqrt{n}}, \quad d(p) = \left[\frac{2p\Gamma(2/p)}{\Gamma(1/p)^2} \right]^p, \quad c(p) = \sqrt{\frac{p d(p)}{2\pi}}. \quad (1.17)$$

Equivalently $d(p) = 4/I_p^p$ and $c(p) = \sqrt{2p/\pi} I_p^{-p/2}$. The inspected entry does not label this formula conjectural. The cubic specialization is credited there to 25 August 2017, updated 23 September 2020; the A218221 equivalent has the same original date and an update of 16 March 2024 [1].

This article derives (1.17) for every fixed $p > 0$, with the logarithmic remainder in Theorem 1.1. The external input is a strong Freud leading-coefficient theorem, directly inspected in Claeys–Krasovsky–Minakov’s explicit restatement [8, Eq. (2.3)], attributed there to Kriecherbauer–McLaughlin [9, Theorem 1.5]. The original 1999 theorem text was not directly inspected; that source boundary is retained throughout. The cubic Dixon and quartic Berg–Valent derivations remain independent checks of the special constants. Neither a missing proof link on OEIS nor a bounded source search establishes absence of an earlier proof.

Avdoshkin–Dymarsky already derived the same formal power-weight saddle geometry [12, Appendix C]; Dymarsky–Smolkin studied the moment power induced by a constant shift of asymptotically linear Lanczos coefficients [13]. Classical associated-Hermite and secant-power families give exact low-degree tests. The analytic and combinatorial ingredients of the Dixon and quartic comparison models are credited where used. The contribution developed here is a complete, explicitly scoped proof chain from uniform endpoint control to the stated relative amplitudes and applications.

1.2 How the proof is organized

Sections 2–4 prove endpoint occupation and product stability without a limit shape. Section 5 then proves the all- p absolute amplitude and its logarithmic rate from Freud norms. Sections 6–8 establish the direct interpolated concentration, regularize harmonic occupation, and integrate it. Section 9 proves the parity extension using signed conditional means, not pair independence. Gamma algebra is in Section 10; Sections 11–13 supply independent classical anchors and examples. Section 15 gives a generic $o(1/\log t)$ pre-rounding envelope and a pure-power $O_p((\log t)^{-3})$ envelope. Numerical checks remain separate from proofs and finite-threshold certificates.

Remark 1.7 ([write] The OEIS entries, quoted, 7 October 2026). Read on 7 October 2026 in the internal format.

- *A290569*, revision #31 (22 August 2017), Ilya Gutkovskiy (8 August 2017): “Square array $A(n,k)$, $n \geq 0$, $k \geq 0$, read by antidiagonals, where column k is the expansion of continued fraction $1/(1 - x/(1 - 2^k x/(1 - 3^k x/(1 - 4^k x/(1 - 5^k x/(1 - \dots))))))$.”; so $A(n, k) = Z_n(h^k)$, the object of this report; Part IV treats its diagonal.
- *A216966*, revision #29 (2 October 2023), Paul D. Hanna (20 September 2012): the column $k = 3$. Its comments contain Kotěšovec’s general formula of 24 September 2020, “ $a(n,s) \sim c(s) \cdot d(s)^n \cdot (n!)^s / \sqrt{n}$, where $d(s) = (2^s \Gamma(2/s) / \Gamma(1/s)^2)^s$, $c(s) = \sqrt{s \cdot d(s) / (2\pi)}$ ”, not labelled conjectural, and its formula line gives the cubic case (25 August 2017, updated 23 September 2020). The same comment stands in *A227887* (revision #28, the column $k = 4$).

- *A218221*, revision #23 (16 March 2024), Hanna (23 October 2012): weights $\binom{h+2}{3}$; Kotěšovec's "a(n) ~ c * d^n * (n!)^3 * n^(5/2)" with $d = 0.241821816937867322064\dots$ and $c = 0.60382458700655692263\dots$ (25 August 2017, updated 16 March 2024).

The write computed these sequences with its own code: all b-file terms of A216966, A218221 and A227887 agree. It also checked numerically that d_p and C_p of (1.5) equal Kotěšovec's $d(s)$ and $c(s)$ at $p = 3, 4$ to 25 digits, and that (1.14) with $Q(x) = x(x+1)(x+2)/6$ gives $\kappa r_3^3 = 0.24182181693786732206\dots$ and amplitude $0.60382458700655692263\dots$, A218221's posted constants. Nothing was submitted to the OEIS.

2 Moment ratios and exact excursion deletion

For the rest of this section assume (1.10). Write

$$u_n(h) = \mathbb{E}_{n,\lambda} N_h, \quad v_n(h) = \mathbb{P}_{n,\lambda}(\max D \geq h),$$

and set $u_N(j) = 0$ when $j > N$.

Lemma 2.1 (An all-length ratio bound). *For every $n \geq 1$ and $0 \leq k \leq n$,*

$$Z_n(\lambda) \geq c^n (n!)^p \geq (cn^p / e^p)^n, \quad \frac{Z_{n-k}(\lambda)}{Z_n(\lambda)} \leq \left(\frac{e^p}{cn^p} \right)^k. \quad (2.1)$$

Proof. Take the finite symmetric Jacobi matrix indexed by $0, \dots, n$, with zero diagonal and entries $J_{h-1,h} = J_{h,h-1} = \sqrt{\lambda_h}$. A closed nearest-neighbor walk from zero of length $2j$, $j \leq n$, is a Dyck path and cannot leave this truncation. It traverses each edge equally often in the two directions. Hence

$$(J^{2j})_{00} = Z_j(\lambda), \quad 0 \leq j \leq n.$$

The spectral theorem, applied to squared eigenvalues, makes these initial moments those of a positive measure on $[0, \infty)$ of mass one. Hölder's inequality gives $Z_{n-k} \leq Z_n^{(n-k)/n}$. The mountain $U^n D^n$ contributes at least $c^n (n!)^p$, and $n! \geq (n/e)^n$. Dividing the moment inequality by Z_n proves the claim, including $k = 0, n$. $\square$

This fixed- n chord bound controls every deletion length at once. No asymptotic one-step ratio is used.

Let $e_{h,k}$ be the total weight of primitive negative excursions of length $2k$ at height h : they start at h , immediately descend, remain in $\{0, \dots, h-1\}$ until the last step, then rise to h . In particular these excursions remain nonnegative. Their k downsteps have heights at most h , so

$$0 \leq e_{h,k} \leq 4^k C^k h^{pk}. \quad (2.2)$$

Lemma 2.2 (Exact deletion and slot identity). *For $1 \leq h \leq n$,*

$$u_n(h) - v_n(h) = \sum_{k=1}^{n-h} e_{h,k} \frac{Z_{n-k}}{Z_n} (u_{n-k}(h) + u_{n-k}(h+1)). \quad (2.3)$$

Proof. Mark a non-last descent across edge h . A later return to h exists; cut out the path from this descent through its first such return. It is a unique primitive negative excursion. Deletion leaves a Dyck path of semilength $n - k$ with a distinguished vertex of height h .

Conversely insert any permitted negative excursion at any vertex of height h in the shorter path. Since that path subsequently ends at zero, its later steps include a descent across h , so the inserted marked descent is non-last. The constructions are inverse, preserve all other descent heights, and multiply weights. The shorter path must reach h , which forces $k \leq n - h$.

For $h > 0$, a path has exactly $N_h + N_{h+1}$ vertices of height h : classify their arrivals as upsteps across edge h or downsteps across edge $h + 1$, using upward/downward balance on edge h . The number of non-last descents is $N_h - \mathbf{1}_{\{\max D \geq h\}}$. Weighted summation gives (2.3). $\square$

Put

$$A = \frac{4Ce^p}{c}, \quad a(n, h) = A(h/n)^p. \quad (2.4)$$

Equations (2.1)–(2.3) imply

$$u_n(h) \leq 1 + \sum_{k=1}^{n-h} a(n, h)^k (u_{n-k}(h) + u_{n-k}(h+1)). \quad (2.5)$$

All quantities in this recursion are nonnegative, a fact used in the deliberate overcounting below.

3 A uniform cone expansion at the lower boundary

Choose constants

$$\alpha = \min\left(\frac{1}{4}, (12A)^{-1/p}\right), \quad \eta = A\alpha^p \leq \frac{1}{12}, \quad r_0 = 3\eta \leq \frac{1}{4}, \quad \kappa_0 = \frac{\alpha}{2(1+\alpha)}. \quad (3.1)$$

We first prove a rough bound in a wider cone and then extract the small error in a narrower one.

Start with $h \leq \alpha n/2$. Expand (2.5) recursively at every pair (N, j) with $j \leq \alpha N$; stop at the first exit from this outer cone. The expansion is finite because N strictly decreases at every edge. Inside the cone the coefficient of a loss k is at most η^k .

For total loss s and ℓ recursive edges, the positive compositions of s supply $\binom{s-1}{\ell-1}$ possibilities, and at each edge there are two height choices, preserving or increasing j . Thus the total coefficient of all chains with loss $s \geq 1$ is bounded by

$$\sum_{\ell=1}^s 2^\ell \binom{s-1}{\ell-1} \eta^s = 2 \cdot 3^{s-1} \eta^s = \frac{2}{3} r_0^s. \quad (3.2)$$

This includes impossible chains and therefore only enlarges a nonnegative upper bound. The constants inserted at internal nodes sum to at most $1 + 2\eta/(1 - 3\eta)$.

A terminal exit has coordinates $(n - s, h + t)$, where $0 \leq t \leq s$. Since $h + t > \alpha(n - s)$ but $h \leq \alpha n/2$, it must have $s > \kappa_0 n$. Every terminal occupation is at most n ; an impossible child has occupation zero. Therefore all exit terms together are at most

$$\frac{2n}{3(1 - r_0)} r_0^{\kappa_0 n} \leq \frac{2}{3(1 - r_0)e\kappa_0 \log(1/r_0)}. \quad (3.3)$$

The last inequality uses $\sup_{x \geq 0} x e^{-bx} = 1/(eb)$. Consequently

$$u_n(h) \leq B_0 := 1 + \frac{2\eta}{1-3\eta} + \frac{2}{3(1-r_0)\epsilon\kappa_0 \log(1/r_0)} \quad (1 \leq h \leq \alpha n/2). \quad (3.4)$$

Now take $n \geq 8/\alpha$ and $h \leq \alpha n/8$. If $k \leq n/2$, then

$$h+1 \leq \alpha n/4 \leq \alpha(n-k)/2,$$

so both child occupations in (2.3) are at most B_0 . For $k > n/2$ use n . With $a = a(n, h) \leq \eta/8^p < 1/4$, summing the geometric bounds gives

$$\begin{aligned} 0 \leq u_n(h) - v_n(h) &\leq \frac{2B_0 a + 2na^{\lfloor n/2 \rfloor + 1}}{1-a} \\ &\leq \frac{8}{3}(B_0 + 1)a. \end{aligned} \quad (3.5)$$

Here $n4^{-\lfloor n/2 \rfloor} \leq 1$ for $n \geq 2$. A path with maximum below h has at most 4^n possible step patterns and weight at most $(Ch^p)^n$. The mountain bound yields

$$1 - v_n(h) \leq [A(h/n)^p]^n = a^n \leq a. \quad (3.6)$$

Finally the identity

$$|N_h - 1| = (N_h - \mathbf{1}_{\{\max D \geq h\}}) + \mathbf{1}_{\{\max D < h\}}$$

holds pathwise. Equations (3.5)–(3.6) prove Theorem 1.4 with the explicit, nonoptimal choices

$$\rho = \alpha/8, \quad n_0 = \lceil 8/\alpha \rceil, \quad C_1 = A \left[1 + \frac{8}{3}(B_0 + 1) \right]. \quad (3.7)$$

All constants depend only on p, c, C . They can become large as $p \downarrow 0$; the theorem fixes $p > 0$ and asserts no uniformity in that limit.

4 Summable changes without a shape theorem

We prove Theorem 1.5. Put $g_h = \log(\tilde{\lambda}_h/\lambda_h)$ and use geometric interpolation

$$\lambda_h(t) = \lambda_h e^{tg_h}, \quad 0 \leq t \leq 1.$$

It stays in the same interval $[ch^p, Ch^p]$. The finite positive path sum can be differentiated without a limiting interchange:

$$\frac{d}{dt} \log Z_n(\lambda(t)) = \sum_{h=1}^n g_h u_{n,t}(h). \quad (4.1)$$

Take $L = \lfloor \rho n \rfloor$. Subtract the absolutely convergent sum $\sum_{h \geq 1} g_h$. The endpoint theorem bounds the low part by

$$\left| \sum_{h \leq L} g_h (u_{n,t}(h) - 1) \right| \leq C_1 K n^{-p} \sum_{h \leq L} h^{p-q}.$$

Since $\sum_h u_{n,t}(h) = n$, the high occupation and deterministic tail are bounded by

$$\left| \sum_{h > L} g_h u_{n,t}(h) \right| \leq \frac{Kn}{(L+1)^q}, \quad \left| \sum_{h > L} g_h \right| \leq K \sum_{h > L} h^{-q}.$$

Their orders are $O(n^{1-q})$. The first power sum has order n^{p-q+1} , $\log n$, or 1 according as $q < p + 1$, $q = p + 1$, or $q > p + 1$. These bounds are uniform in t with precisely the dependence stated in the theorem. Integrating (4.1) and exponentiating proves (1.12).

In particular a fixed finite collection of weight changes contributes exactly its finite product to the leading ratio. However, this proof uses a uniform pointwise power bound on the logarithmic tail. It is not a theorem for an arbitrary ℓ^1 perturbation, and the borderline harmonic case cannot be obtained by simply setting $q = 1$.

4.1 A broader small-perturbation lemma

The pointwise power decay can be relaxed if the logarithmic product has a convergent ordinary sum. This extension will be needed for the Freud comparison, whose error is smaller than harmonic but need not be bounded by $h^{-1-\delta}$.

Lemma 4.1 (Convergent products below the harmonic scale). *Let λ and $\tilde{\lambda}$ both belong to a common power class (1.10). Suppose*

$$f_h = \log(\tilde{\lambda}_h/\lambda_h) = o(1/h), \quad \sum_{h \geq 1} f_h = S$$

converges as an ordinary real series, not necessarily absolutely. Then

$$\frac{Z_n(\tilde{\lambda})}{Z_n(\lambda)} \longrightarrow e^S. \quad (4.2)$$

Proof. Interpolate $\lambda_h(t) = \lambda_h e^{t f_h}$. The fixed power bounds and endpoint constants are uniform for $0 \leq t \leq 1$. With $L = \lfloor \rho n \rfloor$, subtract S from the finite logarithmic derivative. The difference is exactly

$$\sum_{h \leq L} f_h(u_{n,t}(h) - 1) + \sum_{L < h \leq n} f_h u_{n,t}(h) - \sum_{h > L} f_h. \quad (4.3)$$

The final, signed ordinary tail tends to zero by hypothesis. The middle term is at most $n \sup_{h > L} |f_h| = o(1)$, using $h|f_h| \rightarrow 0$ and $n/(L+1) = O(1)$. The first term is at most

$$C_1 n^{-p} \sum_{h \leq L} h^p |f_h|.$$

Given $\varepsilon > 0$, choose a fixed H with $|f_h| \leq \varepsilon/h$ for $h > H$. The finitely many $h \leq H$ contribute $O_H(n^{-p})$, and the rest contributes at most $C_1 \varepsilon n^{-p} \sum_{h \leq L} h^{p-1} \leq C_p \varepsilon$. Letting $\varepsilon \downarrow 0$ proves uniform $o(1)$ for (4.3). Integrate on $[0, 1]$ and exponentiate. $\square$

The condition $f_h = o(1/h)$ is separate from convergence of the sum. In particular, we never infer it merely from a telescoping norm product. Neither this lemma nor Theorem 1.5 is a theorem for arbitrary ℓ^1 changes without a pointwise tail condition.

5 The general absolute amplitude from Freud norms

We now prove Theorem 1.1. Only the endpoint estimate and product stability are needed; the later shape and harmonic-occupation arguments play no role in this absolute baseline proof.

5.1 The precise published coefficient estimate

Fix $p > 0$, set $\beta = 2/p$, and define

$$A_F = B(\beta/2, 1/2)^{1/\beta} = (pI_p)^{p/2}. \quad (5.1)$$

For the fixed, *unnormalized* even measure

$$d\mu(x) = \exp(-|A_F x|^\beta) dx \quad (x \in \mathbb{R}),$$

let $\phi_h(x) = \gamma_h x^h + \dots$ be orthonormal with $\gamma_h > 0$. Claeys–Krasovsky–Minakov [8, Section 2, Eq. (2.3)] explicitly state, for every fixed $\beta > 0$,

$$\begin{aligned} \log \gamma_h = & -\frac{1}{2} \log \pi - \frac{h+1/2}{\beta} \log h + \frac{h}{\beta} + h \log 2 \\ & + \frac{\beta-4}{24\beta h} + O_\beta \left(\frac{1}{h \log^2 h} \right). \end{aligned} \quad (5.2)$$

They attribute this estimate to Kriecherbauer–McLaughlin [9, Theorem 1.5]. We use the exact modern restatement, directly checked in the March 2023 institutional manuscript, current September 2022 arXiv v2, and SISSA postprint. The original 1999 theorem text and proof were not directly accessible in the inspected sources.

The distinction between fixed and varying weights matters. Although [8] also studies an ensemble with varying weight $\exp(-n|y|^\beta)$, its γ_h in (5.2) is the leading coefficient for the fixed weight above, not the varying-weight coefficient χ_h . Equations (2.1)–(2.2) of that source connect them by a change of variables. Likewise the restriction $\beta \geq 1$ attached to a subsequent local polynomial estimate does not restrict the explicit all- β quantifier in (2.3). No analyticity hypothesis on the potential is being silently extended to small β .

5.2 Exact moments and probability normalization

The total mass is

$$M_F = \int_{\mathbb{R}} e^{-|A_F x|^\beta} dx = \frac{2\Gamma(1/\beta)}{\beta A_F} = \frac{p\Gamma(p/2)}{A_F}. \quad (5.3)$$

Let X have probability measure μ/M_F and put $Y = 2X$. Substituting $u = (A_F x)^\beta$ in the positive half-line integral gives the exact even moments

$$\nu_n = \mathbb{E}Y^{2n} = \left(\frac{4}{A_F^2} \right)^n \frac{\Gamma(pn + p/2)}{\Gamma(p/2)}, \quad \nu_0 = 1. \quad (5.4)$$

Let q_h be the squared norm of the degree- h monic polynomial for the law of Y . The monic polynomial for the unnormalized X measure is ϕ_h/γ_h , of norm γ_h^{-2} . Its transformed monic polynomial is $2^h \phi_h(Y/2)/\gamma_h$. Thus

$$q_h = \frac{4^h}{M_F \gamma_h^2}, \quad q_0 = 1. \quad (5.5)$$

In particular $\gamma_0 = M_F^{-1/2}$ handles the initial normalization exactly; no large- h formula is used at zero.

The concrete even probability measure has moments of every order and infinite support. Gram–Schmidt gives positive monic norms and zero diagonal recurrence, with coefficients

$$\tilde{\lambda}_h = q_h/q_{h-1} > 0. \quad (5.6)$$

Finite polynomial orthogonality gives

$$Z_n(\tilde{\lambda}) = \nu_n. \quad (5.7)$$

Indeed multiplication by y in the orthogonal-polynomial basis has these Jacobi coefficients; the coefficient of the degree-zero polynomial in y^{2n} equals its integral, and the closed multiplication walks are the weighted Dyck paths. Equivalently use a symmetric Jacobi truncation at heights $0, \dots, n$: a return walk of length $2n$ cannot go higher. No infinite self-adjoint realization or uniqueness of a representing measure is required. Moment indeterminacy for $\beta < 1$, equivalently $p > 2$, therefore causes no ambiguity in this finite identity for the chosen measure.

5.3 The undifferentiated error and product constant

Substitute (5.2) into (5.5). With $\beta = 2/p$ this gives

$$\log q_h = \log(\pi/M_F) + p(h + 1/2) \log h - ph + \frac{2p-1}{12h} + E_h, \quad E_h = O_p\left(\frac{1}{h \log^2 h}\right). \quad (5.8)$$

For $h \geq 2$, subtract the formula at $h-1$ and then subtract $p \log h$. The smooth leading bracket satisfies

$$(h + 1/2) \log h - (h - 1/2) \log(h - 1) - 1 - \log h = (h - 1/2) \log \frac{h}{h-1} - 1 = O(h^{-2}).$$

The first difference of the explicit $1/h$ term is also $O_p(h^{-2})$. Crucially we do *not* differentiate the unknown error or assume a smooth expansion for it. The triangle inequality alone gives

$$|E_h - E_{h-1}| \leq |E_h| + |E_{h-1}| = O_p\left(\frac{1}{h \log^2 h}\right).$$

Therefore

$$f_h := \log \frac{\tilde{\lambda}_h}{h^p} = O_p\left(\frac{1}{h \log^2 h}\right) = o(1/h). \quad (5.9)$$

This is absolutely summable. Also $\tilde{\lambda}_h/h^p \rightarrow 1$, so positivity at the finitely many earlier indices supplies common two-sided power bounds for $\tilde{\lambda}_h$ and h^p .

The product telescopes *exactly*:

$$\prod_{h=1}^N \frac{\tilde{\lambda}_h}{h^p} = \frac{q_N}{(N!)^p}. \quad (5.10)$$

Applying fixed- p Stirling asymptotics to (5.8) yields its positive limit

$$P = \frac{\pi}{M_F(2\pi)^{p/2}}. \quad (5.11)$$

Lemma 4.1 and (5.7) now give

$$\frac{\nu_n}{Z_n(h^p)} \longrightarrow P. \quad (5.12)$$

The orientation and the normalization at zero both matter: the desired moments are asymptotic to ν_n/P .

5.4 Simplifying the absolute prefactor

Fixed- p Stirling applied to the exact gamma moment gives

$$\nu_n \sim C_\nu \left(\frac{4p^p}{A_F^2} \right)^n (n!)^p n^{-1/2}, \quad C_\nu = \frac{(2\pi)^{(1-p)/2} p^{(p-1)/2}}{\Gamma(p/2)}. \quad (5.13)$$

The shift $pn + p/2$ has not been replaced by pn ; it supplies the essential power of p in C_ν . Equations (5.1) and (5.3) imply

$$\begin{aligned} \frac{4p^p}{A_F^2} &= \frac{4}{I_p^p} = d_p, \\ \frac{C_\nu}{P} &= \frac{(2\pi)^{(1-p)/2} p^{(p-1)/2}}{\Gamma(p/2)} \frac{M_F(2\pi)^{p/2}}{\pi} \\ &= \sqrt{\frac{2}{\pi}} \frac{p^{(p+1)/2}}{A_F} = \sqrt{\frac{2p}{\pi}} I_p^{-p/2} = \mathcal{C}_p. \end{aligned} \quad (5.14)$$

Together with (5.12), this proves the leading equivalent in Theorem 1.1 for every fixed $p > 0$.

5.5 The logarithmic remainder

The same input gives more than qualitative product stability. From (5.8) and Stirling,

$$\sum_{h=1}^L f_h = \log \frac{q_L}{(L!)^p} = \log P + \frac{p-1}{12L} + O_p\left(\frac{1}{L \log^2 L}\right). \quad (5.15)$$

The coefficient is $(2p-1)/12 - p/12$; the next $O_p(L^{-3})$ Stirling term is absorbed. Thus the *signed* tail satisfies

$$\left| \sum_{h>L} f_h \right| = O_p(L^{-1}). \quad (5.16)$$

The bound obtained by summing $|f_h|$ would be only $O_p(1/\log L)$, so this exact telescoping information must be retained.

Interpolate $\lambda_h(t) = h^p e^{t f_h}$ and again take $L = \lfloor \rho n \rfloor$. The logarithmic derivative minus $\log P$ is precisely (4.3). Its third term is $O_p(1/n)$ by (5.16); its middle term is

$$O_p\left(n \sup_{h>L} \frac{1}{h \log^2 h}\right) = O_p((\log n)^{-2}).$$

For the endpoint term use $C_1 n^{-p} \sum_{h \leq L} h^p |f_h|$. Finitely many exceptional heights contribute $O_p(n^{-p})$. Split the remaining sum at $\lfloor \sqrt{n} \rfloor$. Below the split the weaker estimate $|f_h| \leq C/h$ gives $O_p(n^{-p/2})$. Above it, $\log h \geq (\log n)/2$, so (5.9) and $\sum_{h \leq L} h^{p-1} = O_p(n^p)$ give $O_p((\log n)^{-2})$. For fixed $p > 0$, the former bound is smaller eventually. If $L < \sqrt{n}$ at first, this only changes the onset.

All estimates hold uniformly in t . Integration gives

$$\log \frac{\nu_n}{Z_n(h^p)} = \log P + O_p((\log n)^{-2}), \quad Z_n(h^p) = \frac{\nu_n}{P} (1 + O_p((\log n)^{-2})).$$

The exact gamma formula (5.4) has relative Stirling error $O_p(1/n)$. Equations (5.13)–(5.14) therefore prove the full statement (1.6).

The constants inherit the fixed-parameter dependence of the published estimate and the end-point class. They are not supplied effectively. This proof neither strengthens the Freud input nor differentiates its error, and it does not transfer the logarithmic rate to a general harmonic perturbation.

[write] **The trust boundary of Theorem 1.1, 7 October 2026.** Everything in this section is proved here except one input: the strong asymptotic (5.2) for the leading coefficients of the fixed Freud weight, for every $\beta > 0$. The source used it as restated by Claeys, Krasovsky and Minakov [8, Eq. (2.3)], who attribute it to Kriecherbauer and McLaughlin [9, Theorem 1.5]; neither the source nor the write read the 1999 theorem itself. Theorem 1.1, and with it the absolute statements of Theorem 1.6, Corollary 15.2 and Part III's references to the general-power baseline, are therefore proved *relative to that cited theorem*. The relative transfer theorems (Theorems 1.2, 1.3, 1.4, 1.5) do not use it, and the cubic and quartic constants are checked independently in Sections 11 and 12.

6 Direct uniform concentration under vanishing perturbations

The singular statistic $\sum_h N_h/h$ requires both macroscopic shape and boundary control. We now establish the shape directly under the entire interpolated family. A bounded Radon–Nikodym comparison with the pure-power law would be invalid: for instance a nonzero f_1 changes the weight of $(UD)^n$ by e^{nf_1} . Uniform concentration must therefore be proved rather than transferred by a pathwise bounded-density shortcut.

In this section only, assume an arbitrary fixed real sequence $f_h \rightarrow 0$ and put

$$\lambda_h(t) = h^p e^{tf_h}, \quad 0 \leq t \leq 1. \quad (6.1)$$

Interpolate the path heights linearly, with time divided by n and height by n . Thus $y_n(s)$, $0 \leq s \leq 2$, belongs to the compact uniform-topology space

$$\mathcal{K} = \{y : [0, 2] \rightarrow [0, \infty) : y(0) = y(2) = 0, \quad y \text{ is 1-Lipschitz}\}.$$

Every $y \in \mathcal{K}$ satisfies $y(s) \leq \min(s, 2-s) \leq 1$. Compactness follows from uniform boundedness, equicontinuity, and closure.

For $|v| \leq 1$ define the binary entropy

$$\mathcal{E}(v) = -\frac{1+v}{2} \log \frac{1+v}{2} - \frac{1-v}{2} \log \frac{1-v}{2}, \quad 0 \log 0 = 0,$$

and the extended-real functionals

$$S(y) = \int_0^2 \mathcal{E}(y'(s)) \, ds, \quad V(y) = \frac{p}{2} \int_0^2 \log y(s) \, ds, \quad \Phi_p(y) = S(y) + V(y). \quad (6.2)$$

Lipschitz functions are differentiable almost everywhere. A divergent negative logarithmic integral means $V(y) = -\infty$.

If H_j is the upper endpoint height of step j , upward/downward edge balance gives the exact identity

$$\log \frac{W_{\lambda(t)}(D)}{n^{pn}} = \frac{1}{2} \sum_{j=1}^{2n} \left[p \log(H_j/n) + t f_{H_j} \right]. \quad (6.3)$$

The factor $1/2$ is essential: each weighted descent is paired with the corresponding number of ascents across its edge.

6.1 A clipped closed-set upper bound

Fix $\varepsilon \in (0, 1]$ and an integer $M \geq 1$. Set

$$F = \max(0, \sup_h f_h), \quad \omega_M = \sup_{h > M} |f_h| \longrightarrow 0.$$

For $n \geq (M/\varepsilon)e^{F/p}$, every $h \geq 1$ and $0 \leq t \leq 1$ satisfy

$$p \log(h/n) + t f_h \leq p \log \max(h/n, \varepsilon) + \omega_M. \quad (6.4)$$

For $h > M$ the perturbation is bounded by ω_M . For $h \leq M$, the chosen n gives $p \log(h/n) + F \leq p \log \varepsilon$. Thus the finitely many large low-height values are absorbed into the clipping slack, rather than charged an uncontrolled $O(n)$ error.

For a finite partition $Q = (s_i)$ of $[0, 2]$, define

$$S_Q(y) = \sum_i (s_{i+1} - s_i) \mathcal{E} \left(\frac{y(s_{i+1}) - y(s_i)}{s_{i+1} - s_i} \right), \quad V_\varepsilon(y) = \frac{p}{2} \int_0^2 \log \max(y(s), \varepsilon) ds.$$

Both are continuous on $\mathcal{K}$. Nested dyadic partitions give $S_Q \downarrow S$: Jensen's inequality gives monotonicity, and differentiation of Lipschitz functions with bounded convergence identifies the limit. Likewise $V_\varepsilon \downarrow V$ as $\varepsilon \downarrow 0$, by monotone convergence applied to the nonnegative function $-\log \max(y, \varepsilon)$.

Let $B \subseteq \mathcal{K}$ be closed. Divide the $2n$ steps into blocks at rounded times ns_i . There are only polynomially many vectors of block endpoint heights. For a block of length L and displacement d , the bridge count is at most

$$\binom{L}{(L+d)/2} \leq \exp\{L\mathcal{E}(d/L)\};$$

impossible parities give no bridge, and nonnegativity can only decrease the count. For fixed Q , rounding errors in normalized entropy are $o(1)$ uniformly over paths, by continuity of $\mathcal{E}$. The step upper height divided by n differs from $y_n(s)$ by at most $1/n$ within its interval. Since the clipped logarithm is $1/\varepsilon$ -Lipschitz, (6.3)–(6.4) give

$$\limsup_{n \rightarrow \infty} \sup_{t \in [0, 1]} \frac{1}{n} \log \left(n^{-pn} \sum_{D: y_n \in B} W_{\lambda(t)}(D) \right) \leq \sup_{y \in B} (S_Q(y) + V_\varepsilon(y)) + \omega_M. \quad (6.5)$$

The conventions are $\log 0 = -\infty$ and $\sup \emptyset = -\infty$.

First let $M \rightarrow \infty$. Then take nested dyadic partitions and decreasing $\varepsilon \rightarrow 0$. For a decreasing sequence of continuous functions on a compact set, the maxima converge to the supremum of their pointwise upper-semicontinuous limit. Indeed choose maximizing points, take a convergent subsequence, and bound its limiting values by each fixed earlier function; the reverse inequality is immediate. Applying this to the compact set B proves

$$\limsup_{n \rightarrow \infty} \sup_t \frac{1}{n} \log \left(n^{-pn} \sum_{D: y_n \in B} W_{\lambda(t)}(D) \right) \leq \sup_{y \in B} \Phi_p(y). \quad (6.6)$$

It also shows that Φ_p is upper semicontinuous. This is the closed-set large-deviation upper estimate actually needed below.

6.2 The unique calibrated arch

Define y_* increasingly on $[0, 1]$ by

$$s = \int_0^{y_*(s)} \frac{dv}{\sqrt{1 - (v/H_p)^p}}, \quad (6.7)$$

and reflect about $s = 1$. Since $H_p I_p = 1$, its maximum is H_p , and $y_* \in \mathcal{K}$. In the interior,

$$(y'_*)^2 = 1 - (y_*/H_p)^p, \quad y''_* = -\frac{p}{2H_p^p} y_*^{p-1}. \quad (6.8)$$

The second identity extends through the top by continuation. At either endpoint, height divided by the distance to that endpoint tends to one.

Lemma 6.1 (Calibration and uniqueness). *The functional Φ_p has the unique maximizer y_* on $\mathcal{K}$.*

Proof. Set

$$A_*(s) = \mathcal{E}'(y'_*(s)) = \frac{1}{2} \log \frac{1 - y'_*(s)}{1 + y'_*(s)}.$$

Equations (6.8) give

$$A'_*(s) = \frac{p}{2y_*(s)}, \quad A_*(s) = O(1 + |\log s| + |\log(2 - s)|).$$

The entropy and logarithm tangent inequalities are

$$\begin{aligned} \mathcal{E}(y') - \mathcal{E}(y'_*) &\leq A_*(y' - y'_*), \\ \frac{p}{2}(\log y - \log y_*) &\leq \frac{p}{2} \frac{y - y_*}{y_*}. \end{aligned}$$

Integrate on $[d, 2 - d]$ and integrate the first right side by parts. The interior terms cancel. The remaining boundary term $[A_*(y - y_*)]_d^{2-d}$ is $O(d|\log d|)$ because both heights are $O(d)$ at the endpoints.

The tangent terms are integrable: A_* is integrable and $(y - y_*)/y_*$ is bounded near the endpoints. If $V(y) = -\infty$ there is nothing to prove; otherwise taking $d \downarrow 0$ proves $\Phi_p(y) \leq \Phi_p(y_*)$. If $y \neq y_*$, continuity makes them different on a set of positive measure, and the logarithmic tangent deficit has strictly positive integral,

$$\frac{p}{2} \int_0^2 \left[\frac{y}{y_*} - 1 - \log \frac{y}{y_*} \right] ds > 0,$$

possibly infinite. The strict inequality proves uniqueness, including nonsmooth competitors, zero plateaus, and multiple arches. $\square$

For every fixed $\eta_0 > 0$, compactness and upper semicontinuity therefore give a strict gap

$$\sup_{\|y - y_*\|_\infty \geq \eta_0} \Phi_p(y) < \Phi_p(y_*). \quad (6.9)$$

6.3 A matching lower bound with controlled endpoints

We provide the lower bound explicitly because the logarithm is singular at zero. Approximate y_* by positive polygons $z \in \mathcal{K}$ that equal s on $[0, \varepsilon]$ and $2 - s$ on $[2 - \varepsilon, 2]$, and have all interior slopes strictly between -1 and 1 . For example join $(\varepsilon, \varepsilon)$ to $(2\varepsilon, y_*(2\varepsilon))$, use fine chords of y_* on the remaining interior, and reflect at the end. The joining slope is greater than -1 , less than 1 , and tends to 1 as $\varepsilon \downarrow 0$.

These approximants converge uniformly to y_* , their derivatives converge almost everywhere, and they satisfy $z(s) \geq c_0 \min(s, 2 - s)$ with a common $c_0 > 0$ for sufficiently advanced approximants. The bounded entropy and an integrable constant plus $|\log \min(s, 2 - s)|$ dominate the two integrands. Hence

$$\Phi_p(z) \longrightarrow \Phi_p(y_*). \quad (6.10)$$

Fix one polygon. Refine its linear interior pieces into blocks of maximal length d without changing z . Choose lattice times and parity-compatible lattice heights within $O(1)$ of their prescribed values $ns_i, nz(s_i)$. Take deterministic all-up and all-down endpoint pieces of rounded lengths. For large n , the fixed positive slack of every interior slope from ± 1 guarantees feasible bridges.

Let $b_z = \min_{[\varepsilon, 2-\varepsilon]} z > 0$. Every unrestricted bridge in an interior block stays within $2d + O(1/n)$ of z , since both its slopes and those of z have absolute value at most one. Choosing d sufficiently small relative to b_z makes *every* such bridge positive. Their concatenations are consequently admissible Dyck paths. Distinct bridge choices give distinct paths. Stirling's formula for the finitely many binomial factors gives

$$\log \#\{\text{constructed paths}\} = nS(z) + o(n).$$

The endpoint pieces have zero entropy.

On the interior, the logarithm is uniformly Lipschitz and the normalized pure-power log weights approach the *interior contribution* to $nV(z)$, with error $nO_z(d) + o(n)$. The deterministic endpoints contribute separately

$$p \sum_{h=1}^{\lfloor \varepsilon n \rfloor} \log(h/n) + O(\log n), \quad (6.11)$$

whose quotient by n tends to $p \int_0^\varepsilon \log s \, ds$. This is exactly the two-endpoint contribution to $V(z)$, including the factor $1/2$ in (6.3).

The perturbation is uniformly $o(n)$ over all these paths and all t . On the deterministic endpoints its absolute contribution is at most $\sum_{h \leq \varepsilon n + O(1)} |f_h| = o(n)$ by Cesàro convergence. On the interior all heights are at least $c_z n$, so the absolute contribution is at most $n \sup_{h \geq c_z n} |f_h| = o(n)$. It follows that

$$\liminf_{n \rightarrow \infty} \inf_t \frac{1}{n} \log[n^{-pn} Z_n(\lambda(t))] \geq \Phi_p(z) - O_z(d).$$

The limits are taken in order: first $n \rightarrow \infty$ at fixed polygon and refinement, then $d \downarrow 0$, then improve the polygon. Combining with (6.6) for $B = \mathcal{K}$ proves the uniform variational limit

$$\sup_{t \in [0,1]} \left| \frac{1}{n} \log[n^{-pn} Z_n(\lambda(t))] - \Phi_p(y_*) \right| \longrightarrow 0. \quad (6.12)$$

Using (6.6) on the closed complement of an η_0 -ball and the gap (6.9) yields the needed conclusion.

Proposition 6.2 (Uniform fixed-neighborhood concentration). *For every fixed $p > 0$, fixed real sequence $f_h \rightarrow 0$, and fixed $\eta_0 > 0$, there are $c_{\eta_0} > 0$ and n_{η_0} such that*

$$\sup_{0 \leq t \leq 1} \mathbb{P}_{n,t}(\|y_n - y_*\|_\infty \geq \eta_0) \leq e^{-c_{\eta_0} n} \quad (n \geq n_{\eta_0}). \quad (6.13)$$

Only fixed-neighborhood concentration and the displayed closed-set upper estimate have been proved or used. No rate for shrinking neighborhoods, fluctuation theorem, or full lower large-deviation principle on arbitrary open sets is asserted.

7 The regularized harmonic occupation constant

Retain a fixed vanishing sequence f_h and the interpolation (6.1). Since f_h is bounded, the whole family lies in one power class (1.10), so the endpoint theorem applies uniformly in t . This observation allows the next statement to be used for both harmonic and parity-harmonic perturbations.

Proposition 7.1 (Harmonic occupation). *Uniformly for $0 \leq t \leq 1$,*

$$\sum_{h=1}^n \frac{u_{n,t}(h)}{h} = \log n + \gamma + K_p + o(1). \quad (7.1)$$

Proof. For every bounded height function g , edge balance gives

$$\frac{1}{n} \sum_h N_h g(h/n) = \frac{1}{2n} \sum_{j=1}^{2n} g(H_j/n). \quad (7.2)$$

Fix $0 < \varepsilon < \min(\rho, H_p)$ and put $L = \lfloor \varepsilon n \rfloor$. Use $g_\varepsilon(x) = x^{-1} \mathbf{1}_{\{x > \varepsilon\}}$, with value zero at zero. Proposition 6.2 gives

$$\sum_{h > L} \frac{u_{n,t}(h)}{h} \rightarrow \frac{1}{2} \int_0^2 g_\varepsilon(y_*(s)) ds = T_\varepsilon := \int_\varepsilon^{H_p} \frac{dy}{y \sqrt{1 - (y/H_p)^p}} \quad (7.3)$$

uniformly in t .

Here is a justification despite the cutoff's discontinuity. The arch hits height ε at exactly two times. The set $A_\delta = \{s : |y_*(s) - \varepsilon| \leq \delta\}$ has measure tending to zero as $\delta \downarrow 0$. If $\|y_n - y_*\|_\infty + 1/n < \delta/2$, the cutoff indicators agree outside A_δ ; where they are one the reciprocal difference is bounded by a constant times $\varepsilon^{-2}(\|y_n - y_*\|_\infty + 1/n)$. On A_δ the functions are bounded by $1/\varepsilon$. Thus the path functional is uniformly close to its arch value on a sufficiently small fixed neighborhood. Its complement has exponentially small probability and the functional is bounded, proving uniform expectation convergence. There is no lattice mismatch: $h/n > \varepsilon$ is exactly $h > \lfloor \varepsilon n \rfloor$. The two arch branches cancel the $1/2$ in (7.2).

The low-height remainder satisfies, for fixed $p > 0$,

$$\left| \sum_{h \leq L} \frac{u_{n,t}(h) - 1}{h} \right| \leq C_1 n^{-p} \sum_{h \leq L} h^{p-1} \leq C_p \varepsilon^p. \quad (7.4)$$

The power-sum bound is valid also for $0 < p < 1$. The ordinary harmonic number contributes

$$\sum_{h \leq L} \frac{1}{h} = \log n + \log \varepsilon + \gamma + o(1).$$

To evaluate the regularized bulk integral, substitute $v = (y/H_p)^p$:

$$\begin{aligned} T_\varepsilon &= \frac{1}{p} \int_{(\varepsilon/H_p)^p}^1 \frac{dv}{v\sqrt{1-v}} = \frac{2}{p} \operatorname{artanh} \sqrt{1 - (\varepsilon/H_p)^p} \\ &= \log(H_p/\varepsilon) + \frac{2}{p} \log \left(1 + \sqrt{1 - (\varepsilon/H_p)^p} \right). \end{aligned} \quad (7.5)$$

Hence $\log \varepsilon + T_\varepsilon \rightarrow K_p$. More precisely, combining the previous estimates gives

$$\limsup_{n \rightarrow \infty} \sup_t \left| \sum_h \frac{u_{n,t}(h)}{h} - \log n - \gamma - K_p \right| \leq C_p \varepsilon^p + |\log \varepsilon + T_\varepsilon - K_p|.$$

Let $\varepsilon \downarrow 0$ after $n \rightarrow \infty$. This proves (7.1) without selecting an unproved shrinking cutoff. $\square$

The Euler constant has a concrete origin in the low-height harmonic sum. Its positive sign is necessary: it later cancels the Euler constant in the Weierstrass gamma product.

8 Residual estimates and harmonic interpolation

Under the hypotheses of Theorem 1.2, the interpolation $\lambda_h(t) = h^p e^{tf_h}$ has a common power class, and Proposition 7.1 applies. For the residual r_h take $L = \lfloor \rho n \rfloor$ and repeat the proof of summable stability:

$$\begin{aligned} \sum_{h \leq n} r_h u_{n,t}(h) - \sum_{h \geq 1} r_h &= O \left(n^{-p} \sum_{h \leq L} h^{p-1-\delta} + n^{-\delta} \right) \\ &= \begin{cases} O(n^{-\delta}), & 0 < \delta < p, \\ O(n^{-p} \log n), & \delta = p, \\ O(n^{-p}), & \delta > p. \end{cases} \end{aligned} \quad (8.1)$$

The constants are uniform in t and depend on the fixed parameters and class bounds. In every case this is $o(1)$. These rates control only the residual contribution; they are not rates for the harmonic term or for the whole equivalent.

Differentiation of the finite path sum yields

$$\begin{aligned} \frac{d}{dt} \log Z_n(\lambda(t)) &= a \sum_{h=1}^n \frac{u_{n,t}(h)}{h} + \sum_{h=1}^n r_h u_{n,t}(h) \\ &= a(\log n + \gamma + K_p) + \sum_{h \geq 1} r_h + o(1), \end{aligned}$$

uniformly in t . Integrating on $[0, 1]$ and exponentiating proves (1.7). Independently,

$$\sum_{h=1}^N f_h - a \log N \longrightarrow a\gamma + \sum_{h \geq 1} r_h,$$

which proves the product formula. This proof integrates expectations under every interpolated law. It does not attempt an unjustified passage from convergence in probability under one law to exponential moments of an unbounded statistic.

9 Parity cancellation by exact two-step conditioning

We now prove Theorem 1.3. Under (1.8), $|f_h| \leq D_0/h$ after increasing a fixed constant to cover small heights. The weights $\lambda_h(t) = h^p e^{t f_h}$ belong to a common power class and satisfy

$$\left| \log \frac{\lambda_{h+1}(t)}{\lambda_h(t)} \right| \leq p \log(1 + 1/h) + |f_{h+1}| + |f_h| \leq \frac{D}{h}. \quad (9.1)$$

Therefore

$$\frac{|\lambda_{h+1}(t) - \lambda_h(t)|}{\lambda_{h+1}(t) + \lambda_h(t)} \leq \frac{D}{2h}, \quad (9.2)$$

using $\tanh(|x|/2) \leq |x|/2$. Monotonicity of the weights is not needed.

9.1 The signed statistic with a smooth cutoff

Fix $\varepsilon > 0$. Choose a nondecreasing C^1 function $\chi : [0, \infty) \rightarrow [0, 1]$, zero for $x \leq \varepsilon$ and one for $x \geq 2\varepsilon$. Set

$$\psi(x) = \chi(x)/x \ (x > 0), \quad \psi(0) = 0, \quad w_h = \frac{\chi(h/n)}{h} = \frac{1}{n} \psi(h/n), \quad G_n(D) = \sum_{h \geq 1} (-1)^h w_h N_h(D).$$

For fixed ε , both ψ and ψ' are bounded, ψ' is integrable on $[0, \infty)$, and

$$|w_{h+1} - w_h| \leq C_\varepsilon/n^2, \quad w_h \leq 1/(\varepsilon n) \quad \text{when } w_h \neq 0. \quad (9.3)$$

Group the steps into n consecutive pairs. Every pair starts at an even time, hence at an even height k . Condition on all path vertices except the middle vertex of one particular pair. Its endpoints and all external weight factors are fixed. There are four cases:

- A pair UU makes no descent and contributes zero to G_n .
- A pair DD contributes $w_k - w_{k+1}$, of magnitude at most C_ε/n^2 .
- For equal endpoints $k \geq 2$, both UD and DU are nonnegative and feasible. Their local weights are respectively $\lambda_{k+1}(t)$ and $\lambda_k(t)$; their contributions to G_n are $-w_{k+1}$ and w_k .
- At equal endpoints $k = 0$, only UD is feasible. Its contribution $-w_1$ is zero once $n \geq 1/\varepsilon$.

For equal endpoints $k \geq 2$, the exact conditional mean is consequently

$$\frac{\lambda_k(t)w_k - \lambda_{k+1}(t)w_{k+1}}{\lambda_k(t) + \lambda_{k+1}(t)}. \quad (9.4)$$

Its absolute value is bounded by

$$|w_k - w_{k+1}| + |w_k| \frac{|\lambda_k(t) - \lambda_{k+1}(t)|}{\lambda_k(t) + \lambda_{k+1}(t)}.$$

If $w_k \neq 0$ then $k > \varepsilon n$, so (9.2)–(9.3) bound this by C'_ε/n^2 . If $w_k = 0$, the second term vanishes. Thus linearity and the tower property, separately for each pair's conditioning, give

$$\sup_t |\mathbb{E}_{n,t} G_n| \leq C'_\varepsilon/n \longrightarrow 0. \quad (9.5)$$

This is an estimate for a *signed expectation*. The pairs have not been declared independent, and neither a vanishing absolute moment nor concentration of G_n follows from this argument or is used.

9.2 Removing the cutoff

Take $2\varepsilon < \rho$ from Theorem 1.4 and extend $u_{n,t}(h) = 0$ for $h > n$. The low part after subtracting one traversal per edge satisfies

$$\left| \sum_{h \geq 1} (1 - \chi(h/n)) \frac{(-1)^h (u_{n,t}(h) - 1)}{h} \right| \leq C_1 n^{-p} \sum_{h \leq 2\varepsilon n} h^{p-1} \leq C_p \varepsilon^p. \quad (9.6)$$

The high deterministic baseline tends to zero as well. The partial sums of $(-1)^h$ are bounded, and summation by parts gives

$$\begin{aligned} \left| \sum_{h \geq 1} (-1)^h w_h \right| &\leq |w_1| + \sum_{h \geq 1} |w_{h+1} - w_h| \\ &\leq |w_1| + \frac{1}{n} \int_0^\infty |\psi'(x)| dx = O_\varepsilon(n^{-1}). \end{aligned} \quad (9.7)$$

The infinite sum converges because $w_h \rightarrow 0$ and has finite variation. The difference between the full occupation sum and the full alternating harmonic series is exactly the low term in (9.6), plus $\mathbb{E}G_n$, minus the deterministic sum in (9.7). There is no omitted high baseline tail. Let $n \rightarrow \infty$ and then $\varepsilon \downarrow 0$ to obtain

$$\sum_{h=1}^n \frac{(-1)^h u_{n,t}(h)}{h} = -\log 2 + o(1) \quad \text{uniformly for } 0 \leq t \leq 1. \quad (9.8)$$

The direct shape theorem applies since $f_h \rightarrow 0$, so (7.1) remains valid. The residual estimate (8.1) is unchanged. The logarithmic derivative of Z_n is therefore

$$a(\log n + \gamma + K_p) - b \log 2 + \sum_{h \geq 1} r_h + o(1)$$

uniformly in t . Integrating proves Theorem 1.3, and the elementary harmonic and alternating-harmonic sums prove (1.9). Nothing here asserts transfer for arbitrary periodic $1/h$ terms; the proof uses the specific parity structure of two-step blocks.

10 Gamma products for shifted factors and positive polynomials

For $\alpha > -1$, the positive weights $h^{p-1}(h + \alpha)$ have

$$f_h = \log(1 + \alpha/h) = \alpha/h + O(h^{-2}), \quad \prod_{h=1}^N (1 + \alpha/h) = \frac{\Gamma(N + 1 + \alpha)}{\Gamma(1 + \alpha)\Gamma(N + 1)}.$$

Fixed-shift Stirling asymptotics yield $P_f = 1/\Gamma(1 + \alpha)$. Thus

$$\frac{Z_n(h^{p-1}(h + \alpha))}{Z_n(h^p)} \sim \frac{(r_p n)^\alpha}{\Gamma(1 + \alpha)}. \quad (10.1)$$

For $c > -1$, the fully shifted powers similarly give

$$\frac{Z_n((h + c)^p)}{Z_n(h^p)} \sim \frac{(r_p n)^{pc}}{\Gamma(1 + c)^p}. \quad (10.2)$$

More generally, if $c_j > -1$, $b_j > 0$, and $\sum_j b_j = p$, then the weight $\prod_j (h + c_j)^{b_j}$ has $a = \sum_j b_j c_j$ and product constant $\prod_j \Gamma(1 + c_j)^{-b_j}$. The parameters in these statements are fixed as $n \rightarrow \infty$.

For a real positive-at-positive-integers polynomial $Q(x) = \kappa \prod_{j=1}^d (x + c_j)$, positivity excludes a negative-integer c_j . Let $a = \sum_j c_j \in \mathbb{R}$. The logarithm of the *whole positive value* has expansion

$$\log \frac{Q(h)}{\kappa h^d} = \frac{a}{h} + O(h^{-2}).$$

No logarithm of an individual negative linear factor is introduced. The gamma functional equation proves the exact identity

$$\prod_{h=1}^N \frac{Q(h)}{\kappa h^d} = \prod_{j=1}^d \frac{\Gamma(N+1+c_j)}{\Gamma(1+c_j)\Gamma(N+1)}. \quad (10.3)$$

For every fixed shift, real or complex, the gamma quotient is asymptotic to N^{c_j} along positive real N . Therefore the normalized product tends to $\prod_j \Gamma(1+c_j)^{-1}$. Theorem 1.2 proves (1.13). Its product constant is positive by (1.7); hence the gamma product is real and positive too. This argument handles conjugate roots and real shifts below -1 without a branch ambiguity.

For example the cubic $h(h^2 + 1)$ is positive and has shifts $0, i, -i$, so

$$\prod_j \Gamma(1+c_j) = \Gamma(1+i)\Gamma(1-i) = \frac{\pi}{\sinh \pi}, \quad a = 0.$$

Its leading ratio to pure cubic moments is $\sinh \pi / \pi$. The real cubic $h(h - 6/5)(h - 9/5)$ is also positive at every positive integer: both nontrivial factors are negative at $h = 1$, and both are positive for $h \geq 2$. Here $a = -3$ and the positive gamma product is $\Gamma(-1/5)\Gamma(-4/5)$. A quartic example is $h^2(h - 3/2)^2$, with $a = -3$ and gamma product $\Gamma(-1/2)^2 = 4\pi$. Positivity on the integers, rather than positivity of each factor on the whole positive real axis, is the relevant hypothesis.

10.1 Absolute amplitudes and a broader residual corollary

Combining (1.13) with Theorem 1.1 at $p = d$ gives exactly (1.14). Its amplitude is $\mathcal{C}_d r_d^a / \prod_j \Gamma(1+c_j)$, its exponential base is κr_d^d , and its remaining power is $n^{a-1/2}$. This multiplication yields only a relative $o(1)$ error for the general family, because the harmonic transfer has no stated logarithmic rate. For $d = 3, 4$ it gives (1.15) and (1.16); the next two sections recover their pure baselines by independent classical anchors.

The small-product lemma also permits a broader, carefully separated residual hypothesis.

Corollary 10.1 (Convergent residuals after harmonic transfer). *Let $p > 0$, $\kappa > 0$, $a, b \in \mathbb{R}$ be fixed, and suppose $\lambda_h > 0$ with*

$$\log \frac{\lambda_h}{\kappa h^p} = \frac{a}{h} + \frac{b(-1)^h}{h} + r_h, \quad r_h = o(1/h), \quad \sum_{h \geq 1} r_h \text{ convergent.}$$

The residual sum is an ordinary real series and need not converge absolutely. Then

$$Z_n(\lambda) \sim \mathcal{C}_p r_p^a P_f (\kappa d_p)^n (n!)^p n^{a-1/2}, \quad P_f = \exp \left(a\gamma - b \log 2 + \sum_{h \geq 1} r_h \right) > 0. \quad (10.4)$$

Equivalently $P_f = \lim_{N \rightarrow \infty} N^{-a} \prod_{h \leq N} \lambda_h / (\kappa h^p)$.

Proof. First apply Theorem 1.3 to the exact comparison weights $\lambda_h^{(0)} = h^p \exp(a/h + b(-1)^h/h)$, with zero residual. Next compare λ_h/κ to $\lambda_h^{(0)}$ using Lemma 4.1; both are in a common power class and their log ratio is precisely r_h . This contributes $\exp(\sum r_h)$. Multiply by the pure-power equivalent and restore the exact homogeneous factor κ^n . Ordinary harmonic and alternating harmonic summation gives the product limit. $\square$

Thus the original shape/parity proof is not being applied directly to a more irregular residual. The extension is a composition with a separate lemma, and still does not cover arbitrary ℓ^1 residuals without the pointwise $o(1/h)$ condition.

11 The classical Dixon anchor and the absolute cubic law

The parity theorem connects pure cubic weights with an exactly solved classical continued fraction. We track the analytic normalization because three different indices occur: Dyck semilength n , the power $3n + 1$ in a Dixon function, and its factorial $(3n + 1)!$.

11.1 The formal Laplace fraction and signs

Let sm and cm be the analytic-at-zero solutions of

$$\text{sm}' = \text{cm}^2, \quad \text{cm}' = -\text{sm}^2, \quad \text{sm}(0) = 0, \quad \text{cm}(0) = 1.$$

Conrad–Flajolet [16, Theorem 2], on PDF pages 11–12, give the formal Laplace identity

$$\int_0^\infty \text{sm}(u) e^{-u/x} du = \frac{x^2}{1 + \frac{d_1 x^3}{1 + \frac{d_2 x^3}{1 + \ddots}}}, \quad (11.1)$$

where

$$d_{2j-1} = (3j-2)(3j-1)^2, \quad d_{2j} = (3j)^2(3j+1) \quad (j \geq 1). \quad (11.2)$$

The integral is interpreted term by term: $u^m/m!$ maps to x^{m+1} . We do not require a convergent improper integral on the positive real axis.

Substitute $z = -x^3$ in the Dyck S-fraction (1.2). Comparison with (11.1) shows

$$\text{sm}(z) = \sum_{n \geq 0} (-1)^n Z_n(d) \frac{z^{3n+1}}{(3n+1)!}.$$

Following Conrad–Flajolet’s equation (19), define $\text{smh}(z) = -\text{sm}(-z)$. Since $(-1)^n(-1)^{3n+1} = -1$, all signs cancel:

$$\text{smh}(z) = \sum_{n \geq 0} Z_n(d) \frac{z^{3n+1}}{(3n+1)!}. \quad (11.3)$$

The first moments are 1, 4, 160, 20800, 6476800. The package compares the polynomial ODE coefficients with an independent Dyck computation through $n = 48$; the formal source theorem supplies the identity for all n .

11.2 Checking the analytic urn theorem

To establish the complete dominant-singularity statement, use Flajolet–Gabarró–Pekari’s *Analytic urns* [17], Theorems 1–2 and Corollary 1. For their replacement matrix

$$\begin{pmatrix} -1 & 2 \\ 2 & -1 \end{pmatrix},$$

the parameters are $a = b = s = 1$ and $h = a + b + s = 3$. The initial state is $a_0 = 1, b_0 = 0$, with $t_0 = a_0 + b_0 = 1$. The conditions $a, b, s > 0$ and the tenability/divisibility conditions

$$a \mid a_0, \quad b \mid b_0, \quad a \mid (b + s), \quad b \mid (a + s)$$

are all satisfied. Moreover $a_0 = a, b_0 = 0$ is exactly the initial specialization of Theorem 2.

Theorem 1’s auxiliary functions become

$$\delta(u) = (1 - u^3)^{1/3}, \quad J(u) = \int_0^u (1 - v^3)^{-2/3} dv, \quad \psi(J(u)) = \frac{u}{(1 - u^3)^{1/3}}. \quad (11.4)$$

The roots are the analytic determinations at zero. We rename the source’s auxiliary integral J to avoid collision with our I_3 .

Put $y = u/(1 - u^3)^{1/3}$. Direct differentiation gives

$$\frac{dy}{du} = (1 - u^3)^{-4/3}, \quad \frac{dJ}{du} = (1 - u^3)^{-2/3}, \quad 1 + y^3 = (1 - u^3)^{-1}.$$

Hence $\psi(0) = 0$ and $\psi' = (1 + \psi^3)^{2/3}$ near zero. Defining $c = (1 + \psi^3)^{1/3}$ gives the polynomial initial-value problem

$$\psi' = c^2, \quad c' = \psi^2, \quad \psi(0) = 0, \quad c(0) = 1.$$

The pair $(-\text{sm}(-z), \text{cm}(-z))$ satisfies the same equations. Local analytic uniqueness thus identifies $\psi = \text{smh}$, with no untracked rescaling.

11.3 All three dominant poles and their phases

Theorem 2 of [17] gives the radius

$$\rho_D = J(1) = \frac{1}{3}B(1/3, 1/3) = \frac{\Gamma(1/3)^2}{3\Gamma(2/3)}. \quad (11.5)$$

It proves analyticity inside $|z| < \rho_D$, identifies exactly the three singularities $\rho_D, \rho_D\omega, \rho_D\omega^2$ on the circle, where $\omega = e^{2\pi i/3}$, and supplies analytic continuation in sectors of angle larger than π at them, with regularity elsewhere on that circle. These global facts are needed for coefficient transfer; locating a positive real pole alone would not suffice.

The source’s displayed singular expansion (23), whose leading exponent is $-t_0/s$, specializes to

$$\psi(z) = (\rho_D - z)^{-1} \mathcal{A}((\rho_D - z)^3), \quad \mathcal{A}(0) = 1, \quad (11.6)$$

with $\mathcal{A}$ analytic near zero. The residue at ρ_D is therefore -1 in the usual $(z - \rho_D)^{-1}$ convention. The support of (11.3), or the initial-value equations, gives $\psi(\omega z) = \omega\psi(z)$. At $z_j = \rho_D\omega^j$ the principal part is consequently

$$\frac{\omega^{2j}}{\rho_D\omega^j - z}, \quad \text{Res}_{z=z_j} \psi(z) = -\omega^{2j}. \quad (11.7)$$

Its contribution to $[z^m]\psi(z)$ is $\rho_D^{-m-1}\omega^{j(1-m)}$. For $m = 3n + 1$ all three phases are one. The singularity-transfer argument in Corollary 1 of [17] therefore gives

$$[z^{3n+1}]\text{smh}(z) \sim 3\rho_D^{-3n-2}, \quad Z_n(d) \sim 3\Gamma(3n+2)\rho_D^{-3n-2}. \quad (11.8)$$

The factor three counts the equally contributing poles; the extra inverse radius comes from $1/(\rho_D - z)$ rather than $1/(1 - z/\rho_D)$. No sharper source remainder is needed or transferred to the final perturbed family.

The beta integral for I_3 and duplication at $1/3$ show

$$I_3 = \frac{\Gamma(1/3)\sqrt{\pi}}{3\Gamma(5/6)}, \quad \rho_D = 2^{1/3}I_3, \quad r_3 = \frac{2^{2/3}}{I_3} = \frac{4\sqrt{3}\pi}{\Gamma(1/3)^3}. \quad (11.9)$$

For comparison, the real change of variables in (11.4) also gives $\rho_D = \int_0^\infty (1+y^3)^{-2/3} dy$, but that real integral does not replace the three-pole theorem.

11.4 The exact parity product and pure cubic amplitude

With $\kappa_D = 27/8$, the weights (11.2) are

$$d_h = \begin{cases} \kappa_D(h - 1/3)(h + 1/3)^2, & h \text{ odd,} \\ \kappa_D h^2(h + 2/3), & h \text{ even.} \end{cases}$$

They are positive. Their first logarithmic coefficients are $1/3$ on odd heights and $2/3$ on even heights, so

$$\log \frac{d_h}{\kappa_D h^3} = \frac{1/2}{h} + \frac{(1/6)(-1)^h}{h} + O(h^{-2}). \quad (11.10)$$

For an even cutoff $N = 2m$, separating the factors gives the exact product

$$\begin{aligned} \prod_{h=1}^{2m} \frac{d_h}{\kappa_D h^3} &= \frac{\Gamma(m+1/3)\Gamma(m+2/3)^2\Gamma(m+4/3)}{\Gamma(m+1/2)^3\Gamma(m+1)} \\ &\quad \cdot \frac{\Gamma(1/2)^3}{\Gamma(1/3)\Gamma(2/3)^2\Gamma(4/3)}. \end{aligned} \quad (11.11)$$

The first quotient is asymptotic to $m^{1/2}$. Reflection and recurrence reduce the constant on the second line to $9/(4\sqrt{\pi})$. The extra odd-cutoff factor tends to one; hence the full limit is

$$\lim_{N \rightarrow \infty} N^{-1/2} \prod_{h=1}^N \frac{d_h}{\kappa_D h^3} = \frac{9}{4\sqrt{2\pi}}. \quad (11.12)$$

The parity theorem now gives

$$\frac{Z_n(d)}{Z_n(h^3)} \sim \kappa_D^n (r_3 n)^{1/2} \frac{9}{4\sqrt{2\pi}}. \quad (11.13)$$

Finally Stirling's formula yields

$$\frac{\Gamma(3n+2)}{(n!)^3} \sim \frac{3^{3n+3/2}}{2\pi}.$$

Divide (11.8) by (11.13) and use (11.9). The result is

$$Z_n(h^3) \sim \sqrt{\frac{6}{\pi}} I_3^{-3/2} \left(\frac{4}{I_3^3}\right)^n (n!)^3 n^{-1/2}. \quad (11.14)$$

Together with (1.13), this proves (1.15). The intermediate Dixon fraction is genuinely parity-dependent; it is not the S-fraction for A218221 itself.

12 The quartic baseline and parallel amplitudes

The pure quartic equivalent established in Report 199 is

$$Z_n(h^4) \sim \sqrt{\frac{2}{\pi}} r^{4n+2} (n!)^4 n^{-1/2}, \quad r = \frac{\sqrt{2}}{I_4} = \frac{8\sqrt{\pi}}{\Gamma(1/4)^2}. \quad (12.1)$$

For completeness, the leading-order normalization can be recovered directly from the classical Berg–Valent comparison [11] and the summable theorem already proved here. Let $b_n = Z_n(\gamma)$ with $\gamma_h = h^2(h^2 - 1/4)$ and put $K_0 = \Gamma(1/4)^2/(4\sqrt{\pi}) = \sqrt{2}I_4$. Berg–Valent’s equation (3.1) uses the birth–death rates

$$\lambda_j = (4j+1)(4j+2)^2(4j+3), \quad \mu_j = (4j-1)(4j)^2(4j+1), \quad \mu_0 = 0.$$

The alternating S-coefficients $\alpha_{2j+1} = \lambda_j$, $\alpha_{2j} = \mu_j$ satisfy $\alpha_h = 4h^2(4h^2 - 1) = 16\gamma_h$. Their S-to-J contraction has diagonal $\alpha_{2j} + \alpha_{2j+1}$, with $\alpha_0 = 0$, and squared off-diagonal $\alpha_{2j+1}\alpha_{2j+2}$, agreeing with their equation (2.31). Thus their moments s_n satisfy $s_n = 16^n b_n$.

Their Proposition 3.6.1, pages 194–195, gives

$$b_n = \frac{4\Gamma(4n+2)}{16^n K_0^{4n+2}} (1 + O(25^{-n})). \quad (12.2)$$

Both γ_h and h^4 lie between $(3/4)h^4$ and h^4 . Since $\log(\gamma_h/h^4) = \log(1 - 1/(4h^2)) = O(h^{-2})$, Theorem 1.5 gives

$$\frac{b_n}{Z_n(h^4)} \longrightarrow \prod_{h \geq 1} \left(1 - \frac{1}{4h^2}\right) = \frac{2}{\pi}.$$

The product is Euler’s sine product at $1/2$, or (10.3) with shifts $0, 0, -1/2, 1/2$. Stirling’s formula in (12.2) now yields (12.1). This rederives only the leading baseline used here; the finer correction in Report 199 is not needed.

Applying the polynomial relative theorem to (12.1) proves (1.16). A harmonic perturbation changes the subleading problem, so neither the baseline’s first correction nor the comparison’s exponential remainder is carried unchanged into that general family.

13 OEIS applications and a matching interpretation

All identifications in this section use the current displayed definitions or programs as well as exact displayed-prefix checks. Finite prefix agreement alone would not prove a weight identity. Entries were inspected on 4 October 2026; no OEIS or repository record was changed.

13.1 A216966 and common-opener matching triples

The continued fraction for A216966 explicitly uses $\lambda_h = h^3$. Its first terms are

$$1, 1, 9, 297, 24273, 3976209, 1145032281, 530050022073.$$

A perfect matching on $[2n]$ has an opener at the smaller endpoint of each pair and a closer at the larger. Consider ordered triples of matchings having the same opener set. Scanning the vertices gives

a common Dyck path: an opener increments the unmatched-opener count, a closer decrements it, and the count never becomes negative. At a closer of current height h , each of the three matchings independently chooses one of its h available earlier openers. Thus the number of decorations of a fixed path is $\prod_h h^{3N_h}$. Conversely these choices produce exactly one ordered triple. Therefore A216966 counts these common-opener matching triples. This elementary interpretation is the cubic analogue of the matching quadruples for pure quartic moments; it does not require any connectedness condition.

Write $D_3 = r_3^3 = 4/I_3^3$ and $C_3 = \sqrt{3D_3/(2\pi)}$. Equation (11.14) reads

$$\text{A216966}(n) \sim C_3 D_3^n (n!)^3 n^{-1/2}. \quad (13.1)$$

In gamma notation $D_3 = 192\sqrt{3}\pi^3/\Gamma(1/3)^9$. These are exactly the already-posted Kotěšovec constants described in Section 1.

13.2 A218221 and tetrahedral descent weights

A218221's displayed finite-fraction program gives $\lambda_h = \binom{h+2}{3} = h(h+1)(h+2)/6$. Thus $\kappa = 1/6$, $a = 3$, and the gamma product is $\Gamma(1)\Gamma(2)\Gamma(3) = 2$. Its first terms are

$$1, 1, 5, 65, 1725, 81225, 6181125, 710984625.$$

The exact leading consequences are

$$\frac{6^n \text{A218221}(n)}{\text{A216966}(n)} \sim \frac{(r_3 n)^3}{2}, \quad (13.2)$$

$$\text{A218221}(n) \sim \sqrt{\frac{3}{8\pi}} r_3^{9/2} \left(\frac{r_3^3}{6}\right)^n (n!)^3 n^{5/2}. \quad (13.3)$$

The amplitude is equivalently $2^{15/2} 3^{11/4} \pi^4 / \Gamma(1/3)^{27/2} = 0.603824587006556\dots$, matching the posted formula. This is an application of a rigorous cubic baseline plus polynomial transfer. The Dixon weights begin 4, 36, 100, 252, $\dots$; after scaling the first to one they begin 1, 9, 25, $\dots$, whereas A218221's weights begin 1, 4, 10, $\dots$. The parity theorem is a substantive bridge, not an identification of these two S-fractions.

13.3 Lemniscatic and tree examples in degree four

The pure quartic moment sequence is A227887, with $\lambda_h = h^4$. Its first terms are

$$1, 1, 17, 1585, 485729, 372281761.$$

The following three examples are classical elliptic/tree families, used as applications and normalization tests [2].

A144853 identifies its S-fraction coefficients with A187756, namely

$$\lambda_h = \frac{h^2(4h^2 - 1)}{3} = \frac{4}{3}h^2(h - 1/2)(h + 1/2).$$

Here $a = 0$ and the gamma product is $\pi/2$. Therefore

$$\text{A144853}(n) \sim \frac{2}{\pi} \sqrt{\frac{2}{\pi}} r^2 \left(\frac{4}{3}r^4\right)^n (n!)^4 n^{-1/2}. \quad (13.4)$$

Its initial values are 1, 1, 21, 2541, 1023561, 1036809081. A261000 has weights $h^2(4h^2 - 1)$ and is exactly

$$A261000(n) = 3^n A144853(n). \quad (13.5)$$

This also follows from the entries' identical JacobiSD coefficient formulas with their displayed scale factors. Its interpretation is in unordered even-degree bilabeled increasing trees on $2n + 1$ nodes, and its first terms are 1, 3, 189, 68607, 82908441.

A144849 explicitly uses $h A139757(h)/3$, where $A139757(h) = (h + 1)(2h + 1)^2$. Thus

$$\lambda_h = \frac{h(h + 1)(2h + 1)^2}{3} = \frac{4}{3}h(h + 1/2)^2(h + 1).$$

Its shift sum is $a = 2$ and gamma product is $\pi/4$, giving

$$A144849(n) \sim \frac{4}{\pi} \sqrt{\frac{2}{\pi}} r^4 \left(\frac{4}{3}r^4\right)^n (n!)^4 n^{3/2}. \quad (13.6)$$

It begins 1, 6, 336, 77616, 50916096, 76307083776. The entry gives squared sine-lemniscate and increasing bilabeled strict binary tree interpretations, the latter with $4n + 2$ labels. In particular the comparison within this classical family is

$$\frac{A144849(n)}{A144853(n)} \sim 2r^2 n^2. \quad (13.7)$$

This explains the exponent and constant uniformly through the shifts. We do not claim that these classical elliptic asymptotics are new.

13.4 Pure quartic moments are not A338634

Let $M(z) = \sum_{n \geq 0} A227887(n)z^n$. A338634 is the distinct formal transform $A(x)$ satisfying

$$A(x) = M(x/A(x)^2), \quad M(z) = A(zM(z)^2).$$

Its positive-index coefficients are the even free cumulants of the symmetric moment sequence with even moments $Z_n(h^4)$ and vanishing odd moments. For example it begins 1, 1, 15, 1490, 472475, 367254494, rather than the raw-moment prefix above. Report 199 treats that transform and its connected noncrossing-closure interpretation. The transfer theorems in this article concern the raw weighted-Dyck moments Z_n , and must not be applied to A338634 by silently identifying it with $Z_n(h^4)$.

14 Classical low-degree normalization checks

The values $I_1 = 2$ and $I_2 = \pi/2$ give $r_1 = 2$ and $r_2 = 4/\pi$. The following exact families independently test the semilength convention, the harmonic power, and the gamma constant. They are classical benchmarks rather than new targets.

14.1 Associated Hermite moments

For $c > -1$, the normalized associated-Hermite recurrence with coefficient $h + c$ has density

$$\varrho_c(y) = \frac{1}{\sqrt{2\pi}\Gamma(c+1)|D_{-c}(iy)|^2}, \quad Z_n(h+c) = \int_{\mathbb{R}} y^{2n} \varrho_c(y) dy. \quad (14.1)$$

Here D_ν is the parabolic-cylinder function. The normalization follows by rescaling Askey–Wimp’s recurrence $H_{m+1} = 2xH_m - 2(m+c)H_{m-1}$ using $P_m(y) = 2^{-m/2}H_m(y/\sqrt{2}; c)$. The displayed recurrence and orthogonality measure were checked in DLMF 18.30.13–15, which attributes them to Askey–Wimp [18, 19]; the original publisher record, but not its full paper, was inspected. Drake gives the associated-Hermite matching context [20].

DLMF 12.9.1 applies along the imaginary axis and gives

$$\varrho_c(y) \sim \frac{|y|^{2c} e^{-y^2/2}}{\sqrt{2\pi}\Gamma(c+1)} \quad (|y| \rightarrow \infty).$$

For any fixed tail cutoff the compact part of the moment integral is negligible compared with the Gaussian-tail moment. Comparing tails and then evaluating the gamma integral yields

$$Z_n(h+c) \sim \frac{2^{n+c}\Gamma(n+c+1/2)}{\sqrt{\pi}\Gamma(c+1)}.$$

Since $Z_n(h) = (2n-1)!! = 2^n\Gamma(n+1/2)/\sqrt{\pi}$ exactly,

$$\frac{Z_n(h+c)}{Z_n(h)} \sim \frac{(2n)^c}{\Gamma(1+c)}, \quad (14.2)$$

in agreement with (10.1). There is no uniformity claim as $c \downarrow -1$ or as c grows with n .

14.2 The secant-power family

For $\beta = 1 + \alpha > 0$, the exact EGF is

$$\sum_{n \geq 0} Z_n(h(h+\alpha)) \frac{t^{2n}}{(2n)!} = (\sec t)^\beta. \quad (14.3)$$

Deb–Sokal [15, Section 6.3, Eq. (6.24a) at $k = 0$] give the continued-fraction normalization, with attribution to Stieltjes and Rogers. Flajolet’s weighted-path treatment supplies the general correspondence [14]. The exact identity can also be checked from the addition law

$$(\sec(t+u))^\beta = (\sec t)^\beta (\sec u)^\beta (1 - \tan t \tan u)^{-\beta},$$

which yields the Jacobi coefficients $h(h+\beta-1)$.

At $R = \pi/2$ the local expansion is

$$(\sec t)^\beta = R^{-\beta} (1 - t/R)^{-\beta} (1 + O((1 - t/R)^2)).$$

The pole or algebraic singularity at $-R$ contributes equally to an even coefficient; no other singularity is on $|t| = R$. Standard coefficient transfer gives

$$[t^{2n}](\sec t)^\beta \sim \frac{2R^{-2n-\beta}(2n)^{\beta-1}}{\Gamma(\beta)}.$$

Dividing by the $\alpha = 0$ case proves

$$\frac{Z_n(h(h+\alpha))}{Z_n(h^2)} \sim \frac{(4n/\pi)^\alpha}{\Gamma(1+\alpha)}. \quad (14.4)$$

The two nearest singularities and the EGF index $2n$ account for the powers of two. The distinct weights $(h+c)^2$ are not identified with this EGF: $h(h+2c)$ differs from $(h+c)^2$ by c^2 .

The package includes displayed-prefix checks for the associated-Hermite sequences A000698 with its one-term offset, A167872 and A321963, and the appropriate secant-power sequences A000182, A002105, A326328, A395382 and A395383. These identifications are used only as finite normalization checks, with their sign and scale conventions stored explicitly.

15 A safeguarded inverse for factorial-power growth

The absolute harmonic and polynomial formulas are useful for locating a large threshold, but their relative $o(1)$ errors do not justify an unconditional rounding rule. The following lemma states exactly what a leading equivalent supplies.

Theorem 15.1 (Inclusive factorial-power inverse). *Let $(z_n)_{n \geq 0}$ be positive and suppose, for fixed $C > 0$, $B > 0$, $p > 0$, and $q \in \mathbb{R}$,*

$$z_n \sim CB^n(n!)^p n^q. \quad (15.1)$$

Define $N(Y) = \min\{n \geq 0 : z_n \geq Y\}$. Set

$$R = B^{1/p}, \quad b = q + p/2, \quad C_{\log} = \log[C(2\pi)^{p/2}].$$

For sufficiently large Y , write $L = \log Y$ and define

$$t = \frac{L}{pW_0(RL/(pe))}, \quad \xi = t - \frac{b \log t + C_{\log}}{p \log(Rt)}, \quad (15.2)$$

where W_0 is the principal Lambert branch. There exists a nonnegative radius $\eta(Y) = o(1/\log t)$ such that

$$\lceil \xi - \eta(Y) \rceil \leq N(Y) \leq \lceil \xi + \eta(Y) \rceil. \quad (15.3)$$

The two bounding integers are equal or consecutive for all sufficiently large Y . Equivalently, for each fixed $\varepsilon > 0$, eventually

$$\left\lceil \xi - \frac{\varepsilon}{\log t} \right\rceil \leq N(Y) \leq \left\lceil \xi + \frac{\varepsilon}{\log t} \right\rceil. \quad (15.4)$$

The onset may depend on the sequence and ε and is not effective from (15.1) alone.

Proof. Write $z_n = CB^n(n!)^p n^q(1 + e_n)$ with $e_n \rightarrow 0$. Then

$$\frac{z_{n+1}}{z_n} = B(n+1)^p(1+1/n)^q \frac{1+e_{n+1}}{1+e_n} \sim Bn^p \rightarrow \infty.$$

Thus for some n_* the sequence is strictly increasing on $n \geq n_*$ and tends to infinity. No global monotonicity assumption is needed. To ensure that the first crossing belongs to this tail, impose the finite-prefix safeguard

$$Y > \max_{0 \leq n \leq n_*} z_n. \quad (15.5)$$

It holds automatically for all sufficiently large Y , but must be checked or otherwise certified for a concrete threshold.

Stirling's formula gives

$$\log z_n = G(n) + o(1), \quad G(x) = px \log(Rx/e) + b \log x + C_{\log}. \quad (15.6)$$

The $O(1/n)$ Stirling term is absorbed into $o(1)$; the unknown e_n need not have a power rate. Let F be piecewise-linear interpolation of $\log z_n$ on $[n_*, \infty)$. It is continuous and strictly increasing. Since $G''(x) = p/x - b/x^2 = O(1/x)$, its own linear-interpolation error is $O(1/x)$. Interpolation of the discrete $o(1)$ term still tends to zero, so

$$F(x) = G(x) + o(1) \quad (x \rightarrow \infty). \quad (15.7)$$

For Y satisfying (15.5), let x_Y be the unique real solution of $F(x_Y) = L$. The inclusive convention gives the exact equality

$$N(Y) = \lceil x_Y \rceil. \quad (15.8)$$

In particular if $Y = z_m$ then $x_Y = m$ and the first crossing is m .

The Lambert equation in (15.2) is exactly

$$pt \log(Rt/e) = L. \quad (15.9)$$

As $Y \rightarrow \infty$, $t \rightarrow \infty$. Put $d = \xi - t$; the displayed formula gives $d = O(1)$. For $G_0(x) = px \log(Rx/e)$, one has $G'_0(x) = p \log(Rx)$ and $G''_0(x) = p/x$. Taylor expansion over the bounded displacement d yields

$$G_0(t+d) = L + p \log(Rt)d + O(1/t), \quad b \log(t+d) = b \log t + O(1/t).$$

The chosen d cancels the logarithmic and constant terms, proving

$$G(\xi) - L = O(1/t). \quad (15.10)$$

This is the residual of the explicit center, not an estimate for the unknown error in (15.1).

Comparing $F((1 \pm \varepsilon)t)$ with L for fixed small $\varepsilon > 0$ in (15.7)–(15.9) proves $x_Y \sim t$. Since $F(x_Y) = L$, equations (15.7) and (15.10) give $G(x_Y) - G(\xi) = o(1)$. On the interval between x_Y and ξ ,

$$G'(x) = p \log(Rx) + b/x \sim p \log t.$$

The mean-value theorem consequently gives $x_Y - \xi = o(1/\log t)$. Take $\eta(Y) = |x_Y - \xi|$, or any larger radius of the same order. Monotonicity of ceilings and (15.8) prove (15.3) and (15.4). Eventually $2\eta < 1$, so the ceilings differ by at most one. $\square$

The theorem is intentionally qualitative. Multiplying the template by $\exp(1/\log(n+2))$ preserves (15.1), while its logarithmic perturbation shifts the continuous inverse by order $1/(\log t)^2$. Thus an $o(1/(t \log t))$ pre-rounding radius cannot follow from a leading equivalent alone. Nor may one erase the two-ceiling safeguard near an integer boundary. A concrete certification requires an explicit remainder or exact evaluations near the candidate index, together with the finite-prefix check.

For any family in Corollary 10.1, use $C = \mathcal{C}_p r_p^a P_f$, $B = \kappa d_p$, and $q = a - 1/2$. For a degree- d polynomial use $p = d$, $C = \mathcal{C}_d r_d^a / \prod_j \Gamma(1 + c_j)$, and $B = \kappa r_d^d$. The generic finite-prefix safeguard remains part of the first-crossing statement.

For a cubic polynomial in Theorem 1.6, put

$$C = C_Q = \frac{\sqrt{6/\pi} I_3^{-3/2} r_3^a}{\prod_j \Gamma(1 + c_j)}, \quad B = r_3^3 \kappa, \quad p = 3, \quad q = a - 1/2.$$

Then $R = r_3 \kappa^{1/3}$, $b = a + 1$, and $C_{\log} = \log[(2\pi)^{3/2} C_Q]$. For a quartic polynomial, use

$$C = C_Q = \frac{\sqrt{2/\pi} r^{2+a}}{\prod_j \Gamma(1 + c_j)}, \quad B = \kappa r^4, \quad p = 4, \quad q = a - 1/2,$$

so $R = r \kappa^{1/4}$, $b = a + 3/2$, and $C_{\log} = \log(4\pi^2 C_Q)$.

15.1 A sharper logarithmic envelope for pure powers

For unscaled pure weights h^p , the quantitative baseline supplies a smaller asymptotic radius. It still does not supply effective numerical bounds.

Corollary 15.2 (Pure-power logarithmic inverse). *Fix $p > 0$, write $m_n = Z_n(h^p)$, and define $N_p(Y) = \min\{n \geq 0 : m_n \geq Y\}$. In (15.2) take*

$$R = r_p, \quad b = (p-1)/2, \quad C_{\log} = \log[C_p(2\pi)^{p/2}].$$

There are constants $C'_p > 0$ and Y_p such that for $Y \geq Y_p$,

$$\left\lceil \xi - \frac{C'_p}{(\log t)^3} \right\rceil \leq N_p(Y) \leq \left\lceil \xi + \frac{C'_p}{(\log t)^3} \right\rceil. \quad (15.11)$$

The two integers are equal or consecutive eventually. The constants and onset are not supplied effectively.

Proof. Appending UD at height zero injects $\mathcal{D}_n$ into $\mathcal{D}_{n+1}$ without changing a weight, since $\lambda_1 = 1$. For $n \geq 1$, the mountain of semilength $n+1$ is a positive-weight path outside that image. Therefore $m_0 = m_1 = 1$ and $m_{n+1} > m_n$ for $n \geq 1$. For $Y > 1$ there is consequently no finite-prefix ambiguity.

Let F linearly interpolate $\log m_n$ on integers $n \geq 1$. Theorem 1.1 and Stirling give

$$F(x) = G(x) + O_p((\log x)^{-2}), \quad G(x) = px \log(r_p x/e) + \frac{p-1}{2} \log x + C_{\log}.$$

At nonintegers the interpolation error of G is $O_p(1/x)$ because $G''(x) = O_p(1/x)$, so the displayed error is unchanged. If $F(x_Y) = \log Y$, strict increase and the inclusive convention give $N_p(Y) = \lceil x_Y \rceil$, including $Y = m_k$.

The same Lambert equation and bounded correction as in Theorem 15.1 give $G(\xi) - \log Y = O_p(1/t)$ and $x_Y \sim t$. Since G' is asymptotic to $p \log t$ between x_Y and ξ , the mean-value theorem yields

$$|x_Y - \xi| = O_p((\log t)^{-3}).$$

Choose a sufficiently large fixed implied constant and use monotonicity of the ceiling. Once $2C'_p/(\log t)^3 < 1$, the bounding integers differ by at most one. $\square$

This corollary is stated for h^p itself. Scaling by $\kappa < 1$ need not preserve the immediate monotonicity argument, and a concrete scaled-family inverse must retain a finite-prefix safeguard. More importantly, the pure-power logarithmic rate has not been proved for a general harmonic or polynomial perturbation. Those families retain Theorem 15.1's $o(1/\log t)$ radius. Neither enclosure is an unconditional single-ceiling formula.

Remark 15.3 ([write] The inverse and the transseries volume, 7 October 2026). Statement by statement, against the volume named in the Guide (its symbols carry the subscript “vol”).

- (a) *Instance.* The identity (15.8), $N(Y) = \lceil x_Y \rceil$ with x_Y the root of the piecewise-linear interpolation F of $\log z_n$ on the increasing tail, is `p0:thm:staircase(1)` for the admissible interpolation F of `p0:def:three-inverses` (with $A_n = \log z_n$); the brackets (15.3), (15.4) and (15.11) then follow from $|x_Y - \xi| \leq \eta$ by monotonicity of the ceiling, the setting of `p0:thm:staircase(2)`. The same holds in Corollary 15.2.

- (b) *Instance.* The centre t of (15.2) solves (15.9), $pt \log(Rt/e) = L$, which is `p0:prop:factorial-core` with $\kappa_{\text{vol}} = p$, $d_{\text{vol}} = p \log R - p$: there $X = L/(\kappa_{\text{vol}}v)$ with $v = W_0((L/\kappa_{\text{vol}})e^{d_{\text{vol}}/\kappa_{\text{vol}}}) = W_0(RL/(pe))$, exactly t .
- (c) *Formal instance, first coefficient.* With $x = t(1 + E)$ the core part of G becomes $pt[\Lambda E + (1 + E) \log(1 + E) - E]$ with $\Lambda = \log(Rt)$, the form of `p0:thm:core-reversion` ($\Lambda_{\text{vol}} = \log(Rt)$, $h_{\text{vol}}(E) = (1 + E) \log(1 + E) - E$, the remaining terms of G in f_{vol} , with $\log t$ in the coefficient ring); its first coefficient is the displacement $\xi - t = -(b \log t + C_{\log})/(p \log(Rt))$ of (15.2). Only formally, with the source's own residual (15.10). The final mean-value step on the smooth G , between x_Y and ξ , is `p0:thm:backward-error`.

No novelty is claimed for any of these inversions.

16 Exact checks and reproducibility

The companion archive contains the complete TeX source and PDF, portable Python programs, exact integer and rational data, displayed-source fixtures with provenance, numerical diagnostics, a README, and deterministic file manifests. It redistributes no third-party paper PDF. The mathematical proofs are the preceding arguments; finite checks are independent consistency tests, not substitutes for limiting estimates.

The principal moment computation is a height-state walk recurrence. With $v_s(h)$ the total weight of nonnegative prefixes of length s ending at h , start from $v_0(0) = 1$ and apply

$$v_{s+1}(h+1) += v_s(h), \quad v_{s+1}(h-1) += \lambda_h v_s(h) \quad (h \geq 1).$$

Then $Z_n = v_{2n}(0)$. A separate first-return recurrence uses elevated moments $M_{a,n}$ for paths based at height a :

$$M_{a,0} = 1, \quad M_{a,n} = \sum_{k=0}^{n-1} \lambda_{a+1} M_{a+1,k} M_{a,n-1-k}.$$

An independently truncated formal S-fraction checks the same prefixes. Integer scaling is explicit for rational-weight examples; floating-point arithmetic is not used in these equality tests.

The public exact suite includes:

- moment prefixes through $n = 256$ for A216966, A218221, A227887, A144849, A144853, A261000, and the Dixon comparison, together with a separately computed A338634 transform;
- all inspected displayed moment terms, with offsets and signs explicit, and the displayed A187756 weight values; unexamined b-files are not claimed as validated;
- independent walk, first-return, and S-fraction prefix comparisons through $n = 24$;
- exact total-occupation and primitive negative-excursion deletion identities for varied positive rational laws, including irregular laws and positive polynomials with complex or negative nonintegral shifts;
- exact two-step conditional-pair identities, retaining the boundary case where DU is infeasible;
- Dixon ODE-versus-Dyck coefficients through $n = 48$, exact finite product recurrences and gamma-factor algebra, and Gaussian/secant-power formal coefficient checks.

A separate Freud companion constructs exact rational gamma moments for integer parameters $p = 1, \dots, 6$, obtains monic norms through degree 16 by rational Hankel factorization, and compares the resulting recurrence S-fraction with the moments through $n = 16$. It checks the telescoping products, probability and scale normalizations, Gaussian recurrence, and prefactor algebra. Its default receipt records 760 finite instances, including 60 independently checked inclusive thresholds for pure integer powers. Separate high-precision output samples eight fixed powers, including $p = 1/4, 1/2, 3/2$, and records the scaled logarithmic residuals and inverse errors. These diagnostics do not estimate universal constants in the O_p bounds, and the exact algebra does not independently prove the cited Freud coefficient theorem.

The code uses explicit exception checks rather than Python assertions, so its validations remain active under `python -O`. Code, public fixtures, and output files use relative package paths; no private research paths are needed. The README records exact instance counts and the finite ranges, rather than equating a count of tests with a proof.

Numerical files are explicitly labeled diagnostics. They evaluate arch constants, pure-power and one-shift occupation means, gamma products, cubic and quartic amplitudes, and inclusive inverse thresholds. Such computations can expose normalization mistakes and slow finite convergence, but they are not interval-certified bounds, effective asymptotic onsets, or proofs of the claimed limits. The low-power tests deliberately include $p = 1/2$, where finite convergence can be slow.

The rounding warning is visible in a finite example: at the exact integer threshold $Y = A218221(128)$, the numerical center is $\xi \approx 128.000375119743$, so its unguarded ceiling is 129, whereas exact comparison against all earlier generated terms gives the inclusive crossing 128. This demonstrates why the two-ceiling convention matters; it does not determine the theorem's unknown asymptotic radius.

The build validates source and package manifests, regenerates the data and PDF in fresh directories, and makes a deterministic ZIP. A normal replay compares every public member with its recorded release value. The release procedure additionally extracts the actual ZIP and runs the documented builds in both ordinary and optimized Python modes, with the caller's bytecode environment unset, then compares every public member and the complete ZIP bytes. PDF identity is scoped to the documented matching TeX toolchain and font files; arbitrary TeX versions are not asserted to be byte-identical. Guard tests cover the listed malformed inputs, manifests, symlinks, and unsafe destinations; they are a finite rejection suite, not a general security certification.

17 Source inspection and attribution boundaries

The preceding external analytic inputs are specific and checkable. For the general-power baseline, Claeys–Krasovsky–Minakov [8] was inspected directly at the fixed-weight definitions and equation (2.3), including the explicit all- $\beta > 0$ range. The full March 2023 UCLouvain manuscript, September 2022 arXiv v2, and SISSA postprint agree on the needed formula. The publisher record confirms the 2023 publication; its typeset full body was not the inspected copy. The original Kriecherbauer–McLaughlin full body was not retrieved, so its Theorem 1.5 is cited only as explicitly restated in CKM. The varying-weight ensemble coefficient, a later local estimate restricted to $\beta \geq 1$, and the source's separate Hankel-determinant constant problem are not substituted for the fixed-weight leading-coefficient theorem used here.

The Conrad–Flajolet full PDF was available and its Theorem 2, signs, formal-Laplace convention, and equation (19) were inspected. The *Analytic urns* full PDF was inspected at Theorem 1 (PDF

pages 8–9), Theorem 2 and displayed expansion (23) (PDF page 13), and Corollary 1 and its coefficient argument (PDF page 16). Journal and PDF pagination differ. Section 11 checks the urn parameters, hypotheses, local identification, all dominant singularities, and all residue phases explicitly.

The quartic comparison uses the Berg–Valent full-text formulas (2.31), (3.1), the definition of K_0 , and Proposition 3.6.1 on pages 194–195, already inspected for Report 199. Only its stated leading normalization and classical moment theorem are used here. Report 199 is a companion result, not a replacement citation for Berg–Valent.

For low-degree checks, the DLMF formulas for associated Hermite and cylinder functions were inspected; the original Askey–Wimp publisher metadata and abstract were available, but its complete original paper was not audited. Drake’s arXiv paper and the relevant Deb–Sokal and Flajolet full-text formulas were inspected. Stieltjes and Rogers are historical attributions as recorded in those sources, not unexamined theorem statements invoked to justify a general- p amplitude.

Avdoshkin–Dymarsky’s Appendix C already formulates the weighted entropy saddle and equation $-f''/(1 - f^2) = p/(2f)$ for height scaling $h \approx 2nf(t)$. Its parabola for $p = 1$ and sine for $p = 2$ translate into the arch here by $y(s) = 2f(s/2)$. That is prior for the formal geometry; the singular fixed-height occupation regularization and the precise transfer hypothesis in this article are separate steps. Dymarsky–Smolkin’s supplement studies the logarithmic moment correction from a constant shift of asymptotically linear Lanczos coefficients. Its n -power is consistent with $a = 2c$ for $\lambda_h = (h + c)^2$, but the inspected discussion does not supply the exact $\Gamma(1 + c)^{-2}$ amplitude under the present hypotheses. These are bounded statements about the inspected portions, not literature-wide absence claims.

For OEIS, A216966 and A218221 already post the equivalents and authorship dates stated earlier, and A216966 also posts (1.17). The A144849 weight formula is explicit in its continued-fraction definition; A144853 uses A187756’s coefficients. Peter Bala’s associated A144853 note was available as indexed text, while its direct PDF fetch did not yield a usable file in this inspection. Its contents are corroborative, not needed in place of the explicit entry definitions. The elliptic and tree interpretations are known applications, not a claim of new sequence discovery.

Focused literature searches and a bounded read-only screen of the public ProveIt repository sought the sequence identifiers and weighted-Dyck/harmonic topics. Searches also checked the most relevant existing reports; the nearby peak-weight Newton-edge problem concerns a different model. These checks did not establish an exact duplicate of the present full theorem package, but they cannot prove that no equivalent result exists elsewhere. No first-ever proof or priority claim is made.

18 What is reusable and what remains open

The uniform occupation estimate applies to an entire nonsmooth power class, before any arch is known. Its combination with a concrete Freud moment measure and an exact telescoping norm product proves the already-posted absolute formula for all fixed positive powers, without a new saddle fluctuation determinant. It provides a direct way to separate low-height product constants from the macroscopic contribution of a harmonic tail. The direct interpolated concentration avoids the need for a bounded change of measure. The parity argument isolates the cancellation needed by the cubic comparison without imposing independence. These components can be reused separately, with their stated hypotheses.

Several questions remain outside the results proved here:

1. Can the pure-power logarithmic remainder be sharpened to a proved inverse-power correction for every fixed $p > 0$? The current argument controls the published norm error without differentiating it; an asserted smooth recurrence expansion would require additional justification.
2. Can harmonic transfer be given an effective rate, including a quantitative shrinking-cutoff occupation estimate? The present fixed-cutoff limit does not determine a universal rate or a numerical onset.
3. Which residual classes beyond ordinarily convergent series with $r_h = o(1/h)$ admit amplitude transfer, and which oscillatory $1/h$ terms admit cancellation beyond parity? Arbitrary ℓ^1 residuals and general periodic terms require further arguments.
4. Can fluctuation theory for the arch yield controlled subleading amplitudes for genuinely shifted cubic and quartic families? A complete raw inverse-power expansion would require substantially more than the leading transfer result.
5. Can explicit error constants make the two-ceiling inverse a certified finite-threshold method?

[write] **7 October 2026.** Under Vladimir’s standing rule of 4 October 2026 (unproved claims are moved to questions, wrong ones are refuted on the record), the write found no claim of this Part that is unproved as stated and none that is wrong. The one external dependence, the Freud coefficient theorem read only as restated, is disclosed at the end of Section 5; to read the original Kriecherbauer–McLaughlin Theorem 1.5 and confirm the all- $\beta > 0$ statement remains an open task of the same kind as Part III’s Question 1. On Question 1, Part III records a conditional first correction ((42.1)); at $p = 3$ the write finds it numerically consistent (note at the end of Section 42), which is not a proof.

Part II

Quartic Dyck moments and noncrossing cumulants: leading amplitude and first correction for OEIS A338634

Bundle Report 199, archive `Report199.zip`, delivered as `Report199.tex` (22 pages). Delivered title block “Quartic Dyck moments and noncrossing cumulants / Leading amplitude and first correction for OEIS A338634 / Report 199 / 4 October 2026”. Printed as delivered, with [write] remarks and notes; Report 199’s Section k is Section $k + 18$ here, with its statements and equations. Its labels carry `pwd:qt::`; its citation key `oeis` is `oeisA338634`. This Part is self-contained.

Abstract of this Part (as delivered). The implicit continued fraction defining OEIS A338634 is the noncrossing cumulant transform of Dyck paths whose descents from height h carry weight h^4 . We give a concrete interpretation by ordered matching quadruples with a common opener set, and prove the leading equivalent already posted by Václav Kotěšovec together with its first relative correction $-(1 + \pi)/(16n)$. The amplitude comes from a uniform elementary occupation estimate and the exactly solved quartic comparison of Berg and Valent. To identify the correction, we prove a singular-potential limit shape by block counting, an endpoint-safe lower bound, and a calibration argument; a subtracted occupation integral then evaluates to $-\pi/16$. Independently, a cyclic-interval Bonferroni argument gives an expansion to every fixed order in shifted raw moments, with an explicit uniform remainder. This is not an all-orders inverse-power

expansion of the raw moments. We also prove the parity criterion and obtain a second explicit Lambert- W inverse center with a little- o two-ceiling threshold envelope. Exact computations and a self-contained reproducibility package accompany the proofs. No effective asymptotic onset, $O(n^{-2})$ raw-moment remainder, unconditional single-ceiling inverse, or absolute priority claim is made.

19 Definitions and main results

A Dyck path of semilength n starts at height zero, has n upward and n downward unit steps, stays nonnegative, and ends at zero. Let $N_h(D)$ count its descents from h to $h - 1$. Set

$$m_n = \sum_{D: |D|=2n} \prod_{h \geq 1} h^{4N_h(D)}, \quad m_0 = 1, \quad M(z) = \sum_{n \geq 0} m_n z^n. \quad (19.1)$$

The first-return decomposition gives the formal Stieltjes fraction

$$M(z) = \frac{1}{1 - \frac{1^4 z}{1 - \frac{2^4 z}{1 - \frac{3^4 z}{\ddots}}}}. \quad (19.2)$$

Let $A(x) = \sum_{n \geq 0} a_n x^n$, $a_0 = 1$, be determined by

$$A(x) = M(x/A(x)^2), \quad M(z) = A(zM(z)^2). \quad (19.3)$$

These equivalent identities define the sequence A338634 [4]. All power series here are formal; their coefficients grow too rapidly for positive analytic radii of convergence. In the usual symmetric moment notation, the even moments are m_n , the odd moments vanish, and the even free cumulants are a_n .

The first coefficients fix the normalization:

n	0	1	2	3	4	5
m_n	1	1	17	1585	485729	372281761
a_n	1	1	15	1490	472475	367254494

Put

$$I = \int_0^1 \frac{du}{\sqrt{1-u^4}} = \frac{\Gamma(1/4)^2}{4\sqrt{2\pi}}, \quad r = \frac{\sqrt{2}}{I} = \frac{8\sqrt{\pi}}{\Gamma(1/4)^2}, \quad F(x) = \sqrt{\frac{2}{\pi}} r^{4x+2} \frac{\Gamma(x+1)^4}{\sqrt{x}}. \quad (19.4)$$

Numerically $r = 1.0787052023767587 \dots$

Theorem 19.1 (Amplitude and first correction). *As $n \rightarrow \infty$,*

$$\frac{m_n}{F(n)} = 1 - \frac{1+\pi}{16n} + o(n^{-1}), \quad \frac{a_n}{F(n)} = 1 - \frac{1+\pi}{16n} + o(n^{-1}). \quad (19.5)$$

More precisely, if b_n is the Dyck moment with descent weights $\gamma_h = h^2(h^2 - 1/4)$, then

$$\frac{m_n}{b_n} = \frac{\pi}{2} \left(1 - \frac{\pi}{16n} + o(n^{-1}) \right). \quad (19.6)$$

The leading equivalent $a_n \sim F(n)$ is already posted on OEIS by Kotěšovec, dated 12 November 2020. It is a posted formula, not a formula labeled conjectural on the inspected entry. The exact comparison model and its gamma-factor asymptotic are classical results of Berg and Valent [11]. The work needed here is the amplitude transfer from γ_h to h^4 , the signed correction, and the passage to this cumulant sequence. We make no claim that a bounded literature search establishes the first proof of the equivalent.

Theorem 19.2 (Every fixed order in shifted moments). *For each fixed integer $K \geq 0$,*

$$a_n = \sum_{j=0}^K Q_j(n) m_{n-j} + O_K(n^{K+1} m_{n-K-1}), \quad Q_j(n) = [z^j] M(z)^{-2n}. \quad (19.7)$$

The remainder divided by m_n is $O_K(n^{-3K-3})$. In particular,

$$\begin{aligned} Q_0(n) &= 1, & Q_1(n) &= -2n, \\ Q_2(n) &= 2n^2 - 33n, & Q_3(n) &= -\frac{4}{3}n^3 + 66n^2 - \frac{9410}{3}n. \end{aligned} \quad (19.8)$$

The distinction between (19.5) and (19.7) is essential. The latter is a finite expansion using the exact shifted values m_{n-j} , available for every fixed K . It does not assert that $m_n/F(n)$ has an all-orders expansion in powers of $1/n$. It is a sequence-specific rapidly-growing-series calculation of the Bender–Richmond type, with a self-contained tail proof, rather than a new general inversion principle [23, 24, 25].

Section 20 identifies a_n as the number of common-opener matching quadruples whose least noncrossing closure has one block. Section 28 proves that a_n is odd precisely for $n = 0$ or a positive power of two. Section 29 gives an explicit refined inverse: its pre-rounding uncertainty is $o(1/(t \log t))$ on the Lambert- W scale t . The two ceilings are necessary because no separation from integer boundaries is assumed.

Remark 19.3 ([write] The OEIS entry, quoted, and its conjecture proved, 7 October 2026). OEIS A338634, read on 7 October 2026 in the internal format, is at revision #9 (12 November 2020), author Paul D. Hanna (4 November 2020), offset 0; its name is “G.f. $A(x)$ satisfies: $1 = A(x) - x/(A(x) - 2^4 x/(A(x) - 3^4 x/(A(x) - 4^4 x/(A(x) - 5^4 x/(A(x) - 6^4 x/(A(x) - \dots))))))$), a continued fraction relation.” Its formula lines include the unattributed line “For $n > 0$, $a(n)$ is odd iff n is a power of 2 (conjecture).” and Kotěšovec’s “ $a(n) \sim \sqrt{2/\pi} * (8\sqrt{\pi}) / \Gamma(1/4)^2 * (4^n + 2) * (n!)^4 / \sqrt{n}$.” (12 November 2020), which is the leading term of Theorem 19.1. *The parity conjecture is proved* by Theorem 28.1: modulo 2 the Stieltjes fraction collapses to $M = (1 - z)^{-1}$, and $A^2 + A = x$ over $\mathbb{F}_2$ forces $A = 1 + \sum_{j \geq 0} x^{2^j}$; this is the exact posted statement ($n > 0$, with $1 = 2^0$ a power of two). The source did not retrieve the b-file ($0 \leq n \leq 130$, Hanna); the write did, computed a_n for $n \leq 130$ by the Lagrange formula (20.1) from its own m_n , found all 131 terms equal, and found the odd terms for $1 \leq n \leq 130$ exactly at $n = 1, 2, 4, 8, 16, 32, 64, 128$. The moments m_n are A227887, whose b-file also agrees. Nothing was submitted to the OEIS.

20 Matching quadruples and noncrossing closure

A perfect matching on $[2n] = \{1, \dots, 2n\}$ has an *opener* at the smaller endpoint of each edge and a *closer* at the larger endpoint. Consider ordered quadruples of perfect matchings with the same opener set. Scanning the vertices gives a common Dyck path: an opener raises the number of unmatched openers and a closer lowers it. At a closer of current height h , each matching independently chooses

one of its h unmatched earlier openers. Thus a fixed path has exactly $\prod_h h^{4N_h}$ decorations. This proves that m_n counts all such quadruples.

A partition of an ordered set is noncrossing if there are no two distinct blocks containing $a < c$ and $b < d$, respectively, with $a < b < c < d$. Take the least noncrossing partition of $[2n]$ that contains every edge of all four matchings within a block. It can be constructed by first joining the endpoints of every edge and then repeatedly merging any two crossing blocks. This process terminates at the unique least such partition: every merge is forced in any noncrossing coarsening.

This is different from ordinary connectivity of the multigraph formed by the four matchings. For example, four copies of the matching $\{(1, 3), (2, 4)\}$ have two graph components, but their noncrossing closure has a single block. Counting graph-connected quadruples would give the wrong sequence.

Proposition 20.1. *For $n \geq 1$, a_n counts the ordered common-opener matching quadruples on $[2n]$ whose least noncrossing closure has one block. In particular $0 \leq a_n \leq m_n$.*

Proof. Every closure block has even size, since no matching edge leaves it. Restrict the four matchings to one block and standardize the inherited linear order. Their opener sets remain equal, and their noncrossing closure on the block is connected; otherwise the original closure could be refined. Conversely, decorate each block of an even noncrossing partition by such a connected quadruple. The union has the common opener property, and its least noncrossing closure is the chosen partition.

Let c_k count connected decorations of size $2k$. In the block containing vertex 1, the $2k$ vertices leave $2k$ ordered gaps, each independently filled by an arbitrary object. Thus

$$M(z) = 1 + \sum_{k \geq 1} c_k z^k M(z)^{2k}.$$

The formal recursion uniquely determines the c_k and agrees with (19.3); hence $c_k = a_k$. □

The same elementary interpretation works for d ordered common-opener matchings with weights h^d , for any positive integer d . This observation is combinatorial only; it does not extend the quartic asymptotic theorem to a general family of weights. The moment–cumulant decomposition belongs to the established noncrossing framework; broad relations with Jacobi parameters and tree enumerations are discussed in [26, 28].

Lagrange inversion in $u = zA(u)^2$ gives the exact coefficient formula

$$a_n = \frac{1}{1-2n} [z^n] M(z)^{1-2n}, \quad n \geq 1. \quad (20.1)$$

Indeed $x = zM(z)^2$, so the standard change-of-variable coefficient identity applied to $A(x) = M(z)$ gives (20.1). This identity is also useful for exact computation, but by itself does not justify truncating the asymptotic expansion: a uniform bound for the discarded configurations is still needed.

21 Elementary moment bounds and convolution tails

Lemma 21.1. *For $R_n = m_n/m_{n-1}$ and $n \geq 1$,*

$$(n/e)^4 \leq R_n \leq 4n^4, \quad (n!)^4 \leq m_n \leq C_n (n!)^4 \leq 4^n (n!)^4, \quad (21.1)$$

where C_n is the n th Catalan number.

Proof. The mountain $U^n D^n$ has weight $(n!)^4$. The j th descent of any path has height at most $n - j + 1$, so no path has larger weight. This proves the bounds on m_n .

Use the symmetric tridiagonal matrix J with zero diagonal and edge entries $J_{h-1,h} = J_{h,h-1} = h^2$. A closed walk of length $2n$ has an upward and a downward traversal for each edge pair, so $m_n = \langle e_0, J^{2n} e_0 \rangle = \|J^n e_0\|^2$. All statements use finite truncations large enough to contain the paths in question. Cauchy–Schwarz gives $m_n^2 \leq m_{n-1} m_{n+1}$; consequently $R_n \geq m_n^{1/n} \geq (n/e)^4$. Truncate to heights $0, \dots, n$. Its maximum row sum is at most $2n^2$, so its operator norm is at most $2n^2$. Applying this to $J^{n-1} e_0$ gives $m_n \leq 4n^4 m_{n-1}$. $\square$

Lemma 21.2 (Uniform convolution tail). *For every fixed positive integer q ,*

$$\sum_{k=q}^{\lfloor n/2 \rfloor} m_k m_{n-k} = O_q(m_{n-q}). \quad (21.2)$$

Proof. Put $f_k = m_k m_{n-k}$. For $k \leq n/10$, (21.1) yields, for all sufficiently large n ,

$$\frac{f_{k+1}}{f_k} \leq 4e^4 \left(\frac{k+1}{n-k} \right)^4 \leq \frac{1}{2}.$$

This part sums to at most $2m_q m_{n-q}$. In the central range $n/10 \leq k \leq n/2$, the factorial upper bound for m_k and the ratio lower bound for m_n/m_{n-k} give

$$\frac{f_k}{m_n} \leq \frac{(4e^4)^k}{\binom{n}{k}^4} \leq (n+1)^4 \exp\{n[\alpha \log(4e^4) - 4\mathcal{H}(\alpha)]\},$$

where $\alpha = k/n$ and $\mathcal{H}(\alpha) = -\alpha \log \alpha - (1-\alpha) \log(1-\alpha)$. We used $\binom{n}{k} \geq (n+1)^{-1} \exp(n\mathcal{H}(k/n))$. The exponent as a function of α is convex, and is negative at both endpoints $1/10$ and $1/2$; at $1/2$ it equals $2 - 3 \log 2 < 0$. Thus the entire central sum is exponentially small compared with m_n . Finally $m_{n-q} \geq m_n/(4^q n^{4q})$ by the upper ratio bound, absorbing that tail into (21.2). $\square$

22 The shifted moment expansion with a uniform remainder

We prove Theorem 19.2 by inclusion–exclusion over small interval blocks. This supplies the global error control that a formal “one large index” calculation would omit.

View the $2n$ vertices cyclically. Fix $K \geq 1$. For each cyclic interval I of $2k$ vertices, $1 \leq k \leq K$, let E_I be the event that I is exactly one block of the least noncrossing closure. Its internal decoration has a_k choices. Distinct such events can coexist only for disjoint intervals. Conversely, connected decorations on disjoint cyclic intervals coexist with an arbitrary object on the remaining vertices. The standardization on every subset always uses its inherited *linear* order. This argument does not require a matching law invariant under cyclic rotation.

Every nontrivial noncrossing partition has at least two cyclic interval blocks, often called ears. To see this, inspect the cyclic gaps of any block. If only one gap is nonempty, that block is an ear and an innermost block in the remaining gap is another. If at least two gaps are nonempty, each contains an innermost interval block. Hence there is an ear containing at most half the vertices.

A disconnected object with none of the E_I for $k \leq K$ therefore has an ear of semilength between $K + 1$ and $\lfloor n/2 \rfloor$. Its total count is at most

$$2n \sum_{k=K+1}^{\lfloor n/2 \rfloor} a_k m_{n-k} \leq 2n \sum_{k=K+1}^{\lfloor n/2 \rfloor} m_k m_{n-k} = O_K(nm_{n-K-1}). \quad (22.1)$$

The same estimate with $K = 0$ already gives $a_n = m_n + O(nm_{n-1})$, proving Theorem 19.2 for $K = 0$.

For ℓ disjoint small intervals of total semilength $j < n$, the intersection sum is

$$m_{n-j} \frac{2n}{\ell} \binom{2n-2j+\ell-1}{\ell-1} [u^j] (A_{\leq K}(u) - 1)^\ell, \quad A_{\leq K}(u) = 1 + \sum_{k=1}^K a_k u^k. \quad (22.2)$$

Choose the starting vertex in $2n$ ways, order the intervals cyclically, and distribute $2n - 2j$ unused vertices among ℓ possibly empty gaps. Divide by ℓ for the interval chosen to start the cyclic list. Ordered sizes and decorations give the coefficient in (22.2). Adjacent interval blocks are allowed. For $j \leq K$ the truncation of A is irrelevant; for larger j , replacing it by full A is only an upper bound.

Apply Bonferroni through intersection order K . The remainder is bounded by the sum of $(K + 1)$ -fold intersections. Their total semilength lies in the fixed range $K + 1 \leq j \leq K(K + 1)$. Formula (22.2) is $O_K(n^{K+1}m_{n-j})$, and (21.1) makes the sum $O_K(n^{K+1}m_{n-K-1})$. Among retained intersections $\ell \leq K$, the terms with $j > K$ satisfy the same bound. The event of avoiding all small interval blocks differs from connectedness only by (22.1). Thus the coefficient of m_{n-j} for $1 \leq j \leq K$ is

$$P_j(n) = \sum_{\ell=1}^j (-1)^\ell \frac{2n}{\ell} \binom{2n-2j+\ell-1}{\ell-1} [u^j] (A(u) - 1)^\ell. \quad (22.3)$$

For $n > j$, Lagrange inversion in $u = zA(u)^2$ gives

$$[z^j] M(z)^{-2n} = \frac{n}{n-j} [u^j] A(u)^{-2(n-j)}.$$

Expanding the last power gives exactly (22.3). This proves (19.7), including its uniform remainder. Applying the lower ratio bound $K + 1$ times shows $n^{K+1}m_{n-K-1}/m_n = O_K(n^{-3K-3})$. The displayed polynomials (19.8) follow by finite expansion of $M(z)^{-2n}$, using $m_1 = 1, m_2 = 17, m_3 = 1585$.

In particular,

$$\frac{a_n}{m_n} = 1 + O(n^{-3}), \quad \frac{a_n}{m_n} = 1 - 2n \frac{m_{n-1}}{m_n} + O(n^{-6}). \quad (22.4)$$

After the moment asymptotic is known, $m_{n-1}/m_n \sim r^{-4}n^{-4}$, so $a_n/m_n = 1 - 2r^{-4}n^{-3} + o(n^{-3})$. This last consequence concerns the cumulant-to-moment ratio; it does not add higher raw-moment corrections to (19.5).

23 The classical quartic comparison and its normalization

Let b_n be the Dyck moments with weights $\gamma_h = h^2(h^2 - 1/4)$, and put

$$K_0 = \frac{\Gamma(1/4)^2}{4\sqrt{\pi}} = \sqrt{2}I. \quad (23.1)$$

Berg and Valent [11, Eq. (3.1), Prop. 3.6.1] study the birth and death rates

$$\lambda_j = (4j+1)(4j+2)^2(4j+3), \quad \mu_j = (4j-1)(4j)^2(4j+1), \quad \mu_0 = 0.$$

Their moment sequence s_n has contracted Jacobi coefficients $\lambda_j + \mu_j$ and $\lambda_j \mu_{j+1}$, in their notation (2.31). To track the normalization directly, the alternating Stieltjes coefficients are

$$\alpha_{2j+1} = \lambda_j, \quad \alpha_{2j} = \mu_j, \quad \alpha_h = 4h^2(4h^2 - 1) = 16\gamma_h.$$

The standard contraction of an S-fraction has diagonal entries $\alpha_{2j} + \alpha_{2j+1}$ (with $\alpha_0 = 0$) and squared off-diagonal entries $\alpha_{2j+1}\alpha_{2j+2}$. These agree term by term with the birth–death coefficients above. Scaling every descent weight by 16 multiplies every moment of semilength n by 16^n , so $b_n = 16^{-n}s_n$.

Their Proposition 3.6.1, on pages 194–195, gives directly

$$s_n = \frac{4\Gamma(4n+2)}{K_0^{4n+2}}(1 + O(25^{-n})), \quad b_n = \frac{4\Gamma(4n+2)}{16^n K_0^{4n+2}}(1 + O(25^{-n})). \quad (23.2)$$

This is the classical source input in our proof. It avoids any dependence on a separately normalized integral density. Chen–Ismail’s model [21], reproduced in [22], supplies a corroborating equivalent comparison recurrence, not a replacement of γ_h by h^4 .

Stirling expansion in (23.2) yields

$$\frac{\pi}{2}b_n = F(n) \left(1 - \frac{1}{16n} + O(n^{-2}) \right). \quad (23.3)$$

For clarity, the $1/n$ term in $\log \Gamma(4n+2)$ after its leading terms is $13/48$, while that in $4 \log \Gamma(n+1)$ is $1/3 = 16/48$. Their difference is $-1/16$; the remaining constants give precisely (19.4). The $O(n^{-2})$ in (23.3) pertains to the solved comparison b_n , not to the target raw moments m_n .

24 Uniform occupation near the lower boundary

The amplitude comparison requires more than an exponential growth rate. We prove an occupation estimate that is uniform over a whole bounded class of quartic weights.

For any positive weights $w = (w_h)_{h \geq 1}$ satisfying

$$\frac{3}{4}h^4 \leq w_h \leq h^4, \quad (24.1)$$

write $Z_n = \sum_D \prod_h w_h^{N_h(D)}$, and use its normalized weighted law $\mathbb{P}_{n,w}$ and expectation $\mathbb{E}_{n,w}$. Let $H(D)$ be the maximum height. Define

$$u_n(h) = \mathbb{E}_{n,w} N_h, \quad p_n(h) = \mathbb{P}_{n,w}(H \geq h), \quad u_N(j) = 0 \quad (j > N).$$

Theorem 24.1 (Uniform lower-edge occupation). *Uniformly over all weights in (24.1),*

$$\mathbb{E}_{n,w} |N_h - 1| \leq 15552(h/n)^4, \quad n \geq 64, \quad 1 \leq h \leq n/64. \quad (24.2)$$

24.1 An all-length deletion ratio

Finite symmetric Jacobi matrices with off-diagonal entries $\sqrt{w_h}$ give log-convexity of Z_n , just as in Lemma 21.1. Convexity of $\log Z_n$ and $Z_0 = 1$ imply, for all $0 \leq k \leq n$,

$$Z_{n-k} \leq Z_n^{(n-k)/n}.$$

The mountain and $n! \geq (n/e)^n$, $e < 3$, give

$$Z_n \geq (\frac{3}{4})^n (n!)^4 \geq (n^4/108)^n, \quad \frac{Z_{n-k}}{Z_n} \leq (108/n^4)^k. \quad (24.3)$$

This is a fixed- n convexity chord bound. Multiplying less well-adapted one-step bounds would introduce a factorial loss that we do not need.

24.2 Exact negative excursion deletion

Let $e_{h,k}$ be the total weight of paths of length $2k$ that start at h , immediately descend, stay in $\{0, \dots, h-1\}$ until their final step, and finally rise to h . These are primitive negative excursions at height h . Every descent has height at most h , and there are at most C_{k-1} such paths, whence

$$0 \leq e_{h,k} \leq 4^k h^{4k}. \quad (24.4)$$

Every descent across edge h except the last is followed by a first return to h . Delete this entire negative excursion. Conversely, insert any such excursion at any vertex of height h in a shorter Dyck path. All other descent heights are unchanged. For $h > 0$, the number of height- h vertices equals $N_h + N_{h+1}$: arrivals at height h are either upward traversals of edge h or downward traversals of edge $h+1$. Thus the marked bijection gives the exact identity

$$u_n(h) - p_n(h) = \sum_{k=1}^{n-h} e_{h,k} \frac{Z_{n-k}}{Z_n} (u_{n-k}(h) + u_{n-k}(h+1)). \quad (24.5)$$

The remaining path must reach h , explaining the cutoff $n-h$. Empty sums are allowed. With $a(n, h) = 432(h/n)^4$, (24.3)–(24.4) imply

$$u_n(h) \leq 1 + \sum_{k=1}^{n-h} a(n, h)^k (u_{n-k}(h) + u_{n-k}(h+1)). \quad (24.6)$$

24.3 A bounded occupation cone

We first show

$$u_n(h) \leq 12 \quad (1 \leq h \leq n/16). \quad (24.7)$$

Expand (24.6) recursively while the current pair (N, j) lies in the outer cone $j \leq N/8$. Stop a branch on leaving this cone and bound its terminal value $u_N(j)$ by $N \leq n$. The expansion is finite since N decreases at every recursion. Inside the cone,

$$a(N, j) \leq \eta := \frac{432}{8^4} = \frac{27}{256}, \quad \theta := 3\eta = \frac{81}{256} < \frac{1}{3}.$$

For a chain with $q \geq 1$ steps and total size decrease $s = k_1 + \dots + k_q$, its coefficient is at most η^s . There are at most $2^q \binom{s-1}{q-1}$ choices: a composition of s , and at each step either retaining or increasing the height index. Since

$$\sum_{q=1}^s 2^q \binom{s-1}{q-1} = 2 \cdot 3^{s-1},$$

the total contribution of the constant terms at internal nodes is at most

$$1 + \sum_{s \geq 1} 2 \cdot 3^{s-1} \eta^s = 1 + \frac{2\eta}{1-3\eta} = \frac{229}{175} < \frac{4}{3}. \quad (24.8)$$

An exit chain starting with $h \leq n/16$ terminates at $(n-s, h+t)$ for $0 \leq t \leq q \leq s$. The exit condition $8(h+t) > n-s$ and $8h \leq n/2$ imply $9s > n/2$. Thus the sum of all terminal contributions is bounded by

$$\frac{2n}{3} \frac{\theta^{\lfloor n/18 \rfloor + 1}}{1-\theta} \leq \frac{2}{3(1-\theta)} n \theta^{n/18} < 9. \quad (24.9)$$

For the final inequality, $2/[3(1-\theta)] < 1$ and the maximum of $x\theta^{x/18}$ for real $x \geq 0$ is $18/\lceil e \log(1/\theta) \rceil < 9$. A child with $j > N$ contributes zero. Counting unrestricted compositions overcounts only nonnegative terms, so is harmless. Equations (24.8)–(24.9) imply $u_n(h) < 11$, proving the safe bound (24.7).

24.4 Proof of the occupation theorem

Assume $n \geq 64$, $h \leq n/64$. For $k \leq n/2$,

$$h+1 \leq n/32 \leq (n-k)/16,$$

so both child occupations in (24.5) are at most 12. For $k > n/2$ use the bound n for each. With $a = 432(h/n)^4 < 1/4$,

$$\begin{aligned} 0 \leq u_n(h) - p_n(h) &\leq \frac{24a}{1-a} + \frac{2na^{\lfloor n/2 \rfloor + 1}}{1-a} \\ &\leq \frac{26a}{1-a} < 35a. \end{aligned} \quad (24.10)$$

Here $n4^{-\lfloor n/2 \rfloor} \leq 1$ for $n \geq 2$. Paths with maximum below h have at most 4^n possible shapes and every descent weight at most h^4 , giving by (24.3)

$$1 - p_n(h) \leq (432(h/n)^4)^n = a^n \leq a.$$

Pathwise, $|N_h - 1| = (N_h - \mathbf{1}_{\{H \geq h\}}) + \mathbf{1}_{\{H < h\}}$. Combining these estimates proves (24.2) with $36 \cdot 432 = 15552$.

[write] Part I's counterpart, 7 October 2026. This section is the case $p = 4$, with the class $\frac{3}{4}h^4 \leq w_h \leq h^4$ and the explicit constant 15552, of Part I's uniform endpoint estimate (Theorem 1.4, proved in Sections 2–3 by the same deletion identity and cone expansion). It is kept because its explicit constants and range are cited by the later sections of this Part.

25 Amplitude transfer by logarithmic interpolation

For $0 \leq t \leq 1$, define

$$f_h = -\log\left(1 - \frac{1}{4h^2}\right), \quad w_h(t) = \gamma_h e^{tf_h}, \quad Z_n(t) = Z_n(w(t)). \quad (25.1)$$

Then $Z_n(0) = b_n$, $Z_n(1) = m_n$, and (24.1) holds uniformly in t . Since each partition function is a finite positive sum, differentiation is exact:

$$\frac{d}{dt} \log Z_n(t) = \sum_{h=1}^n f_h u_{n,t}(h), \quad 0 < f_h \leq \frac{1}{3h^2}. \quad (25.2)$$

Put $L = \lfloor n/64 \rfloor$. The occupation theorem gives

$$\left| \sum_{h \leq L} f_h (u_{n,t}(h) - 1) \right| \leq \frac{15552}{3n^4} \sum_{h \leq L} h^2 = O(n^{-1}).$$

Since $\sum_h u_{n,t}(h) = n$, the two tails satisfy

$$\sum_{h > L} f_h u_{n,t}(h) \leq \frac{n}{3(L+1)^2} = O(n^{-1}), \quad \sum_{h > L} f_h \leq \frac{1}{3L} = O(n^{-1}).$$

In the second sum the deterministic baseline extends over *all* $h > L$, including $h > n$. Hence, uniformly in t ,

$$\frac{d}{dt} \log Z_n(t) = \sum_{h \geq 1} f_h + O(n^{-1}). \quad (25.3)$$

Euler's sine product at $1/2$ evaluates the convergent sum:

$$\prod_{h \geq 1} \left(1 - \frac{1}{4h^2}\right) = \frac{2}{\pi}, \quad \sum_{h \geq 1} f_h = \log(\pi/2). \quad (25.4)$$

Integrating (25.3) gives

$$\log \frac{m_n}{b_n} = \log(\pi/2) + O(n^{-1}), \quad \log \frac{m_n}{Z_n(t)} = (1-t) \log(\pi/2) + O(n^{-1}), \quad (25.5)$$

with a uniform error. In particular (23.3) already proves $m_n/F(n) = 1 + O(n^{-1})$. Identifying its coefficient requires signed occupation information at macroscopic heights.

A useful consequence of the same uniform estimate is a change-of-measure bound. Since $w_h(t) \leq h^4$ pathwise, for every event B ,

$$\mathbb{P}_{n,t}(B) \leq \frac{m_n}{Z_n(t)} \mathbb{P}_{n,1}(B) \leq C_0 \mathbb{P}_{n,1}(B) \quad (25.6)$$

for all sufficiently large n , with a constant C_0 independent of t and B . Thus concentration for the quartic law transfers uniformly along the entire interpolation. A pointwise limit in t alone would not justify the integration used later.

26 The quartic limit shape with singular potential

For a Dyck path of length $2n$, linearly interpolate its heights and set

$$y_n(s) = \frac{\text{height at step } ns}{n}, \quad 0 \leq s \leq 2.$$

The path belongs to the compact set $\mathcal{K}$ of nonnegative 1-Lipschitz functions on $[0, 2]$ that vanish at both endpoints. Every $y \in \mathcal{K}$ satisfies $y(s) \leq \min(s, 2-s) \leq 1$. Define the binary slope entropy

$$E(v) = -\frac{1+v}{2} \log \frac{1+v}{2} - \frac{1-v}{2} \log \frac{1-v}{2}, \quad -1 \leq v \leq 1,$$

with $0 \log 0 = 0$, and the functionals

$$S(y) = \int_0^2 E(y'(s)) \, ds, \quad P(y) = 2 \int_0^2 \log y(s) \, ds, \quad \Phi(y) = S(y) + P(y). \quad (26.1)$$

The logarithmic integral is understood in the extended Lebesgue sense and may be $-\infty$. Its coefficient follows from an exact identity: if H_j is the larger endpoint height of step j , upward and downward crossings of each edge balance, so

$$\log \frac{\prod_h h^{4N_h}}{n^{4n}} = 2 \sum_{j=1}^{2n} \log(H_j/n). \quad (26.2)$$

The following argument is an elementary sample-path entropy proof, specialized to a singular logarithmic potential and exact Dyck endpoints. It sits within the classical sample-path large-deviation tradition [30]; no external theorem with unchecked singular-potential hypotheses is invoked.

26.1 The weighted upper bound

For a finite partition $Q = (0 = s_0 < \dots < s_M = 2)$ put

$$S_Q(y) = \sum_i (s_i - s_{i-1}) E\left(\frac{y(s_i) - y(s_{i-1})}{s_i - s_{i-1}}\right).$$

These functionals are continuous in the uniform topology. For successive dyadic refinements Q_j , concavity and differentiation of Lipschitz functions imply $S_{Q_j} \downarrow S$: the averages of y' converge almost everywhere to y' , and E is bounded and continuous. For $0 < \varepsilon \leq 1$, put

$$P_\varepsilon(y) = 2 \int_0^2 \log \max(y(s), \varepsilon) \, ds.$$

These are continuous and decrease to P as $\varepsilon \downarrow 0$. Thus Φ is upper semicontinuous on $\mathcal{K}$.

Let $B \subseteq \mathcal{K}$ be closed. Round the partition times ns_i to integer step indices. There are at most $(n+1)^{M-1}$ possible vectors of block-end heights. A block of length L and displacement d has $\binom{L}{(L+d)/2}$ unrestricted bridges when parity is feasible; their number is at most $\exp(LE(d/L))$. Restricting to nonnegative paths or to B can only reduce this number. For fixed Q , time rounding changes the normalized block entropy by $o(1)$ uniformly, using uniform continuity of E .

On each step interval, H_j/n differs from the interpolated height by at most $1/n$. Since $x \mapsto \log \max(x, \varepsilon)$ is $1/\varepsilon$ -Lipschitz, (26.2) is at most $nP_\varepsilon(y_n) + O(1/\varepsilon)$. Within each feasible endpoint

vector, take the largest cutoff potential among paths actually in B . The common entropy bound and this supremum give

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log \left(n^{-4n} \text{weight}_\beta \{y_n \in B\} \right) \leq \sup_{y \in B} (S_Q(y) + P_\varepsilon(y)), \quad \beta_h = h^4. \quad (26.3)$$

The polynomial number of vectors contributes only $O(M \log n) = o(n)$ to the logarithm.

Choose nested dyadic Q_j and $\varepsilon_j \downarrow 0$. On the compact set B , a decreasing sequence of continuous functions has maxima converging to the maximum of its pointwise upper-semicontinuous limit. Indeed, choose maximizing points, pass to a convergent subsequence, and bound by each fixed earlier function before sending its index to infinity. This argument also covers a limit of $-\infty$; the empty case is immediate. Therefore

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log \left(n^{-4n} \text{weight}_\beta \{y_n \in B\} \right) \leq \sup_{y \in B} \Phi(y). \quad (26.4)$$

26.2 Calibration of the unique maximizer

Let $H = 1/I$. Define y_* on $[0, 1]$ by

$$s = \int_0^{y_*(s)} \frac{dv}{\sqrt{1 - (v/H)^4}}, \quad (26.5)$$

and reflect about $s = 1$. Since $HI = 1$, the arch reaches H at time 1. It belongs to $\mathcal{K}$, is positive on $(0, 2)$, and satisfies

$$(y'_*)^2 = 1 - (y_*/H)^4, \quad y''_* = -2y_*^3/H^4. \quad (26.6)$$

The second equation extends continuously across the top. At the endpoints, $y_*(s)/s \rightarrow 1$ and $y_*(s)/(2-s) \rightarrow 1$, respectively.

Set

$$V(s) = E'(y'_*(s)) = \frac{1}{2} \log \frac{1 - y'_*(s)}{1 + y'_*(s)}, \quad 0 < s < 2.$$

Direct differentiation gives

$$V'(s) = \frac{2}{y_*(s)}, \quad V(s) = O(1 + |\log s| + |\log(2-s)|). \quad (26.7)$$

For every $y \in \mathcal{K}$, concavity gives the tangent inequalities

$$E(y') - E(y'_*) \leq V(y' - y'_*), \quad 2(\log y - \log y_*) \leq \frac{2(y - y_*)}{y_*}.$$

Integrate first on $[\delta, 2 - \delta]$ and integrate by parts. Equation (26.7) cancels the interior terms. The remaining boundary expression $[V(y - y_*)]_\delta^{2-\delta}$ tends to zero, because both paths are bounded by $\min(s, 2-s)$ and $\delta |\log \delta| \rightarrow 0$. All tangent terms are integrable: V is integrable, slopes are bounded, and $|y - y_*|/y_*$ is bounded near each endpoint. If $P(y) = -\infty$ the conclusion is automatic; otherwise passage to the endpoints is justified and gives $\Phi(y) \leq \Phi(y_*)$.

The logarithmic tangent deficit is

$$2 \int_0^2 \left[\frac{y}{y_*} - 1 - \log \frac{y}{y_*} \right] ds.$$

It is strictly positive (or infinite) if the continuous functions differ, since then they differ on an interval of positive measure. The entropy deficit is also nonnegative. Thus y_* is the unique maximizer. Compactness and upper semicontinuity imply, for every fixed $\eta > 0$,

$$\sup_{\|y-y_*\|_\infty \geq \eta} \Phi(y) < \Phi(y_*). \quad (26.8)$$

As a normalization check, the identity $E(v) - vE'(v) = \log 2 - \frac{1}{2} \log(1 - v^2)$ gives $E(y'_*) + 2 \log y_* = y'_* V + \log(2H^2)$. Since $y_* V$ vanishes at the endpoints, integration and (26.7) give

$$\Phi(y_*) = -4 + 2 \log(2H^2) = 4 \log r - 4. \quad (26.9)$$

This agrees with the exponential rate of the classical factorial template.

26.3 A lower bound that retains the endpoints

We now prove the matching lower bound without assuming an unbounded-potential form of a general large-deviation theorem. Approximate y_* by positive polygonal paths $z \in \mathcal{K}$ that equal s on $[0, \varepsilon]$ and $2 - s$ on $[2 - \varepsilon, 2]$, and have all interior slopes strictly between -1 and 1 .

One construction replaces the initial segment by the line to $(\varepsilon, \varepsilon)$, joins it to $(2\varepsilon, y_*(2\varepsilon))$, then uses fine chords of y_* , with a reflected end modification. For small ε , $\varepsilon < y_*(2\varepsilon) < 2\varepsilon$, so the joining slope is strictly feasible. Interior chords are also strictly feasible. These approximations converge uniformly, their derivatives converge almost everywhere, and their heights are bounded below by $c \min(s, 2 - s)$ with a common $c > 0$ along sufficiently fine approximations. Boundedness of entropy and an integrable logarithmic envelope give

$$\Phi(z) \longrightarrow \Phi(y_*). \quad (26.10)$$

Fix such a polygon z . Refine its interior linear pieces into blocks of maximum width δ , leaving z and $S(z)$ unchanged. Round block times to lattice times and choose endpoint heights within $O(1)$ of $nz(s_i)$ with the required parity. These choices are feasible for every sufficiently large n , since the fixed interior slopes are separated from ± 1 . On the initial and final pieces choose the deterministic all-up and all-down paths, taking rounded integer lengths and their exact joining heights. Interior endpoint rounding stays $O(1)$.

In each interior block take all bridges with the chosen endpoints. Each lies within $2\delta + O(1/n)$ of z , by the unit Lipschitz bound. Choose δ small compared with the positive minimum of z on $[\varepsilon, 2 - \varepsilon]$; then all these bridges stay positive, and their concatenations are valid Dyck paths. Stirling's formula gives logarithm of their number $nS(z) + o(n)$, because there are a fixed finite number of blocks and the endpoint pieces have zero entropy. On the interior, normalized log weights differ from the integral contribution to $nP(z)$ by $o_\delta(n) + o(n)$, uniformly over all bridges.

The deterministic endpoint contributions in (26.2) are exactly

$$4 \sum_{j=1}^{\lfloor \varepsilon n \rfloor} \log(j/n) + O(\log n).$$

Dividing by n gives $4 \int_0^\varepsilon \log s \, ds$ in the limit, precisely the missing endpoint potential. Consequently

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log(n^{-4n} m_n) \geq \Phi(z) - o_\delta(1).$$

First refine $\delta \downarrow 0$, then let the polygon approach y_* as in (26.10). We obtain

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log(n^{-4n} m_n) \geq \Phi(y_*). \quad (26.11)$$

All limits in the lattice construction were taken with the polygon and finite partition fixed before refinement. No favorable subsequence of n is required.

Theorem 26.1 (Fixed-neighborhood concentration). *For every fixed $\eta > 0$ there is $c_\eta > 0$ such that, for all sufficiently large n ,*

$$\mathbb{P}_{n,1}(\|y_n - y_*\|_\infty \geq \eta) \leq e^{-c_\eta n}. \quad (26.12)$$

Convergence in probability to y_ is uniform for the interpolation laws $\mathbb{P}_{n,t}$, $0 \leq t \leq 1$.*

Proof. Apply (26.4) to the closed complement of the uniform η -ball, use the strict gap (26.8), and divide by the lower bound (26.11). The uniform change-of-measure estimate (25.6) proves the last assertion. $\square$

This theorem concerns each fixed neighborhood. It does not supply a rate for neighborhoods shrinking with n , a fluctuation law, or quantitative concentration constants.

[write] **Part I's counterpart, 7 October 2026.** This limit shape is the case $p = 4$ of Part I's arch concentration (Section 6: closed-set upper bound, calibration and endpoint-safe lower bound), written here for the interpolation of this Part. It is kept because the correction of Section 27 cites it by its own labels.

27 The subtracted occupation limit and first correction

Fix $0 < \varepsilon < H$ and put $L_n = \lfloor \varepsilon n \rfloor$. For every bounded height function g , equal upward and downward edge crossings give the exact identity

$$\frac{1}{n} \sum_h N_h g(h/n) = \frac{1}{2n} \sum_{j=1}^{2n} g(H_j/n). \quad (27.1)$$

Take $g_\varepsilon(x) = x^{-2} \mathbf{1}_{\{x > \varepsilon\}}$. Uniform convergence to y_* implies that the right side tends in probability to

$$T_\varepsilon = \frac{1}{2} \int_0^2 y_*(s)^{-2} \mathbf{1}_{\{y_*(s) > \varepsilon\}} ds = \int_\varepsilon^H \frac{dy}{y^2 \sqrt{1 - (y/H)^4}}. \quad (27.2)$$

The cutoff discontinuity is harmless: $y_* = \varepsilon$ at exactly two times, a set of measure zero, and edge heights differ from the interpolated path by at most $1/n$. Since $g_\varepsilon \leq \varepsilon^{-2}$, the same limit holds in expectation. The convergence is uniform in t by (25.6) and Theorem 26.1. Replacing the cutoff by L_n/n does not affect the limit, by the same bounded convergence argument.

Now $f_h = 1/(4h^2) + O(h^{-4})$ and $\sum_h u_{n,t}(h) = n$, so for fixed ε , uniformly in t ,

$$n \sum_{h > L_n} f_h u_{n,t}(h) \longrightarrow \frac{1}{4} T_\varepsilon, \quad n \sum_{h > L_n} f_h \longrightarrow \frac{1}{4\varepsilon}. \quad (27.3)$$

The first expansion error is $O_\varepsilon(n^{-2})$. The second limit follows from the integral test. Its sum is over *all* heights $h > L_n$, not just those attainable by a path of semilength n . Omitting the deterministic tail above n would alter the correction constant.

For $\varepsilon < 1/64$, Theorem 24.1 gives

$$n \left| \sum_{h \leq L_n} f_h(u_{n,t}(h) - 1) \right| \leq C n^{-3} \sum_{h \leq L_n} h^2 \leq C' \varepsilon^3, \quad (27.4)$$

uniformly in t and sufficiently large n . This is the endpoint estimate needed for the singular statistic: it controls occupation *after subtracting one crossing of each low edge*.

Combining (25.2), (27.3), and (27.4), we first let $n \rightarrow \infty$ with ε fixed and then let $\varepsilon \downarrow 0$. It follows uniformly in t that

$$n \left(\frac{d}{dt} \log Z_n(t) - \sum_{h \geq 1} f_h \right) \rightarrow C_*, \quad C_* := \frac{1}{4} \lim_{\varepsilon \downarrow 0} (T_\varepsilon - 1/\varepsilon). \quad (27.5)$$

More explicitly, the supremum over t of the error has limsup at most $C' \varepsilon^3$ plus the error in the limiting regularized integral; this tends to zero. Thus no interchange of a singular cutoff limit with an uncontrolled expectation is involved.

To evaluate the integral, put $y = Hu$:

$$C_* = \frac{1}{4H} \lim_{a \downarrow 0} \left(\int_a^1 \frac{du}{u^2 \sqrt{1-u^4}} - \frac{1}{a} \right).$$

The elementary identity

$$\frac{d}{du} \frac{\sqrt{1-u^4}}{u} = -\frac{1}{u^2 \sqrt{1-u^4}} - \frac{u^2}{\sqrt{1-u^4}}$$

shows that the regularized bracket equals $-J$, where

$$J = \int_0^1 \frac{u^2 du}{\sqrt{1-u^4}} = \frac{\Gamma(3/4)\Gamma(1/2)}{\Gamma(1/4)}, \quad I = \frac{\Gamma(1/4)\Gamma(1/2)}{4\Gamma(3/4)}, \quad IJ = \frac{\pi}{4}.$$

Since $H = 1/I$,

$$C_* = -\frac{IJ}{4} = -\frac{\pi}{16}. \quad (27.6)$$

Integrating the uniform limit (27.5) over $0 \leq t \leq 1$ and using (25.4) yields

$$\log(m_n/b_n) = \log(\pi/2) - \frac{\pi}{16n} + o(n^{-1}).$$

Exponentiation proves (19.6). Combining it with (23.3) proves the raw-moment formula in (19.5), and (22.4) proves the formula for a_n . This completes the proof of Theorem 19.1.

28 Parity

Theorem 28.1. *For $n > 0$, a_n is odd if and only if n is a power of two. Also $a_0 = 1$ is odd.*

Proof. Modulo 2, h^4 equals 1 for odd h and 0 for even h . Every Dyck path reaching height 2 has zero weight, so the only surviving path is $(UD)^n$. Hence $M(z) = (1-z)^{-1}$ in $\mathbb{F}_2[[z]]$. Since $x = zM(z)^2$ and $A(x) = M(z)$, we obtain

$$A(x)^2 + A(x) = x.$$

The unique solution with constant term 1 is

$$A(x) = 1 + \sum_{j \geq 0} x^{2^j}.$$

For example, uniqueness follows recursively from the coefficient equation: the coefficient at odd degree greater than one is zero, the coefficient at degree one is one, and an even-degree coefficient equals that at half its degree. This proves the assertion. $\square$

This is the parity criterion labeled conjectural on the inspected OEIS entry. It is a side consequence of the exact transform, independent of the asymptotic proof.

29 A second Lambert W center for threshold inversion

For real $Y > 0$, let

$$N(Y) = \min\{n \geq 1 : a_n \geq Y\}.$$

Theorem 19.1 implies $a_{n+1}/a_n \sim r^4 n^4$, so the sequence tends to infinity and is eventually strictly increasing. No monotonicity assertion about an unspecified finite initial segment is needed below.

For large Y define

$$\begin{aligned} L &= \log Y, & t &= \frac{L}{4W_0(rL/(4e))}, & D &= 4 \log(rt), \\ C &= \log(4\sqrt{2}\pi^{3/2}r^2), & B &= \frac{3}{2} \log t + C, & d &= \frac{13 - 3\pi}{48}, \\ \delta_0 &= -\frac{B}{D}, & \delta_1 &= -\frac{2\delta_0^2 + \frac{3}{2}\delta_0 + d}{tD}, & \xi(Y) &= t + \delta_0 + \delta_1. \end{aligned} \quad (29.1)$$

Here W_0 is the principal real Lambert function, satisfying $W_0(u)e^{W_0(u)} = u$ for $u > 0$.

Theorem 29.1 (Shrinking two-ceiling envelope). *There is a positive function $\omega(Y) \rightarrow 0$ such that, for all sufficiently large Y ,*

$$\left\lceil \xi(Y) - \frac{\omega(Y)}{t \log t} \right\rceil \leq N(Y) \leq \left\lceil \xi(Y) + \frac{\omega(Y)}{t \log t} \right\rceil. \quad (29.2)$$

Equivalently, for every $\varepsilon > 0$ the same bounds hold with ε in place of $\omega(Y)$ once Y is sufficiently large depending on ε .

Proof. Write $c = (1 + \pi)/16$. Stirling's formula and Theorem 19.1 give

$$\begin{aligned} \log F(x) &= 4x \log(rx/e) + \frac{3}{2} \log x + C + \frac{1}{3x} + O(x^{-3}), \\ \log a_n &= g(n) + o(n^{-1}), \quad g(x) = h(x) + \frac{3}{2} \log x + C + d/x, \quad h(x) = 4x \log(rx/e), \end{aligned} \quad (29.3)$$

where $d = 1/3 - c$. By the definition of t , $h(t) = L$, and

$$h'(t) = D, \quad h''(t) = 4/t, \quad h'''(t) = -4/t^2.$$

Also $g'(x) = 4 \log(rx) + 3/(2x) - d/x^2 > 0$ for large x .

We have $\delta_0 \rightarrow -3/8$ and $\delta_1 = O(1/(t \log t))$. For bounded s , Taylor expansion yields uniformly

$$g(t+s) = L + Ds + B + \frac{2s^2 + \frac{3}{2}s + d}{t} + O(t^{-2}).$$

Take $s = \delta_0 + \delta_1$. The terms $D\delta_0 + B$ cancel. Replacing s by δ_0 in the quadratic bracket changes its contribution by $O(|\delta_1|/t) = O(t^{-2})$, and $D\delta_1$ cancels the bracket at δ_0 . Thus

$$g(\xi(Y)) = L + O(t^{-2}). \quad (29.4)$$

The large smooth root x_Y of $g(x_Y) = L$ therefore satisfies $x_Y - \xi(Y) = O(1/(t^2 \log t))$.

To preserve integer boundary cases, write $r_n = \log a_n - g(n)$ and set

$$E_0(t) = \sup_{n \geq \lceil t/2 \rceil} |nr_n|.$$

This tends to zero, and $|r_n| \leq 2E_0(t)/t$ in that range. Choose fixed $K'_0 > 0$ with $|g(\xi) - L| \leq K'_0/t^2$ for large Y , and set

$$\eta(t) = E_0(t) + (K'_0 + 1)/t, \quad q(Y) = \frac{8\eta(t)}{tD}.$$

Then $q = o(1/(t \log t))$. On $[\xi - q, \xi + q]$ we have $g' \geq D/2$ for large Y . Hence

$$g(\xi + q) - L \geq \frac{4\eta}{t} - \frac{K'_0}{t^2} > \frac{2E_0(t)}{t}, \quad g(\xi - q) - L \leq -\frac{4\eta}{t} + \frac{K'_0}{t^2} < -\frac{2E_0(t)}{t}.$$

Monotonicity of g shows that every integer $n \geq \xi + q$ has $a_n \geq Y$, while every $\lceil t/2 \rceil \leq n < \xi - q$ has $a_n < Y$. Smaller integers are also below Y for large Y : eventual monotonicity deals with the growing interval below $\lceil t/2 \rceil$, and the finitely many earlier values are eventually dominated by Y . It follows that $[\xi - q] \leq N(Y) \leq \lceil \xi + q \rceil$. Taking $\omega(Y) = q(Y)t \log t$ proves (29.2). $\square$

The uncertainty shrinks, but its onset and the function ω are not made effective. If both ceilings agree, their common value is the threshold; near an integer boundary they may differ by one. An unconditional formula $N(Y) = \lceil \xi(Y) \rceil$ is not a consequence of the theorem.

Remark 29.2 ([write] The inverse and the transseries volume, 7 October 2026). Statement by statement, against the volume named in the Guide (its symbols carry the subscript “vol”).

- (a) *Instance.* For Y beyond the early values, $N(Y)$ is the integer staircase of `p0:def:three-inverses` with $A_n = a_n$ and n_1 the onset of eventual increase, so `p0:thm:staircase(1)` applies.
- (b) *Instance.* The centre t of (29.1), with $h(t) = 4t \log(rt/e) = L$, is `p0:prop:factorial-core` with $\kappa_{\text{vol}} = 4$, $d_{\text{vol}} = 4 \log r - 4$: there $X = L/(4v)$, $v = W_0((L/4)e^{\log r - 1}) = W_0(rL/(4e))$.
- (c) *Formal instance, first coefficient checked.* With $x = t(1+E)$, $h(t(1+E)) - h(t) = 4t[\Lambda E + (1+E) \log(1+E) - E]$ with $\Lambda = \log(rt) = D/4$; dividing the rest of g by $4t$ gives an equation of the form of `p0:thm:core-reversion` ($\Lambda_{\text{vol}} = \log(rt)$, $h_{\text{vol}}(E) = (1+E) \log(1+E) - E$, coefficients in a ring containing $\log t$). Its first coefficient $e_1 = -B/(4\Lambda) = -B/D$ is the source's δ_0 ; the write did not compare δ_1 with the volume's second coefficient.
- (d) *Analogue.* Theorem 29.1 is, as stated and proved, an analogue of `p0:thm:staircase(2)`: a two-ceiling bracket proved directly at the integers from the smooth model g , not deduced from an admissible interpolation.

No novelty is claimed for any of these inversions.

30 Exact computations and reproducibility

The companion archive contains this TeX source and PDF, exact integer data, portable Python code, a README, and file manifests. It contains no third-party paper PDFs. Exact checks use explicit exceptions rather than removable Python assertions; running under `python -O` does not suppress them.

The raw moments are computed by a height-state dynamic program. If $v_s(h)$ is the total weight of nonnegative prefixes of length s ending at height h , start with $v_0(0) = 1$ and apply

$$v_{s+1}(h+1) += v_s(h), \quad v_{s+1}(h-1) += h^4 v_s(h) \quad (h \geq 1).$$

Then $m_n = v_{2n}(0)$. The comparison weights are stored in the integer scaling $4\gamma_h = h^2(4h^2 - 1)$, so their moments are $4^n b_n$ and all stored values remain integers.

For exact powers $G(z) = M(z)^e$, $G(0) = 1$, the identity $MG' = eM'G$ gives

$$g_k = \frac{1}{k} \sum_{j=1}^k ((e+1)j - k) m_j g_{k-j}. \quad (30.1)$$

Together with (20.1), this computes cumulants with exact integer divisibility checks. Separate first-block and finite continued-fraction computations check the defining transform on a finite range. In particular the archive records:

- exact quartic raw moments through $n = 500$, integer-scaled comparison moments through 500, and cumulants through 160;
- equality to the fourteen displayed OEIS coefficients, at indices 0 through 13, and the parity pattern through 160;
- independent enumeration of common-opener matching quadruples and their least noncrossing closures through $n = 3$;
- exact negative-excursion deletion identities for several rational weight laws, and finite occupation checks within the theorem's range;
- symbolic shifted-moment polynomials through degree 8, cyclic-placement identities, and exact algebraic checks of the Stirling and inverse cancellations.

The OEIS b-file was not successfully retrieved in the source inspection. The claim is comparison to the fourteen displayed terms, not a comparison to an unexamined b-file.

Finite matching enumeration gives

n	all common-opener quadruples	one noncrossing-closure block
1	1	1
2	17	15
3	1585	1490

The enumeration treats closure explicitly, rather than substituting ordinary graph connectivity. Exact finite checks are independent consistency tests; they do not prove an asymptotic remainder, establish its numerical onset, or replace any argument in the preceding sections.

A separate optional numerical program evaluates ratios, regularized integrals, and the inverse model at high precision. These outputs are labeled diagnostics. For example, the predicted limits are

$$n \left(\frac{m_n}{(\pi/2)b_n} - 1 \right) \rightarrow -\frac{\pi}{16} = -0.196349540849 \dots,$$

$$n \left(\frac{m_n}{F(n)} - 1 \right) \rightarrow -\frac{1+\pi}{16} = -0.258849540849 \dots$$

For a few exact raw moments, the numerical values are:

n	$n(m_n/((\pi/2)b_n) - 1)$	$n(m_n/F(n) - 1)$
20	-0.192995705	-0.256298484
80	-0.195499050	-0.258208800
160	-0.195923282	-0.258528922
500	-0.196212917	-0.258746889

These rounded values are diagnostics, not certified asymptotic error bounds. A visually small finite residual would not justify replacing the little- o term by $O(n^{-2})$.

The build procedure validates the manifest, replays the exact outputs, and rebuilds the PDF with a fixed date and deterministic metadata. The release archive is extracted and rebuilt in fresh directories in ordinary and optimized Python modes, comparing the public output members byte for byte. PDF reproducibility is scoped to the documented matching TeX toolchain; it is not a claim of identical bytes across arbitrary TeX installations. Input and manifest rejection tests document the malformed cases actually checked, rather than a general security guarantee. Full commands and file roles are in the archive README.

31 Scope of the source and overlap review

The source review underlying this article was bounded. OEIS A338634 and the sibling entries A158119 and A338635 were inspected on 4 October 2026. The latter concern different weight systems and are not additional asymptotic theorems of this article. The current A338634 display attributes its leading equivalent to Kotěšovec and labels its parity criterion conjectural. The absence of a proof link in that display does not establish absence of a published proof.

The full Berg–Valent text was available; the inspected portions include (2.31), (3.1), the definition of K_0 , and Proposition 3.6.1 on pages 194–195. Those are the exact external theorem and normalization used in Section 23. Dai–Ismail–Wang [22] was inspected at its reproduced recurrence and weight, especially (1.5), (1.6), (2.15), and (2.21). The direct Chen–Ismail PDF request did not return a usable full text, so that article is cited as historical context corroborated by the later primary paper, not as a full-text audit here.

For cumulant prior, the Bożejko–Hasebe full text [27] was available, and its relevant recurrence, Remark 3.5, and final open problems were inspected. The arXiv text of Defant–Lee [28] was inspected at its introduction, moment–cumulant formulas, theorem scope, and distribution applications. The historical square-weight question in the former is not represented as still open in 2026. Młotkowski’s abstract and search-visible introduction [26] were inspected, but its complete theorem suite was not audited. These sources establish broader prior context, not an h^4 amplitude theorem used without proof.

Bender’s and Bender–Richmond’s abstracts [23, 24] and the relevant definitions, theorem scope, and Remark 52 of Borinsky [25] were inspected. The original Bender–Richmond full hypotheses were

not independently checked; the explicit proof in Section 22 removes the need to invoke them. Borinsky identifies the faster regime $nf_{n-1} = o(f_n)$ with Bender–Richmond-type methods, and (21.1) puts these moments in that regime. The related Anti-SEQ discussion [29] was also inspected, including its distinction between set/sequence/cycle decompositions and implicit noncrossing settings.

For the limit-shape method, the primary publisher and Math-Net records for Mogul’skii [30] were inspected. The complete specialized entropy proof is supplied here. An institutional abstract of Disanto–Munarini [31] covers related local-height questions for weighted Dyck models with exponents in $\{0, 1, 2\}$; its full text was not audited, and that observation does not exclude earlier quartic work.

A bounded public-repository screen of Vladimir Reshetnikov’s [ProveIt repository](#) checked the current root tree, relevant complete filename trees in Analysis, Combinatorics, Oeis, SetTheory, and the changed Algebra subtree. A saved 64-report README corpus and a broader local collection of 585 README files were searched for the three sequence identifiers. This was not a full-body search of every repository file. Filename or identifier absence is not a novelty test, and earlier cumulant and inversion frameworks remain relevant prior work.

32 What the argument makes reusable

The results have several uses beyond recognizing one integer sequence. First, the matching model turns an implicit divergent series into a nonnegative enumerative class. It makes the distinction between noncrossing closure and graph connectivity concrete and gives independently enumerable finite objects.

Second, the occupation theorem is uniform over all weights between $(3/4)h^4$ and h^4 . It supplies an elementary boundary estimate before a limit shape is known. Its combination with a summable logarithmic weight perturbation explains why the leading amplitude is the Euler product $\pi/2$, while the first finite-size correction is governed by a regularized macroscopic occupation integral. The theorem and interpolation proof have precise stated ranges; they are not promoted here to a general perturbation theorem.

Third, the Bonferroni proof separates a cumulant transform from the asymptotic analysis of the raw moments. For a fixed precision goal it uses finitely many low moments and exact shifted high moments, with a proved remainder. This is useful even when the raw moments have no known inverse-power expansion. Finally, the threshold theorem translates a little- o coefficient estimate into a rounding-safe statement, preserving the exceptional behavior near integers rather than hiding it in a single ceiling.

33 Further questions

Several natural questions remain genuinely outside the proved scope.

1. Does $m_n/F(n)$ have a complete inverse-power expansion? In particular, can the next coefficient be obtained with a proved $O(n^{-2})$ or sharper remainder? This would require more than the fixed-cutoff convergence used here.
2. Can one prove quantitative concentration for shrinking neighborhoods of y_* , a fluctuation law, or a controlled correction to the macroscopic occupation measure? The present theorem gives exponential concentration only for each fixed neighborhood.

3. How far can the elementary occupation argument and amplitude transfer be extended to other polynomial weights or more general summable perturbations? Any such extension needs its own endpoint estimates and uniform change-of-measure argument; no general parameter family is asserted here.
4. Can the asymptotic constants and threshold onset be made numerically effective? The current two-ceiling envelope has an existential little- o radius, not a certified numerical error bound for a given Y .
5. Can alternative combinatorial descriptions make the coefficients or congruences more transparent, beyond the parity result and common-opener interpretation?

[write] **7 October 2026.** Under Vladimir’s standing rule of 4 October 2026, the write found no claim of this Part that is unproved as stated and none that is wrong. One recorded conjecture is settled: A338634’s parity criterion, proved by Theorem 28.1 (Remark 19.3). On Question 1, Part III’s relative theorem gives a second route to the first correction (Section 40.2), but no $O(n^{-2})$ remainder; the question stands. The write recomputed the diagnostic table of Section 30 from its own exact m_n and b_n : all printed entries agree.

The package’s exact and numerical checks provide test cases for these questions, while the proved statements remain those explicitly formulated above.

Part III

First relative corrections for power weighted Dyck moments: three regimes and cumulative perturbations

Bundle Report 202, archive Report202.zip, delivered as Report202.tex (27 pages). Delivered title block “First relative corrections for power weighted Dyck moments / Three regimes and cumulative perturbations / Report 202 / 4 October 2026”. Printed as delivered, with [write] remarks and notes; Report 202’s Section k is Section $k + 33$ here, with its statements and equations, and its appendices keep the letters A and B. Its labels carry `pwd:rc:.` It cites Reports 201 and 199, which are Parts I and II here; it is self-contained through its appendices.

Abstract of this Part (as delivered). For positive descent weights asymptotic to κh^p , we prove first relative corrections under summable logarithmic changes. A tail bh^{-q} has three regimes: a regularized occupation integral when $1 < q < p + 1$, a logarithmically enhanced endpoint term when $q = p + 1$, and an absolutely summable edge correction when $q > p + 1$. We derive uniform one-step ratios from free energy and convexity, sharpen the endpoint occupation estimate, and prove local occupation convergence on compact height intervals strictly below the limiting arch top. This local result yields a cumulative-tail transfer theorem which covers genuinely oscillatory perturbations without a pointwise asymptotic coefficient. The consequences include exact-base relative inverse displacements with safeguarded integer ceilings and unconditional first corrections for all positive zero-harmonic quartic and quadratic polynomial weights. Complete endpoint and concentration foundations are included. Classical comparison models and prior general-power formulas are credited separately from the arguments developed here. A deterministic TeX, PDF, and code package separates exact finite checks from numerical diagnostics. We make no first-proof claim, effective onset claim, all-orders claim, or new unconditional general-power absolute coefficient claim.

34 Results and notation

For a Dyck path D of semilength n , let $N_h(D)$ count descents from h to $h-1$. The path has $2n$ nearest-neighbor steps, stays nonnegative, and begins and ends at zero. For positive weights λ_h define

$$W_\lambda(D) = \prod_{h \geq 1} \lambda_h^{N_h(D)}, \quad Z_n(\lambda) = \sum_{D \in \mathcal{D}_n} W_\lambda(D), \quad Z_0 = 1. \quad (34.1)$$

Thus $\sum_h N_h = n$ and $Z_n(\kappa\lambda) = \kappa^n Z_n(\lambda)$. First-return decomposition gives the classical formal Stieltjes fraction [14]

$$\sum_{n \geq 0} Z_n(\lambda) z^n = \frac{1}{1 - \frac{\lambda_1 z}{1 - \frac{\lambda_2 z}{1 - \frac{\lambda_3 z}{\ddots}}}}.$$

No analytic convergence of this ordinary generating series is assumed.

Fix $p, \kappa > 0$. Throughout the main comparison results,

$$\lambda_h / (\kappa h^p) \longrightarrow 1, \quad \tilde{\lambda}_h = \lambda_h e^{f_h}. \quad (34.2)$$

Write

$$I_p = \int_0^1 \frac{du}{\sqrt{1-u^p}} = \frac{1}{p} B(1/p, 1/2), \quad H_p = I_p^{-1}, \quad d_p = 4H_p^p. \quad (34.3)$$

All limits fix p ; no uniformity as p approaches zero or varies is asserted. If $\sum_h |f_h| < \infty$, then

$$P = \exp\left(\sum_{h \geq 1} f_h\right) = \prod_{h \geq 1} \frac{\tilde{\lambda}_h}{\lambda_h} > 0.$$

Theorem 34.1 (Three relative correction regimes). *Under (34.2) the following hold.*

(i) *If $1 < q < p+1$ and $f_h = bh^{-q} + o(h^{-q})$, then*

$$\frac{Z_n(\tilde{\lambda})}{Z_n(\lambda)} = P \left[1 + bJ_{p,q} n^{1-q} + o(n^{1-q}) \right], \quad (34.4)$$

where

$$J_{p,q} = \lim_{\varepsilon \downarrow 0} \left[\int_\varepsilon^{H_p} \frac{y^{-q} dy}{\sqrt{1-(y/H_p)^p}} - \frac{\varepsilon^{1-q}}{q-1} \right], \quad (34.5)$$

$$= H_p^{1-q} \frac{q-1-p/2}{p(q-1)} B\left(1 - \frac{q-1}{p}, \frac{1}{2}\right). \quad (34.6)$$

Both arguments of the beta function in (34.6) are positive.

(ii) *If $f_h = bh^{-p-1} + o(h^{-p-1})$, then*

$$\frac{Z_n(\tilde{\lambda})}{Z_n(\lambda)} = P \left[1 + \frac{2b}{d_p} n^{-p} \log n + o(n^{-p} \log n) \right]. \quad (34.7)$$

(iii) If $|f_h| \leq C_f h^{-q}$ for some $q > p + 1$, then

$$\frac{Z_n(\tilde{\lambda})}{Z_n(\lambda)} = P [1 + Dn^{-p} + o(n^{-p})], \quad D = \frac{2}{\kappa d_p} \sum_{h \geq 1} (\tilde{\lambda}_h - \lambda_h). \quad (34.8)$$

The series defining D is absolutely convergent.

A zero displayed coefficient yields the corresponding little-o conclusion only; it does not identify a next nonzero term.

For $p > 1$ put $J_p = J_{p,2}$. In particular

$$J_p = \frac{I_p(1 - p/2)}{p} B(1 - 1/p, 1/2), \quad J_2 = 0, \quad J_4 = -\frac{\pi}{4}. \quad (34.9)$$

The zero occurs more generally at $q = 1 + p/2$ within the first regime. The sign of $J_{p,q}$ changes there. This cancellation makes the theorem useful even when a naive tail estimate would suggest a nonzero correction.

Theorem 34.2 (Cumulative rather than pointwise transfer). *Let $p > 1$. Suppose (34.2) holds and*

$$|f_h| \leq C_f h^{-2}, \quad S_N := \sum_{h=1}^N f_h = S + \frac{A}{N} + o(N^{-1}). \quad (34.10)$$

Then

$$\frac{Z_n(\tilde{\lambda})}{Z_n(\lambda)} = e^S \left[1 - \frac{A J_p}{n} + o(n^{-1}) \right]. \quad (34.11)$$

It is unnecessary that $h^2 f_h$ converge. The envelope in (34.10), however, is an essential stated hypothesis of this theorem.

Section 39.1 provides an explicit oscillatory example. Section 41 gives relative inverse displacements for these laws. Section 40 obtains absolute first corrections for quadratic and quartic weights with vanishing coefficient of the next-to-leading power.

34.1 Relation to the companion reports and prior formulas

Report 201 treats harmonic amplitude changes and the general positive-power leading equivalent. The endpoint deletion proof and uniform arch foundations from that companion are reproduced, with the two-sequence uniformity made explicit, in Appendices A–B. This makes the present proofs independent of access to unpublished working notes. Report 199 established the pure-quartic correction. Section 40.2 recovers it directly from the present relative theorem and the correctly normalized Berg–Valent comparison [11].

The general leading formula

$$Z_n(h^p) \sim F_p(n), \quad F_p(n) = \sqrt{\frac{p d_p}{2\pi}} d_p^n (n!)^p n^{-1/2} \quad (34.12)$$

was posted by V. Kotěšovec in A216966 on 24 September 2020 [1]. Report 201 derives it using the precise coefficient estimate restated by Claeys, Krasovsky, and Minakov [8, Eq. (2.3)], attributed

there to Kriecherbauer and McLaughlin [9]. The original KM theorem text was not directly inspected. Formula (34.12) is context, not an input to our relative theorems: their weaker free energy is proved in Appendix B. No general absolute first correction is inferred by differentiating a norm remainder.

Avdoshkin and Dymarsky already derived the formal power-weight entropy saddle and arch equation [12, Appendix C]. Their height convention $h \approx 2nf(t)$ translates to ours by $y(s) = 2f(s/2)$. The appendices here supply the endpoint occupation control and uniform concentration needed for the correction theorems; the arch geometry itself is not claimed as a new discovery.

35 Uniform prerequisites and one step ratios

Interpolate geometrically,

$$\lambda_h(t) = \lambda_h e^{tf_h}, \quad 0 \leq t \leq 1. \quad (35.1)$$

Whenever $f_h \rightarrow 0$, the family belongs to one fixed two-sided power class

$$ch^p \leq \lambda_h(t) \leq Ch^p \quad (h \geq 1, 0 \leq t \leq 1), \quad (35.2)$$

and $\lambda_h(t)/(\kappa h^p) \rightarrow 1$ uniformly in t . Define

$$u_{n,t}(h) = \mathbb{E}_{n,t} N_h, \quad v_{n,t}(h) = \mathbb{P}_{n,t}(\max D \geq h),$$

and extend $u_{n,t}(h) = 0$ for $h > n$.

The two prerequisites needed below are stated here for reference; their complete proofs are in the appendices.

Lemma 35.1 (Endpoint and arch foundations). *For a whole fixed class (35.2), there exist $C_1, \rho, n_0 > 0$, depending only on p, c, C , such that*

$$\mathbb{E}_{n,t} |N_h - 1| \leq C_1 (h/n)^p \quad (n \geq n_0, 1 \leq h \leq \rho n). \quad (35.3)$$

For any family satisfying (35.2) and uniform asymptotic equivalence to κh^p , its linearly interpolated path $y_n(s)$, with time and height divided by n , concentrates uniformly exponentially in every fixed uniform neighborhood of a unique arch y_ on $[0, 2]$. The arch is symmetric about $s = 1$, reaches H_p , and on its increasing half satisfies*

$$y'_*(s) = \sqrt{1 - (y_*(s)/H_p)^p}. \quad (35.4)$$

Moreover,

$$\log Z_n(\lambda(t)) = pn \log n + n\{\log(\kappa d_p) - p\} + o(n) \quad (35.5)$$

uniformly in t .

This statement applies without monotonicity of the weights or continuity of the family parameter. For the interpolation the differentiability used later follows simply because each partition function is a finite positive sum.

Lemma 35.2 (Uniform one step ratios). *Under the family hypotheses of Lemma 35.1,*

$$R_n(t) := \frac{Z_{n-1}(\lambda(t))}{Z_n(\lambda(t))} = \frac{1 + o(1)}{\kappa d_p n^p} \quad (35.6)$$

uniformly in t .

Proof. Put $L_n(t) = \log Z_n(\lambda(t))$. Finite Jacobi spectral positivity, proved in Appendix A, implies discrete convexity of L_n . For fixed $0 < \varepsilon < 1$ and $r = \lfloor \varepsilon n \rfloor$,

$$\frac{L_n - L_{n-r}}{r} \leq L_n - L_{n-1} \leq \frac{L_{n+r} - L_n}{r}.$$

Insert (35.5). Subtract $p \log n + \log(\kappa d_p)$. The two limiting errors are, respectively,

$$p \left[-\frac{1-\varepsilon}{\varepsilon} \log(1-\varepsilon) - 1 \right], \quad p \left[\frac{1+\varepsilon}{\varepsilon} \log(1+\varepsilon) - 1 \right].$$

Both tend to zero as $\varepsilon \downarrow 0$. At each fixed ε , the $o(n)/r$ errors tend to zero uniformly in t , and rounding r is harmless. Exponentiating proves (35.6). Convexity, rather than differentiation of an unknown asymptotic error, is what allows the one-step conclusion. $\square$

Proposition 35.3 (Strengthened endpoint cone). *After decreasing the fixed ρ if necessary,*

$$u_{n,t}(h) - 1 = 2\lambda_h(t)R_n(t) + O((h/n)^{2p}) \quad (1 \leq h \leq \rho n) \quad (35.7)$$

uniformly in t and sufficiently large n . In particular, for every fixed h ,

$$n^p(u_{n,t}(h) - 1) \longrightarrow \frac{2\lambda_h(t)}{\kappa d_p} \quad (35.8)$$

uniformly in t .

Proof. Let $e_{h,k}(t)$ be the weight sum of primitive negative excursions of length $2k$ at height h , staying nonnegative. Exact deletion gives

$$u_{n,t}(h) - v_{n,t}(h) = \sum_{k=1}^{n-h} e_{h,k}(t) \frac{Z_{n-k}(\lambda(t))}{Z_n(\lambda(t))} [u_{n-k,t}(h) + u_{n-k,t}(h+1)]. \quad (35.9)$$

The bijection and the uniform bound

$$e_{h,k}(t)Z_{n-k}/Z_n \leq a^k, \quad a = A_0(h/n)^p \quad (35.10)$$

are proved in Appendix A. Choose ρ so small that $a \leq 1/4$ and for $k \leq n/2$ both child heights lie in the rough bounded-occupation cone of that appendix. Then their occupations are bounded by a fixed B_0 .

The $k = 1$ weight is exactly $e_{h,1} = \lambda_h(t)$. The endpoint bound at size $n - 1$ gives

$$u_{n-1,t}(h) + u_{n-1,t}(h+1) = 2 + O((h/n)^p),$$

using $h+1 \leq 2h$. Since $R_n \lambda_h(t) = O((h/n)^p)$ by (35.6), replacing this bracket by 2 costs $O((h/n)^{2p})$.

For $2 \leq k \leq n/2$ use B_0 , and for larger k use the trivial bound n . The remaining sum is at most

$$2B_0 \sum_{k \geq 2} a^k + 2n \sum_{k > n/2} a^k = O(a^2).$$

Indeed $na^{\lfloor n/2 \rfloor + 1} = O(a^2)$ uniformly for $a \leq 1/4$ and $n \geq 2$. The mountain/counting bound in the appendix gives $1 - v_{n,t}(h) \leq a^n \leq a^2$. This proves (35.7); multiply by n^p at fixed h and apply (35.6) for (35.8). $\square$

36 The bulk correction and its regularization

Along (35.1), finite differentiation gives

$$\frac{d}{dt} \log Z_n(\lambda(t)) = \sum_{h=1}^n f_h u_{n,t}(h). \quad (36.1)$$

Subtracting $\log P = \sum_{h \geq 1} f_h$, and extending u by zero, centers the occupation at one descent per edge. To prove Theorem 34.1(i), it suffices to show

$$n^{q-1} \sum_{h \geq 1} f_h [u_{n,t}(h) - 1] \longrightarrow b J_{p,q} \quad (36.2)$$

uniformly in t .

Choose $0 < \varepsilon < \min(\rho, H_p)$ and $L = \lfloor \varepsilon n \rfloor$. Since $|f_h| \leq C_f h^{-q}$ for all h , the endpoint bound implies

$$n^{q-1} \left| \sum_{h \leq L} f_h (u_{n,t}(h) - 1) \right| \leq C n^{q-1-p} \sum_{h \leq L} h^{p-q} \leq C' \varepsilon^{p-q+1}. \quad (36.3)$$

The exponent is positive precisely in the stated regime $q < p + 1$.

Write $f_h = b h^{-q} + r_h$ and $\omega(L) = \sup_{h > L} h^q |r_h| \rightarrow 0$. Because $\sum_h u_{n,t}(h) = n$,

$$\begin{aligned} n^{q-1} \sum_{h > L} |r_h| u_{n,t}(h) &\leq \omega(L) \frac{n^q}{(L+1)^q} \longrightarrow 0, \\ n^{q-1} \sum_{h > L} |r_h| &\leq C_q \omega(L) n^{q-1} L^{1-q} \longrightarrow 0 \end{aligned}$$

for each fixed ε , uniformly in t . No regularity of the remainder sequence is used.

For every bounded height test g , up/down edge balance gives the exact identity

$$\frac{1}{n} \sum_h N_h g(h/n) = \frac{1}{2n} \sum_{j=1}^{2n} g(H_j/n), \quad (36.4)$$

where H_j is the upper endpoint of step j . Apply this with $g(y) = y^{-q} \mathbf{1}_{y > L/n}$. Uniform arch concentration yields

$$n^{q-1} \sum_{h > L} h^{-q} u_{n,t}(h) \longrightarrow \frac{1}{2} \int_0^2 y_*(s)^{-q} \mathbf{1}_{y_*(s) > \varepsilon} ds = \int_\varepsilon^{H_p} \frac{y^{-q} dy}{\sqrt{1 - (y/H_p)^p}}. \quad (36.5)$$

Here is the cutoff justification. The test is bounded for fixed ε ; its discontinuity has two time preimages on the arch, a null set. For any uniformly convergent deterministic sequence of paths, the edge height and interpolated path differ by at most $1/n$, so dominated convergence gives the displayed limit. Compactness makes this convergence uniform in a sufficiently small fixed neighborhood of y_* . Uniform concentration and boundedness then give uniform convergence of expectations. The same argument accommodates $L/n \rightarrow \varepsilon$.

The integral test also gives

$$n^{q-1} \sum_{h > L} h^{-q} \longrightarrow \frac{\varepsilon^{1-q}}{q-1}.$$

Thus the limiting error in (36.2) is bounded by (36.3) plus the error in replacing the regularized integral at ε by its limit.

To evaluate that limit using ordinary convergent integrals, set $a = \varepsilon/H_p$ and $G(u) = u^{1-q}\sqrt{1-u^p}$. Direct differentiation gives

$$G'(u) = \frac{(1-q)u^{-q} + (q-1-p/2)u^{p-q}}{\sqrt{1-u^p}}.$$

Since $G(1) = 0$, integration from a to 1 gives

$$\int_a^1 \frac{u^{-q} du}{\sqrt{1-u^p}} = \frac{G(a)}{q-1} + \frac{q-1-p/2}{q-1} \int_a^1 \frac{u^{p-q} du}{\sqrt{1-u^p}}.$$

Now $G(a) - a^{1-q} = O(a^{p-q+1}) \rightarrow 0$, and the remaining integral equals $p^{-1}B(1 - (q-1)/p, 1/2)$ in the limit. This proves (34.5)–(34.6) and (36.2).

Integrating (36.1) in t therefore gives

$$\log \frac{Z_n(\tilde{\lambda})}{Z_n(\lambda)} = \log P + bJ_{p,q}n^{1-q} + o(n^{1-q}).$$

Since $q > 1$, the square of the vanishing correction is smaller order, so exponentiation proves (34.4). A beta function with first argument $(1-q)/p$ could alternatively describe this coefficient by analytic continuation; it would not be a convergent improper beta integral. Formula (34.6) avoids that ambiguity.

For $p = 4, q = 2$, ordinary positive-argument beta integrals give

$$I_4 = \frac{\Gamma(1/4)\sqrt{\pi}}{4\Gamma(3/4)}, \quad \int_0^1 \frac{u^2 du}{\sqrt{1-u^4}} = \frac{\Gamma(3/4)\sqrt{\pi}}{\Gamma(1/4)}.$$

Their product is $\pi/4$, proving $J_4 = -\pi/4$. At $p = 2$ the factor $1 - p/2$ in (34.9) vanishes directly.

37 Critical and endpoint dominated corrections

We now prove the remaining two regimes. The common centered derivative is

$$T_n(t) = \frac{d}{dt} \log Z_n(\lambda(t)) - \log P = \sum_{h \geq 1} f_h(u_{n,t}(h) - 1). \quad (37.1)$$

Take $L = \lfloor \rho n \rfloor$ for the strengthened cone.

37.1 The critical power

Suppose $f_h = bh^{-p-1} + o(h^{-p-1})$. Then

$$\sum_{h > L} |f_h|u_{n,t}(h) + \sum_{h > L} |f_h| = O(n^{-p})$$

uniformly in t , using $\sum_h u_{n,t}(h) = n$. By (35.7),

$$\begin{aligned} \sum_{h \leq L} f_h(u_{n,t}(h) - 1) &= 2R_n(t) \sum_{h \leq L} f_h \lambda_h(t) + O\left(n^{-2p} \sum_{h \leq L} h^{2p} |f_h|\right) \\ &= 2R_n(t) \sum_{h \leq L} f_h \lambda_h(t) + O(n^{-p}). \end{aligned}$$

For every fixed $p > 0$, $\sum_{h \leq L} h^{p-1} = O(n^p)$, which justifies the last error. The hypotheses imply $h f_h \lambda_h(t) \rightarrow \kappa b$ uniformly in t . A split at fixed large M , followed by the harmonic sum estimate, gives

$$\sum_{h \leq L} f_h \lambda_h(t) = \kappa b \log n + o(\log n)$$

uniformly in t , including $b = 0$. Consequently (35.6) yields

$$T_n(t) = \frac{2b}{d_p} n^{-p} \log n + o(n^{-p} \log n)$$

uniformly. Integrate and exponentiate to obtain (34.7).

37.2 Summable edge differences

Suppose $|f_h| \leq C_f h^{-q}$ for $q > p + 1$. The high-height terms in (37.1) are now $O(n^{1-q}) = o(n^{-p})$. For $h \leq L$, the endpoint bound gives

$$n^p |f_h(u_{n,t}(h) - 1)| \leq C |f_h| h^p, \quad \sum_{h \geq 1} |f_h| h^p < \infty.$$

Combine this summable majorant with (35.8). First truncate to $h \leq M$, where convergence is uniform in t , and then let $M \rightarrow \infty$. It follows that

$$n^p T_n(t) \rightarrow \frac{2}{\kappa d_p} \sum_{h \geq 1} f_h \lambda_h(t)$$

uniformly in t . The series on the right converges absolutely and uniformly because $\lambda_h(t) \leq C h^p$. Hence termwise integration is justified, and

$$\int_0^1 f_h \lambda_h e^{t f_h} dt = \tilde{\lambda}_h - \lambda_h,$$

with the same identity at $f_h = 0$. Also $|\tilde{\lambda}_h - \lambda_h| \leq C h^p |f_h| = O(h^{p-q})$, so its series converges absolutely. Integration and exponentiation prove (34.8).

For example, changing only λ_1 gives

$$\frac{Z_n(\tilde{\lambda})}{Z_n(\lambda)} = \frac{\tilde{\lambda}_1}{\lambda_1} \left[1 + \frac{2(\tilde{\lambda}_1 - \lambda_1)}{\kappa d_p} n^{-p} + o(n^{-p}) \right].$$

The factor two is fixed by the two vertex-arrival types in the deletion identity; it is not obtained from a fitted numerical coefficient.

38 Local occupation below the arch top

Weak occupation convergence sufficed for the pointwise-tail theorem. The cumulative theorem requires more: a uniform local limit for single-height expected occupation.

Theorem 38.1 (Local occupation density). *Let a family of positive weights satisfy (35.2) and $\lambda_h(t)/(\kappa h^p) \rightarrow 1$ uniformly in t . For every compact $K \subset (0, H_p)$,*

$$\sup_{\substack{t \in [0,1] \\ h/n \in K}} |u_{n,t}(h) - U_p(h/n)| \rightarrow 0, \quad U_p(x) = \frac{1}{\sqrt{1 - (x/H_p)^p}}. \quad (38.1)$$

There is no claim at the arch top, no rate claim, and no monotonicity assumption on the weights.

38.1 Long negative excursions

Fix $b < H_p$ and $\delta > 0$. We show that a primitive negative excursion from $h \leq bn$ of semilength $k > \delta n$ forces a fixed-neighborhood concentration failure. Choose $\zeta > 0$ so that $b + \zeta < H_p$ and

$$\frac{\zeta}{v_0} < \delta, \quad v_0 = \sqrt{1 - ((b + \zeta)/H_p)^p} > 0.$$

If a rescaled path is within ζ of y_* , its excursion interval $[s_1, s_2]$ lies in the sublevel set $y_* \leq h/n + \zeta$. The interval is connected and therefore lies within one of the two disjoint components of that sublevel set. The arch is monotone there, with absolute slope at least v_0 . At both endpoints its height differs from h/n by at most ζ . Thus $s_2 - s_1 \leq 2\zeta/v_0 < 2\delta$, contradicting the excursion length $2k/n > 2\delta$. This argument also covers $h/n < \zeta$.

There are at most n marked descents in a path. Uniform exponential concentration therefore gives

$$\mathbb{E}_{n,t}\{\text{non-last descents at edge } h \text{ with excursion } k > \delta n\} \leq ne^{-c\delta n}, \quad h \leq bn, \quad (38.2)$$

for large n , uniformly in t . By exact deletion, this is precisely the long- k part of (35.9). Bounding the event avoids separately overestimating the factors of a long excursion.

38.2 Catalan contraction on a compact cone

Fix $0 < a \leq b < H_p$ and choose b_1 strictly between b and H_p . Put $M_h(t) = \max_{j \leq h} \lambda_j(t)$. Uniform asymptotic equivalence and the fixed power class give

$$M_h(t)/(\kappa h^p) \rightarrow 1 \quad (38.3)$$

uniformly: large j obey the asymptotic bound and $j^p \leq h^p$, the finitely many small j contribute $o(h^p)$, and $j = h$ gives the lower bound.

Choose $\tau > 0$ and $0 < \delta < 1/2$ sufficiently small that

$$r = \frac{1 + \tau}{1 - \tau} \frac{b_1^p}{d_p(1 - \delta)^p} < \frac{1}{4}. \quad (38.4)$$

This is possible since $d_p = 4H_p^p$. Let $C_j = (j + 1)^{-1} \binom{2j}{j}$ be the Catalan numbers. There are C_{k-1} unrestricted primitive negative excursions of semilength k ; the floor at zero can only remove some. Equations (35.6) and (38.3) imply, for sufficiently large h, N , $h \leq b_1 N$ and $k \leq \delta N$,

$$e_{h,k}(t) \frac{Z_{N-k}}{Z_N} \leq C_{k-1} M_h(t)^k [\kappa d_p (1 - \tau) (N - k)^p]^{-k} \leq C_{k-1} r^k =: c_k. \quad (38.5)$$

Starting at (n, h) with $an \leq h \leq bn$, recursively expand the inequality from (35.9) at nodes (N, j) with $N \geq n/2$ and $j \leq b_1 N$. Retain only $k \leq \delta N$ in the recursion, using (38.2) to bound the omitted part. Throughout the expanded domain, $j \geq h \geq an$ and $N \geq n/2$, so the large-height and large-size bounds are uniform. The forcing at each expanded node is at most $1 + ne^{-cn/2}$. Each retained edge decreases N by k and either leaves j fixed or increases it by one. Stop at the first exit and bound terminal occupations by n .

For a branch with total decrease s , height increases by at most s . An exit through $j > b_1 N$ requires $bn + s > b_1(n - s)$ and hence $s > (b_1 - b)n/(1 + b_1)$. An exit through $N < n/2$ requires $s > n/2$. Thus every exit has $s \geq c_0 n$ for some fixed $c_0 > 0$.

Let $C(z) = \sum_{k \geq 1} c_k z^k = (1 - \sqrt{1 - 4rz})/2$. The two child choices at every recursive step give a total branch coefficient bounded by

$$\sum_{\ell \geq 0} [2C(1)]^\ell = \frac{1}{\sqrt{1 - 4r}}.$$

Choose $\theta > 0$ with $re^\theta < 1/4$. Branches whose total decrease is at least $c_0 n$ have coefficient sum at most

$$\frac{e^{-\theta c_0 n}}{1 - 2C(e^\theta)}.$$

All coefficients are nonnegative; enlarging to all formal branches is legitimate. Consequently

$$u_{n,t}(h) \leq \frac{1 + o(1)}{\sqrt{1 - 4r}} + O(ne^{-\theta c_0 n}), \quad an \leq h \leq bn, \quad (38.6)$$

uniformly in t .

38.3 Identification of the limit

Fix $x \in (0, H_p)$ and any sequence $n \rightarrow \infty$, $h/n \rightarrow x$, with arbitrary $t \in [0, 1]$. First fix $x < b < b_1 < H_p$ and the small parameters above, take the upper limit in (38.6), and then let $b, b_1 \downarrow x$ and $\tau, \delta \downarrow 0$. This yields

$$\limsup u_{n,t}(h) \leq [1 - 4x^p/d_p]^{-1/2} = U_p(x).$$

Let $U_-(x)$ be the liminf over all such sequences. It is finite and nonnegative by (38.6). Concentration gives $v_{n,t}(h) \rightarrow 1$. For each fixed k , every primitive excursion is feasible for large h and stays within k levels of h . Uniform asymptotic equivalence gives

$$e_{h,k}(t) = (1 + o(1))C_{k-1}(\kappa h^p)^k, \quad Z_{n-k}/Z_n = (1 + o(1))(\kappa d_p n^p)^{-k}.$$

Keep only $1 \leq k \leq K_0$ in the nonnegative deletion identity. Both child sequences still have scaled height tending to x , so taking the liminf gives

$$U_-(x) \geq 1 + 2U_-(x) \sum_{k=1}^{K_0} C_{k-1}(x^p/d_p)^k.$$

The coefficient on the right is less than one after multiplication by two. Rearrange, then let $K_0 \rightarrow \infty$ to obtain $U_-(x) \geq U_p(x)$. This proves sequential convergence. If convergence in (38.1) failed on a compact set, a failing sequence would have a subsequence with $h/n \rightarrow x \in K$, contradicting this result and continuity of U_p . The theorem follows.

39 Cumulative tail transfer and an oscillatory example

We prove Theorem 34.2. Along the interpolation, let $0 < \varepsilon < \min(\rho, H_p)$ be fixed. The endpoint bound gives

$$n \left| \sum_{h \leq \varepsilon n} f_h(u_{n,t}(h) - 1) \right| \leq C n^{1-p} \sum_{h \leq \varepsilon n} h^{p-2} \leq C' \varepsilon^{p-1}. \quad (39.1)$$

This tends to zero as $\varepsilon \downarrow 0$ because $p > 1$.

For the interior heights introduce the signed discrete measures

$$\mu_n = n \sum_{h \geq 1} f_h \delta_{h/n}.$$

On every compact interval $[a, b] \subset (0, \infty)$ their total variations are uniformly bounded: $n \sum_{a \leq h \leq bn} |f_h| = O(1)$. For fixed $a < x < y < b$, the cumulative hypothesis gives

$$\mu_n((x, y]) = n(S_{[ny]} - S_{[nx]}) \longrightarrow A(1/y - 1/x) = -A \int_x^y z^{-2} dz.$$

Endpoint atoms are $O(n^{-1})$. Step-function approximation and the common variation bound therefore imply convergence against every continuous test on compact intervals away from zero:

$$\mu_n \Longrightarrow -Ax^{-2} dx. \quad (39.2)$$

For $\delta > 0$ with $\varepsilon < H_p - \delta$, Theorem 38.1 and (39.2) yield

$$n \sum_{\varepsilon n < h < (H_p - \delta)n} f_h u_{n,t}(h) \longrightarrow -A \int_{\varepsilon}^{H_p - \delta} x^{-2} U_p(x) dx \quad (39.3)$$

uniformly in t . Indeed the integral of the fixed continuous function U_p converges by (39.2), and the varying-test error $u_{n,t} - U_p$ is bounded by its uniform supremum times the common total variation.

The upper region is controlled without a local theorem at the top. Above $H_p + \delta$,

$$n \sum_{h > (H_p + \delta)n} |f_h| u_{n,t}(h) \leq \frac{C}{(H_p + \delta)^2} \frac{1}{n} \sum_{h > (H_p + \delta)n} u_{n,t}(h) \longrightarrow 0.$$

This follows from weak occupation convergence (36.4) and uniform concentration. In the band $|h/n - H_p| \leq \delta$, the same argument bounds the absolute contribution by $C/(H_p - \delta)^2$ times its normalized occupation mass. The limiting mass is

$$\frac{1}{2} \text{meas}\{s \in [0, 2] : |y_*(s) - H_p| \leq \delta\},$$

which tends to zero as $\delta \downarrow 0$. The boundary levels have null time preimages, so the indicator convergence is justified as in (36.5). The density U_p has an integrable square-root singularity at its top. First take $n \rightarrow \infty$ and then $\delta \downarrow 0$ in (39.3) to obtain

$$n \sum_{h > \varepsilon n} f_h u_{n,t}(h) \longrightarrow -A \int_{\varepsilon}^{H_p} x^{-2} U_p(x) dx$$

uniformly in t . Separately,

$$n \sum_{h > \varepsilon n} f_h = n(S - S_{[\varepsilon n]}) \longrightarrow -\frac{A}{\varepsilon}.$$

Subtracting this baseline, using (39.1), and then letting $\varepsilon \downarrow 0$ gives

$$n \left[\frac{d}{dt} \log Z_n(\lambda(t)) - S \right] \longrightarrow -AJ_p$$

uniformly. Integrating and exponentiating proves (34.11). The top-band estimate, rather than a claim of local convergence at H_p , is the key remaining uniform-integrability step.

39.1 A genuinely oscillatory telescoping perturbation

Set

$$a_0 = 1, \quad a_h = 1 + \frac{1}{3h} + \frac{(-1)^h}{4h^2} \quad (h \geq 1), \quad f_h = \log \frac{a_h}{a_{h-1}}. \quad (39.4)$$

Every a_h is positive, since $a_h \geq 1 + 1/(3h) - 1/(4h^2) > 0$. Telescoping gives $S_N = \log a_N = 1/(3N) + O(N^{-2})$, so $S = 0$ and $A = 1/3$. For $h \geq 2$,

$$\log a_h = \frac{1}{3h} + \frac{(-1)^h/4 - 1/18}{h^2} + O(h^{-3}).$$

Subtracting the expression at $h - 1$ yields

$$h^2 f_h = -\frac{1}{3} + \frac{(-1)^h}{2} + O(h^{-1}). \quad (39.5)$$

Thus $h^2 f_h$ tends to $1/6$ along even indices and $-5/6$ along odd indices. It has no single pointwise limiting coefficient, although $|f_h| \leq Ch^{-2}$. For any $p > 1$ and any base weights asymptotic to κh^p ,

$$\frac{Z_n(\lambda_h a_h / a_{h-1})}{Z_n(\lambda_h)} = 1 - \frac{J_p}{3n} + o(n^{-1}).$$

At $p = 4$ the correction is $\pi/(12n)$. At $p = 2$ the conclusion is $1 + o(n^{-1})$, with no next term claimed. This example is within the cumulative theorem and outside the pointwise $q = 2$ hypothesis of Theorem 34.1(i).

40 Polynomial comparisons and absolute low degree corrections

40.1 Equal next to leading coefficients

Let Q, R be real polynomials of the same degree $d > 1$, positive at every positive integer, with common leading coefficient $\kappa > 0$. Suppose

$$\frac{Q(h)}{\kappa h^d} = 1 + \frac{a}{h} + \frac{c_Q}{h^2} + O(h^{-3}), \quad \frac{R(h)}{\kappa h^d} = 1 + \frac{a}{h} + \frac{c_R}{h^2} + O(h^{-3}).$$

Then $\log(Q(h)/R(h)) = (c_Q - c_R)h^{-2} + O(h^{-3})$ and

$$\frac{Z_n(Q)}{Z_n(R)} = P_{Q,R} \left[1 + \frac{(c_Q - c_R)J_d}{n} + o(n^{-1}) \right]. \quad (40.1)$$

If $Q(x) = \kappa \prod_{j=1}^d (x + \alpha_j)$ and $R(x) = \kappa \prod_{j=1}^d (x + \beta_j)$, counted with multiplicity, then equality of the next-to-leading coefficients is $\sum_j \alpha_j = \sum_j \beta_j$, and

$$P_{Q,R} = \prod_{j=1}^d \frac{\Gamma(1 + \beta_j)}{\Gamma(1 + \alpha_j)}. \quad (40.2)$$

Indeed the finite product through N equals the displayed constant times $\prod_j \Gamma(N + 1 + \alpha_j) / \Gamma(N + 1 + \beta_j)$; the latter tends to one by the fixed-shift gamma ratio and equal shift sums. Positivity excludes roots at positive integers and hence poles in the displayed gamma factors. Complex conjugate roots or individually negative real shifted factors cause no ambiguity: the full expression is the limit of positive real finite products.

For example, for $|a| < 1$,

$$\frac{Z_n(h^p(1 - a^2/h^2))}{Z_n(h^p)} = \frac{\sin(\pi a)}{\pi a} \left[1 - \frac{a^2 J_p}{n} + o(n^{-1}) \right] \quad (p > 1),$$

with the prefactor interpreted continuously at $a = 0$.

40.2 The quartic baseline from Berg and Valent

Let $m_n = Z_n(h^4)$ and $b_n = Z_n(h^2(h^2 - 1/4))$. The relative theorem gives

$$\frac{b_n}{m_n} = \frac{2}{\pi} \left[1 + \frac{\pi}{16n} + o(n^{-1}) \right], \quad \frac{m_n}{b_n} = \frac{\pi}{2} \left[1 - \frac{\pi}{16n} + o(n^{-1}) \right]. \quad (40.3)$$

Here $f_h = \log(1 - 1/(4h^2)) = -1/(4h^2) + O(h^{-4})$, the product is $2/\pi$, and $J_4 = -\pi/4$.

For the absolute normalization put

$$K_0 = \frac{\Gamma(1/4)^2}{4\sqrt{\pi}} = \sqrt{2}I_4, \quad r = \frac{2}{K_0} = \frac{8\sqrt{\pi}}{\Gamma(1/4)^2}.$$

Berg and Valent's equation (3.1) gives birth–death rates

$$\ell_j = (4j + 1)(4j + 2)^2(4j + 3), \quad m_j^{\text{BD}} = (4j - 1)(4j)^2(4j + 1), \quad m_0^{\text{BD}} = 0.$$

The alternating Stieltjes coefficients $\alpha_{2j+1} = \ell_j$, $\alpha_{2j} = m_j^{\text{BD}}$ obey $\alpha_h = 4h^2(4h^2 - 1) = 16h^2(h^2 - 1/4)$. S-to-J contraction gives diagonal $\alpha_{2j} + \alpha_{2j+1}$ and squared off-diagonal $\alpha_{2j+1}\alpha_{2j+2}$, matching their equation (2.31). Their moments are therefore $16^n b_n$. Proposition 3.6.1, pages 194–195, yields [11]

$$b_n = \frac{4\Gamma(4n + 2)}{16^n K_0^{4n+2}} [1 + O(25^{-n})]. \quad (40.4)$$

With

$$F_4(n) = \sqrt{\frac{2}{\pi}} r^{4n+2} (n!)^4 n^{-1/2}, \quad (40.5)$$

ordinary shifted Stirling gives

$$\frac{\pi b_n}{2F_4(n)} = 1 - \frac{1}{16n} + O(n^{-2}).$$

For explicit bookkeeping, the logarithmic n^{-1} coefficient of $\Gamma(4n+2)$ is $13/48$, while that of $(n!)^4$ is $1/3 = 16/48$. Their difference is $-1/16$. Combining with (40.3) recovers the pure quartic result of Report 199,

$$m_n = F_4(n) \left[1 - \frac{1+\pi}{16n} + o(n^{-1}) \right]. \quad (40.6)$$

The proof uses the classical comparison moment and a relative correction; it does not infer a recurrence correction from Freud norm errors.

[write] **A second route, 7 October 2026.** Equation (40.6) is Part II's Theorem 19.1 for the moments, reached by a second route: Part II integrates its own subtracted occupation limit (Section 27), whereas here the general relative theorem with $J_4 = -\pi/4$ is applied. Both use the same Berg–Valent normalization (Part II's Section 23, Part I's Section 12).

Corollary 40.1 (All positive zero harmonic quartics). *Let $Q(h) = \kappa(h^4 + bh^2 + ch + d)$, with $\kappa > 0$, be positive at all positive integers. Then*

$$Z_n(Q) = \kappa^n P_Q F_4(n) \left[1 - \frac{(1+\pi)/16 + \pi b/4}{n} + o(n^{-1}) \right], \quad (40.7)$$

where

$$P_Q = \prod_{h \geq 1} \frac{Q(h)}{\kappa h^4} = \frac{1}{\prod_{j=1}^4 \Gamma(1 + \alpha_j)} > 0, \quad Q(x)/\kappa = \prod_{j=1}^4 (x + \alpha_j).$$

Proof. The missing cubic coefficient means $\sum_j \alpha_j = 0$ and $\log(Q(h)/(\kappa h^4)) = b/h^2 + O(h^{-3})$. Equations (40.1), (34.9), and (40.6) give (40.7). Formula (40.2) gives the product. The remainder is little- $o(n^{-1})$; no $O(n^{-2})$ remainder for general Q follows. $\square$

40.3 The quadratic secant anchor

For clarity the classical pure-quadratic identification is proved formally. On polynomials in z set

$$L = (1 + z^2) \frac{d}{dz} + z, \quad Lz^h = hz^{h-1} + (h+1)z^{h+1}.$$

The matrix in the monomial basis is symmetric with off-diagonal entry h between levels $h-1$ and h . Therefore the constant coefficient of $L^{2n}1$ is $Z_n(h^2)$ by the finite weighted-walk identity. The formal function

$$\Psi(t, z) = \frac{1}{\cos t - z \sin t}$$

satisfies $\partial_t \Psi = L\Psi$ and $\Psi(0, z) = 1$. Formal uniqueness coefficient by coefficient in t proves

$$\sum_{n \geq 0} \frac{Z_n(h^2)}{(2n)!} t^{2n} = \sec t. \quad (40.8)$$

This is a classical continued-fraction correspondence, also given by Deb and Sokal [15, Sec. 6.3, Eq. (6.24a) at $\alpha = k = 0$], and by the earlier weighted-path theory [14]. No unbounded operator exponential is assumed.

Put $R = \pi/2$. After removing its singularities at $\pm R$,

$$\sec t - \frac{1}{R-t} - \frac{1}{R+t}$$

is analytic in $|t| < 3R$. Cauchy's estimate on $|t| = 2R$ therefore gives

$$[t^{2n}] \sec t = 2R^{-2n-1} [1 + O(4^{-n})].$$

Together with the central-binomial Stirling expansion,

$$\frac{(2n)!}{(n!)^2} = \frac{4^n}{\sqrt{\pi n}} \left[1 - \frac{1}{8n} + O(n^{-2}) \right],$$

this yields

$$Z_n(h^2) = F_2(n) \left[1 - \frac{1}{8n} + O(n^{-2}) \right], \quad F_2(n) = \frac{4}{\pi^{3/2}} \left(\frac{16}{\pi^2} \right)^n (n!)^2 n^{-1/2}. \quad (40.9)$$

Corollary 40.2 (All positive zero harmonic quadratics). *Let $Q(h) = \kappa(h^2 + b)$, where $\kappa > 0$ and $b > -1$. Then*

$$Z_n(Q) = \kappa^n P_b F_2(n) \left[1 - \frac{1}{8n} + o(n^{-1}) \right], \quad (40.10)$$

where

$$P_b = \prod_{h \geq 1} (1 + b/h^2) = \begin{cases} \sinh(\pi\sqrt{b})/(\pi\sqrt{b}), & b > 0, \\ 1, & b = 0, \\ \sin(\pi\sqrt{-b})/(\pi\sqrt{-b}), & -1 < b < 0. \end{cases}$$

Proof. The condition $b > -1$ is exactly positivity on positive integers. The logarithmic ratio is $b/h^2 + O(h^{-4})$. Since $J_2 = 0$, the relative theorem gives $Z_n(Q)/(\kappa^n Z_n(h^2)) = P_b[1 + o(n^{-1})]$. Multiply by (40.9). Euler's sine product, with its imaginary-argument version, evaluates P_b . The coefficient $-1/8$ is independent of b ; the first nonzero b -dependent correction has not been established here. $\square$

40.4 OEIS normalization anchors

A227887 is the raw pure-quartic moment sequence $Z_n(h^4)$, beginning

$$1, 1, 17, 1585, 485729, 372281761.$$

A144853 has descent weights $h^2(4h^2 - 1)/3 = (4/3)h^2(h^2 - 1/4)$, while A261000 has weights $h^2(4h^2 - 1)$ [2]. Their first terms are respectively

$$1, 1, 21, 2541, 1023561, 1036809081 \quad \text{and} \quad 1, 3, 189, 68607, 82908441.$$

Thus $A261000(n) = 3^n A144853(n)$ exactly, and

$$\begin{aligned} A144853(n) &= \frac{2}{\pi} \left(\frac{4}{3} \right)^n F_4(n) \left[1 - \frac{1}{16n} + o(n^{-1}) \right], \\ A261000(n) &= \frac{2}{\pi} 4^n F_4(n) \left[1 - \frac{1}{16n} + o(n^{-1}) \right]. \end{aligned}$$

In these particular classical models (40.4) even supplies an $O(n^{-2})$ remainder after the displayed first correction. These are normalization checks and consequences of classical elliptic formulas, not claims to new elliptic asymptotics.

A338634 is a distinct cumulant transform of the pure-quartic moments. Its prefix is

$$1, 1, 15, 1490, 472475, \dots,$$

which differs from the A227887 prefix. The theorems here concern the raw moments Z_n ; no cumulant transfer for a general polynomial Q is asserted. We assign no OEIS identifier to an unverified moment family.

41 Relative inverse displacements and integer thresholds

Let $F, \tilde{F}$ be the continuous piecewise-linear interpolations of $\log Z_n(\lambda)$ and $\log Z_n(\tilde{\lambda})$ on the non-negative integers. By Lemma 35.2, all one-step slopes are asymptotic to $p \log n$ and hence strictly positive on a tail. For sufficiently large Y , let $x, \tilde{x}$ be the unique tail solutions

$$F(x) = \log(Y/P), \quad \tilde{F}(\tilde{x}) = \log Y. \quad (41.1)$$

The base inverse x is exact for the base sequence's log-linear interpolation. It is not an explicit Lambert- W approximation. The division by P is part of the comparison and cannot be omitted.

Theorem 41.1 (Exact base relative displacement). *Suppose one of the proved logarithmic ratio laws has the form*

$$\log \frac{Z_n(\tilde{\lambda})}{Z_n(\lambda)} = \log P + Ds(n) + o(s(n)), \quad (41.2)$$

where $s(n)$ is n^{1-q} , $n^{-p} \log n$, or n^{-p} in its respective regime. Then

$$\tilde{x} - x = -\frac{Ds(x)}{p \log x} + o\left(\frac{s(x)}{\log x}\right). \quad (41.3)$$

The cumulative theorem also has this conclusion with $s(x) = x^{-1}$, $P = e^S$, and $D = -AJ_p$.

Proof. Each scale is positive on a tail, tends to zero, and obeys $s(n + \theta)/s(n) \rightarrow 1$ uniformly for $0 \leq \theta \leq 1$. Linear interpolation of the endpoint values in (41.2) therefore gives

$$\tilde{F}(z) - F(z) - \log P = Ds(z) + o(s(z))$$

uniformly as real $z \rightarrow \infty$. The maximum of the two normalized endpoint errors tends to zero, and both endpoint scales are asymptotic to the scale within that interval.

At $z = x$ the residual is $\tilde{F}(x) - \log Y = Ds(x) + o(s(x)) = o(1)$. Every slope on $[x - 1, x + 1]$ is $(p + o(1)) \log x$, so first $\tilde{x} \in (x - 1, x + 1)$, and then the lower slope bound gives $|\tilde{x} - x| = O(s(x)/\log x) = o(1)$. If $D = 0$, the same argument gives little- o directly. A secant crossing one or more interpolation breakpoints is a convex combination of neighboring slopes. Hence

$$\tilde{F}(\tilde{x}) - \tilde{F}(x) = (\tilde{x} - x)(p + o(1)) \log x = -Ds(x) + o(s(x)),$$

which proves (41.3). No derivative of the unknown remainder or differentiability at an integer breakpoint is assumed. $\square$

The three explicit displacements are

$$\begin{aligned} 1 < q < p + 1 : \quad \tilde{x} - x &= -\frac{bJ_{p,q}}{p} \frac{x^{1-q}}{\log x} + o(x^{1-q}/\log x), \\ q = p + 1 : \quad \tilde{x} - x &= -\frac{2b}{pd_p} x^{-p} + o(x^{-p}), \\ q > p + 1 : \quad \tilde{x} - x &= -\frac{2}{p\kappa d_p} \left[\sum_{h \geq 1} (\tilde{\lambda}_h - \lambda_h) \right] \frac{x^{-p}}{\log x} + o(x^{-p}/\log x). \end{aligned}$$

In the critical case the logarithm in the correction cancels the logarithm in the inverse slope. In every case a zero coefficient gives only the stated smaller-order displacement.

Define the inclusive integer threshold

$$\tilde{N}(Y) = \min\{n \geq 0 : Z_n(\tilde{\lambda}) \geq Y\}.$$

For sufficiently large Y , its value lies in the increasing tail and the finite earlier values are all below Y . Therefore $\tilde{N}(Y) = \lceil \tilde{x} \rceil$, including equality at an exact integer-indexed moment. Formula (41.3) implies existence of a positive function $\eta(x) \rightarrow 0$ such that

$$\left\lceil x - \frac{Ds(x)}{p \log x} - \eta(x) \frac{s(x)}{\log x} \right\rceil \leq \tilde{N}(Y) \leq \left\lceil x - \frac{Ds(x)}{p \log x} + \eta(x) \frac{s(x)}{\log x} \right\rceil. \quad (41.4)$$

The ceilings eventually coincide or are consecutive. Neither η nor the onset is effective here. A tiny real displacement does not force equality of the two integer thresholds near a boundary, and it does not justify an unconditional single ceiling.

Remark 41.2 ([write] The inverse and the transseries volume, 7 October 2026). Against the volume named in the Guide (its symbols carry the subscript “vol”).

- (a) *Instance.* The identity $\tilde{N}(Y) = \lceil \tilde{x} \rceil$ used for (41.4) is `p0:thm:staircase(1)` for the piecewise-linear interpolation $\tilde{F}$ of $\log Z_n(\tilde{\lambda})$, an admissible interpolation in the sense of `p0:def:three-inverses` on the increasing tail; the bracket (41.4) then follows from the displacement estimate by monotonicity of the ceiling, the setting of `p0:thm:staircase(2)`.
- (b) *Analogue.* Theorem 41.1 divides a residual by a slope, as `p0:thm:backward-error` does, but with secant slopes of a piecewise-linear function rather than a derivative; it is an analogue of that theorem, not an instance.
- (c) *Not compared.* The base inverse x is exact for the base sequence’s interpolation and is not an explicit Lambert or core solution, so the volume’s core theorems do not apply to it as stated.

No novelty is claimed for any of these inversions.

42 What the present results do not yet establish

The cumulative theorem can use information inaccessible to a pointwise coefficient theorem, but it still requires both parts of (34.10). This distinction matters in the Freud comparison underlying Report 201. The directly inspected CKM restatement [8, Eq. (2.3)] supplies, after the established norm telescoping and Stirling calculation, a cumulative coefficient $(p-1)/12$, but its common recurrence envelope is only $O(1/(h \log^2 h))$. That does not imply $O(h^{-2})$. Differentiating an unspecified norm error is not legitimate, and qualitative local occupation convergence does not repair the missing envelope.

A sharper normalized leading-coefficient bound reported in Mastroianni and Milovanović [10, Theorem 2.4.10] suggests a route for $p \geq 4$, but original-primary verification and source-image verification of that stronger statement are outside the cleared source basis of this report. We therefore record only the following conditional implication, without presenting it as a proved general-power asymptotic: if the Freud comparison has both

$$|f_h| = O(h^{-2}), \quad \sum_{h \leq N} f_h = \log P + \frac{p-1}{12N} + o(N^{-1}),$$

then cumulative transfer and the explicit gamma moment would yield

$$\frac{Z_n(h^p)}{F_p(n)} = 1 + \frac{C_p}{n} + o(n^{-1}), \quad C_p = \frac{p^2 - 6p + 2}{24p} + \frac{p-1}{12} J_p. \quad (42.1)$$

For the bookkeeping, the comparison moment is

$$V_n = \left(\frac{4}{s^2}\right)^n \frac{\Gamma(pn + p/2)}{\Gamma(p/2)}, \quad s = (pI_p)^{p/2}.$$

Its Stirling coefficient relative to the factorial template is $B_2(p/2)/(2p) - p/12 = (p^2 - 6p + 2)/(24p)$, where $B_2(z) = z^2 - z + 1/6$. Dividing by the hypothetical cumulative comparison ratio adds $(p-1)J_p/12$, with the sign in (42.1). The formula is a conditional algebraic target. For $p = 4$ it agrees with the independently proved coefficient $-(1 + \pi)/16$. Such agreement and numerical fits do not verify the missing source premise for other p . No unconditional new $p \geq 4$ or cubic absolute coefficient is claimed.

Several concrete questions remain.

1. Can the required Freud envelope be verified directly in primary sources, with every endpoint of its exponent range checked?
2. When $J_{p,q} = 0$, what additional tail information determines the next nonzero correction? The present theorem only gives a little- o improvement.
3. What is the occupation scale and limiting law in a shrinking neighborhood of H_p ? Our local theorem deliberately stops at compact intervals below it.
4. Can the cumulative theorem be extended under weaker pointwise envelopes by proving a quantitative discrete-variation bound for occupation?
5. Which higher corrections involve only tail coefficients and which involve finite-height data? The endpoint regime shows that finite modifications already matter at order n^{-p} .
6. Can explicit error bounds and an effective onset make the two-ceiling inverse enclosure into a practical certified threshold algorithm? Direct integer comparisons are required when refining a smooth absolute inverse beyond the curvature error of log-linear interpolation.

[write] **The conditional coefficient as an open question, 7 October 2026.** Under Vladimir's standing rule of 4 October 2026, the coefficient C_p of (42.1) is not a theorem: it is conditional on a Freud envelope $|f_h| = O(h^{-2})$ with the stated cumulative expansion, which neither the source nor the write has verified in a primary source (Mastroianni–Milovanović is a secondary lead only). It stands as an open question, Question 1 above. What is known: at $p = 2$ it gives $-1/8$, the proved coefficient (40.9), and at $p = 4$ it gives $-(1 + \pi)/16$, the proved coefficient (40.6); both checked by the write ($J_2 = 0$, $J_4 = -\pi/4$). At $p = 3$ it gives $C_3 = -\frac{7}{72} + \frac{1}{6}J_3 = -0.19798885356\dots$, $J_3 = -0.6045997880\dots$; the write computed $Z_n(h^3)$ exactly for $n \leq 400$ and found $n(Z_n(h^3)/F_3(n) - 1) = -0.19776\dots, -0.19787\dots, -0.19793\dots$ at $n = 100, 200, 400$, with Richardson values $2e(2n) - e(n)$ equal to $-0.1979892\dots$ ($n = 100$) and $-0.1979889\dots$ ($n = 200$). This agreement to about six digits is numerical evidence for the conditional formula at $p = 3$, not a proof, and it says nothing about the envelope itself.

43 Reproduction and source boundaries

The accompanying package contains the article’s TeX and PDF, deterministic build instructions, exact rational/integer fixtures, executable finite identities, and separately labeled multiprecision diagnostics. Exact checks test finite mechanisms such as Dyck recurrences, marked-excursion deletion, Catalan branch coefficients, telescoping products, and the secant operator. Numerical checks illustrate convergence and inversion scales. Neither type replaces the proofs or supplies an effective asymptotic error enclosure.

Runtime validation uses explicit exceptions and remains active under optimized Python. The package records complete member hashes and distinguishes source files from generated results. Rebuilds from actual ZIP extraction are required to agree member by member and as a whole archive under the documented toolchain; PDF byte identity across different TeX distributions is not asserted. The README specifies commands, dependencies, and the final finite-check counts. No private paths, third-party PDFs, or remote fetches are needed for a normal reproduction.

The mathematical dependence on public sources is narrow. The formal Dyck/continued-fraction interpretation is classical [14]; the quadratic anchor is reproved here, with its prior secant identification explicitly acknowledged [15]. The quartic absolute normalization uses Berg–Valent’s stated moment asymptotic with the factor 16^n checked explicitly [11]. The general-power leading formula and Freud sources are cited for context and for the scope limitation, not smuggled into a stronger absolute theorem. The companion Report 201 supplies the origin of the appendix arguments, which are reproduced here so they can be checked without access to the companion. Report 199 supplies the prior report-level quartic result, recovered here as an application rather than claimed anew.

The OEIS examples are normalization anchors. A144853’s S-fraction coefficients are identified through A187756, and the exact scaling to A261000 is consistent with their displayed JacobiSD coefficient formulas. A227887 is also identified as column four of the power-weight array A290569. No exhaustive literature search, priority certification, or claim that the classical elliptic formulas are new is made.

A Exact deletion and the uniform endpoint bound

[write] **Part I’s counterpart, 7 October 2026.** This appendix is Part I’s Sections 2–3 almost word for word (Lemmas 2.1–2.2 and the proof of Theorem 1.4), with the constant A renamed A_0 , the interpolation parameter suppressed, and one addition: the Cauchy–Schwarz consequence $Z_j^2 \leq Z_{j-1}Z_{j+1}$ (discrete log-convexity), which Lemma 35.2 uses. The batch-111 placement proposed printing it once; it is printed again because that addition and the references of Section 35 rest on it.

This appendix reproduces the endpoint argument of Report 201 with its constants, so the present correction theorems have a complete portable foundation. The parameter t is suppressed; all bounds depend only on the common power class. For the rest of this section assume (35.2). Write

$$u_n(h) = \mathbb{E}_{n,\lambda} N_h, \quad v_n(h) = \mathbb{P}_{n,\lambda}(\max D \geq h),$$

and set $u_N(j) = 0$ when $j > N$.

Lemma A.1 (An all-length ratio bound). *For every $n \geq 1$ and $0 \leq k \leq n$,*

$$Z_n(\lambda) \geq c^n (n!)^p \geq (cn^p/e^p)^n, \quad \frac{Z_{n-k}(\lambda)}{Z_n(\lambda)} \leq \left(\frac{e^p}{cn^p} \right)^k. \quad (\text{A.1})$$

Proof. Take the finite symmetric Jacobi matrix indexed by $0, \dots, n$, with zero diagonal and entries $J_{h-1,h} = J_{h,h-1} = \sqrt{\lambda_h}$. A closed nearest-neighbor walk from zero of length $2j$, $j \leq n$, is a Dyck path and cannot leave this truncation. It traverses each edge equally often in the two directions. Hence

$$(J^{2j})_{00} = Z_j(\lambda), \quad 0 \leq j \leq n.$$

The spectral theorem, applied to squared eigenvalues, makes these initial moments those of a positive measure on $[0, \infty)$ of mass one. Hölder's inequality gives $Z_{n-k} \leq Z_n^{(n-k)/n}$. The same finite spectral representation, with truncation at $j+1$, gives $Z_j^2 \leq Z_{j-1}Z_{j+1}$ by Cauchy–Schwarz; hence $\log Z_j$ is discrete convex. The mountain $U^n D^n$ contributes at least $c^n (n!)^p$, and $n! \geq (n/e)^n$. Dividing the moment inequality by Z_n proves the claim, including $k=0, n$. $\square$

This fixed- n chord bound controls every deletion length at once. No asymptotic one-step ratio is used.

Let $e_{h,k}$ be the total weight of primitive negative excursions of length $2k$ at height h : they start at h , immediately descend, remain in $\{0, \dots, h-1\}$ until the last step, then rise to h . In particular these excursions remain nonnegative. Their k downsteps have heights at most h , so

$$0 \leq e_{h,k} \leq 4^k C^k h^{pk}. \quad (\text{A.2})$$

Lemma A.2 (Exact deletion and slot identity). *For $1 \leq h \leq n$,*

$$u_n(h) - v_n(h) = \sum_{k=1}^{n-h} e_{h,k} \frac{Z_{n-k}}{Z_n} (u_{n-k}(h) + u_{n-k}(h+1)). \quad (\text{A.3})$$

Proof. Mark a non-last descent across edge h . A later return to h exists; cut out the path from this descent through its first such return. It is a unique primitive negative excursion. Deletion leaves a Dyck path of semilength $n-k$ with a distinguished vertex of height h .

Conversely insert any permitted negative excursion at any vertex of height h in the shorter path. Since that path subsequently ends at zero, its later steps include a descent across h , so the inserted marked descent is non-last. The constructions are inverse, preserve all other descent heights, and multiply weights. The shorter path must reach h , which forces $k \leq n-h$.

For $h > 0$, a path has exactly $N_h + N_{h+1}$ vertices of height h : classify their arrivals as upsteps across edge h or downsteps across edge $h+1$, using upward/downward balance on edge h . The number of non-last descents is $N_h - \mathbf{1}_{\{\max D \geq h\}}$. Weighted summation gives (A.3). $\square$

Put

$$A_0 = \frac{4Ce^p}{c}, \quad a(n, h) = A_0(h/n)^p. \quad (\text{A.4})$$

Equations (A.1)–(A.3) imply

$$u_n(h) \leq 1 + \sum_{k=1}^{n-h} a(n, h)^k (u_{n-k}(h) + u_{n-k}(h+1)). \quad (\text{A.5})$$

All quantities in this recursion are nonnegative, a fact used in the deliberate overcounting below.

A.1 A rough bound followed by the endpoint error

Choose constants

$$\alpha = \min\left(\frac{1}{4}, (12A_0)^{-1/p}\right), \quad \eta = A_0\alpha^p \leq \frac{1}{12}, \quad r_0 = 3\eta \leq \frac{1}{4}, \quad \kappa_0 = \frac{\alpha}{2(1+\alpha)}. \quad (\text{A.6})$$

We first prove a rough bound in a wider cone and then extract the small error in a narrower one.

Start with $h \leq \alpha n/2$. Expand (A.5) recursively at every pair (N, j) with $j \leq \alpha N$; stop at the first exit from this outer cone. The expansion is finite because N strictly decreases at every edge. Inside the cone the coefficient of a loss k is at most η^k .

For total loss s and ℓ recursive edges, the positive compositions of s supply $\binom{s-1}{\ell-1}$ possibilities, and at each edge there are two height choices, preserving or increasing j . Thus the total coefficient of all chains with loss $s \geq 1$ is bounded by

$$\sum_{\ell=1}^s 2^\ell \binom{s-1}{\ell-1} \eta^s = 2 \cdot 3^{s-1} \eta^s = \frac{2}{3} r_0^s. \quad (\text{A.7})$$

This includes impossible chains and therefore only enlarges a nonnegative upper bound. The constants inserted at internal nodes sum to at most $1 + 2\eta/(1 - 3\eta)$.

A terminal exit has coordinates $(n - s, h + t)$, where $0 \leq t \leq s$. Since $h + t > \alpha(n - s)$ but $h \leq \alpha n/2$, it must have $s > \kappa_0 n$. Every terminal occupation is at most n ; an impossible child has occupation zero. Therefore all exit terms together are at most

$$\frac{2n}{3(1 - r_0)} r_0^{\kappa_0 n} \leq \frac{2}{3(1 - r_0)e\kappa_0 \log(1/r_0)}. \quad (\text{A.8})$$

The last inequality uses $\sup_{x \geq 0} x e^{-bx} = 1/(eb)$. Consequently

$$u_n(h) \leq B_0 := 1 + \frac{2\eta}{1 - 3\eta} + \frac{2}{3(1 - r_0)e\kappa_0 \log(1/r_0)} \quad (1 \leq h \leq \alpha n/2). \quad (\text{A.9})$$

Now take $n \geq 8/\alpha$ and $h \leq \alpha n/8$. If $k \leq n/2$, then

$$h + 1 \leq \alpha n/4 \leq \alpha(n - k)/2,$$

so both child occupations in (A.3) are at most B_0 . For $k > n/2$ use n . With $a = a(n, h) \leq \eta/8^p < 1/4$, summing the geometric bounds gives

$$\begin{aligned} 0 \leq u_n(h) - v_n(h) &\leq \frac{2B_0 a + 2na^{\lfloor n/2 \rfloor + 1}}{1 - a} \\ &\leq \frac{8}{3}(B_0 + 1)a. \end{aligned} \quad (\text{A.10})$$

Here $n4^{-\lfloor n/2 \rfloor} \leq 1$ for $n \geq 2$. A path with maximum below h has at most 4^n possible step patterns and weight at most $(Ch^p)^n$. The mountain bound yields

$$1 - v_n(h) \leq [A_0(h/n)^p]^n = a^n \leq a. \quad (\text{A.11})$$

Finally the identity

$$|N_h - 1| = (N_h - \mathbf{1}_{\{\max D \geq h\}}) + \mathbf{1}_{\{\max D < h\}}$$

holds pathwise. Equations (A.10)–(A.11) prove the endpoint part of Lemma 35.1 with the explicit, nonoptimal choices

$$\rho = \alpha/8, \quad n_0 = \lceil 8/\alpha \rceil, \quad C_1 = A_0 \left[1 + \frac{8}{3}(B_0 + 1) \right]. \quad (\text{A.12})$$

All constants depend only on p, c, C . They can become large as $p \downarrow 0$; the theorem fixes $p > 0$ and asserts no uniformity in that limit.

B Uniform arch concentration and free energy

[write] **Part I's counterpart, 7 October 2026.** This appendix adapts Part I's Section 6 to the scaled family $\kappa h^p e^{g_h(t)}$ with uniformly vanishing g_h , which Lemma 35.1 needs, and its final subsection evaluates the free energy $\Phi_p(y_*) = \log d_p - p$, which Part I does not print. It is kept in full for these two reasons; the closed-set bound, the calibration and the lower bound are Part I's arguments.

This appendix adapts the direct concentration proof of Report 201 to the family needed here. Suppose

$$\lambda_h(t) = \kappa h^p e^{g_h(t)}, \quad \sup_{h,t} |g_h(t)| < \infty, \quad \sup_t |g_h(t)| \longrightarrow 0 \quad (h \rightarrow \infty).$$

These conditions follow from the fixed power class and uniform asymptotic equivalence in Lemma 35.1. No continuity in t is needed. In particular geometric interpolation has $g_h(t) = \log(\lambda_h/(\kappa h^p)) + t f_h$. A bounded pathwise density comparison with the pure-power measure would in general be false because finite-height changes can be repeated n times; the argument therefore treats the family directly.

Interpolate the path heights linearly, with time divided by n and height by n . Thus $y_n(s)$, $0 \leq s \leq 2$, belongs to the compact uniform-topology space

$$\mathcal{K} = \{y : [0, 2] \rightarrow [0, \infty) : y(0) = y(2) = 0, \quad y \text{ is 1-Lipschitz}\}.$$

Every $y \in \mathcal{K}$ satisfies $y(s) \leq \min(s, 2-s) \leq 1$. Compactness follows from uniform boundedness, equicontinuity, and closure.

For $|v| \leq 1$ define the binary entropy

$$\mathcal{E}(v) = -\frac{1+v}{2} \log \frac{1+v}{2} - \frac{1-v}{2} \log \frac{1-v}{2}, \quad 0 \log 0 = 0,$$

and the extended-real functionals

$$S(y) = \int_0^2 \mathcal{E}(y'(s)) \, ds, \quad V(y) = \frac{p}{2} \int_0^2 \log y(s) \, ds, \quad \Phi_p(y) = S(y) + V(y). \quad (\text{B.1})$$

Lipschitz functions are differentiable almost everywhere. A divergent negative logarithmic integral means $V(y) = -\infty$.

If H_j is the upper endpoint height of step j , upward/downward edge balance gives the exact identity

$$\log \frac{W_{\lambda(t)}(D)}{\kappa^n n^{pn}} = \frac{1}{2} \sum_{j=1}^{2n} \left[p \log(H_j/n) + g_{H_j}(t) \right]. \quad (\text{B.2})$$

The factor $1/2$ is essential: each weighted descent is paired with the corresponding number of ascents across its edge.

B.1 A clipped closed-set upper bound

Fix $\varepsilon \in (0, 1]$ and an integer $M \geq 1$. Set

$$F = \max(0, \sup_{h,t} g_h(t)), \quad \omega_M = \sup_{h > M, t} |g_h(t)| \longrightarrow 0.$$

For $n \geq (M/\varepsilon)e^{F/p}$, every $h \geq 1$ and $0 \leq t \leq 1$ satisfy

$$p \log(h/n) + g_h(t) \leq p \log \max(h/n, \varepsilon) + \omega_M. \quad (\text{B.3})$$

For $h > M$ the perturbation is bounded by ω_M . For $h \leq M$, the chosen n gives $p \log(h/n) + F \leq p \log \varepsilon$. Thus the finitely many large low-height values are absorbed into the clipping slack, rather than charged an uncontrolled $O(n)$ error.

For a finite partition $Q = (s_i)$ of $[0, 2]$, define

$$S_Q(y) = \sum_i (s_{i+1} - s_i) \mathcal{E} \left(\frac{y(s_{i+1}) - y(s_i)}{s_{i+1} - s_i} \right), \quad V_\varepsilon(y) = \frac{p}{2} \int_0^2 \log \max(y(s), \varepsilon) ds.$$

Both are continuous on $\mathcal{K}$. Nested dyadic partitions give $S_Q \downarrow S$: Jensen's inequality gives monotonicity, and differentiation of Lipschitz functions with bounded convergence identifies the limit. Likewise $V_\varepsilon \downarrow V$ as $\varepsilon \downarrow 0$, by monotone convergence applied to the nonnegative function $-\log \max(y, \varepsilon)$.

Let $B \subseteq \mathcal{K}$ be closed. Divide the $2n$ steps into blocks at rounded times ns_i . There are only polynomially many vectors of block endpoint heights. For a block of length L and displacement d , the bridge count is at most

$$\binom{L}{(L+d)/2} \leq \exp\{L\mathcal{E}(d/L)\};$$

impossible parities give no bridge, and nonnegativity can only decrease the count. For fixed Q , rounding errors in normalized entropy are $o(1)$ uniformly over paths, by continuity of $\mathcal{E}$. The step upper height divided by n differs from $y_n(s)$ by at most $1/n$ within its interval. Since the clipped logarithm is $1/\varepsilon$ -Lipschitz, (B.2)–(B.3) give

$$\limsup_{n \rightarrow \infty} \sup_{t \in [0,1]} \frac{1}{n} \log \left(\kappa^{-n} n^{-pn} \sum_{D: y_n \in B} W_{\lambda(t)}(D) \right) \leq \sup_{y \in B} (S_Q(y) + V_\varepsilon(y)) + \omega_M. \quad (\text{B.4})$$

The conventions are $\log 0 = -\infty$ and $\sup \emptyset = -\infty$.

First let $M \rightarrow \infty$. Then take nested dyadic partitions and decreasing $\varepsilon \rightarrow 0$. For a decreasing sequence of continuous functions on a compact set, the maxima converge to the supremum of their pointwise upper-semicontinuous limit. Indeed choose maximizing points, take a convergent subsequence, and bound its limiting values by each fixed earlier function; the reverse inequality is immediate. Applying this to the compact set B proves

$$\limsup_{n \rightarrow \infty} \sup_t \frac{1}{n} \log \left(\kappa^{-n} n^{-pn} \sum_{D: y_n \in B} W_{\lambda(t)}(D) \right) \leq \sup_{y \in B} \Phi_p(y). \quad (\text{B.5})$$

It also shows that Φ_p is upper semicontinuous. This is the closed-set large-deviation upper estimate actually needed below.

B.2 The unique calibrated arch

Define y_* increasingly on $[0, 1]$ by

$$s = \int_0^{y_*(s)} \frac{dv}{\sqrt{1 - (v/H_p)^p}}, \quad (\text{B.6})$$

and reflect about $s = 1$. Since $H_p I_p = 1$, its maximum is H_p , and $y_* \in \mathcal{K}$. In the interior,

$$(y'_*)^2 = 1 - (y_*/H_p)^p, \quad y''_* = -\frac{p}{2H_p^p} y_*^{p-1}. \quad (\text{B.7})$$

The second identity extends through the top by continuation. At either endpoint, height divided by the distance to that endpoint tends to one.

Lemma B.1 (Calibration and uniqueness). *The functional Φ_p has the unique maximizer y_* on $\mathcal{K}$.*

Proof. Set

$$A_*(s) = \mathcal{E}'(y'_*(s)) = \frac{1}{2} \log \frac{1 - y'_*(s)}{1 + y'_*(s)}.$$

Equations (B.7) give

$$A'_*(s) = \frac{p}{2y_*(s)}, \quad A_*(s) = O(1 + |\log s| + |\log(2 - s)|).$$

The entropy and logarithm tangent inequalities are

$$\begin{aligned} \mathcal{E}(y') - \mathcal{E}(y'_*) &\leq A_*(y' - y'_*), \\ \frac{p}{2}(\log y - \log y_*) &\leq \frac{p}{2} \frac{y - y_*}{y_*}. \end{aligned}$$

Integrate on $[d, 2 - d]$ and integrate the first right side by parts. The interior terms cancel. The remaining boundary term $[A_*(y - y_*)]_d^{2-d}$ is $O(d|\log d|)$ because both heights are $O(d)$ at the endpoints.

The tangent terms are integrable: A_* is integrable and $(y - y_*)/y_*$ is bounded near the endpoints. If $V(y) = -\infty$ there is nothing to prove; otherwise taking $d \downarrow 0$ proves $\Phi_p(y) \leq \Phi_p(y_*)$. If $y \neq y_*$, continuity makes them different on a set of positive measure, and the logarithmic tangent deficit has strictly positive integral,

$$\frac{p}{2} \int_0^2 \left[\frac{y}{y_*} - 1 - \log \frac{y}{y_*} \right] ds > 0,$$

possibly infinite. The strict inequality proves uniqueness, including nonsmooth competitors, zero plateaus, and multiple arches. $\square$

For every fixed $\eta_0 > 0$, compactness and upper semicontinuity therefore give a strict gap

$$\sup_{\|y - y_*\|_\infty \geq \eta_0} \Phi_p(y) < \Phi_p(y_*). \quad (\text{B.8})$$

B.3 A matching lower bound with controlled endpoints

We provide the lower bound explicitly because the logarithm is singular at zero. Approximate y_* by positive polygons $z \in \mathcal{K}$ that equal s on $[0, \varepsilon]$ and $2 - s$ on $[2 - \varepsilon, 2]$, and have all interior slopes strictly between -1 and 1 . For example join $(\varepsilon, \varepsilon)$ to $(2\varepsilon, y_*(2\varepsilon))$, use fine chords of y_* on the remaining interior, and reflect at the end. The joining slope is greater than -1 , less than 1 , and tends to 1 as $\varepsilon \downarrow 0$.

These approximants converge uniformly to y_* , their derivatives converge almost everywhere, and they satisfy $z(s) \geq c_0 \min(s, 2 - s)$ with a common $c_0 > 0$ for sufficiently advanced approximants. The bounded entropy and an integrable constant plus $|\log \min(s, 2 - s)|$ dominate the two integrands. Hence

$$\Phi_p(z) \longrightarrow \Phi_p(y_*). \quad (\text{B.9})$$

Fix one polygon. Refine its linear interior pieces into blocks of maximal length d without changing z . Choose lattice times and parity-compatible lattice heights within $O(1)$ of their prescribed values $ns_i, nz(s_i)$. Take deterministic all-up and all-down endpoint pieces of rounded lengths. For large n , the fixed positive slack of every interior slope from ± 1 guarantees feasible bridges.

Let $b_z = \min_{[\varepsilon, 2-\varepsilon]} z > 0$. Every unrestricted bridge in an interior block stays within $2d + O(1/n)$ of z , since both its slopes and those of z have absolute value at most one. Choosing d sufficiently small relative to b_z makes *every* such bridge positive. Their concatenations are consequently admissible Dyck paths. Distinct bridge choices give distinct paths. Stirling's formula for the finitely many binomial factors gives

$$\log \#\{\text{constructed paths}\} = nS(z) + o(n).$$

The endpoint pieces have zero entropy.

On the interior, the logarithm is uniformly Lipschitz and the normalized pure-power log weights approach the *interior contribution* to $nV(z)$, with error $nO_z(d) + o(n)$. The deterministic endpoints contribute separately

$$p \sum_{h=1}^{\lfloor \varepsilon n \rfloor} \log(h/n) + O(\log n), \quad (\text{B.10})$$

whose quotient by n tends to $p \int_0^\varepsilon \log s \, ds$. This is exactly the two-endpoint contribution to $V(z)$, including the factor $1/2$ in (B.2).

The perturbation is uniformly $o(n)$ over all these paths and all t . On the deterministic endpoints its absolute contribution is at most $\sum_{h \leq \varepsilon n + O(1)} \sup_t |g_h(t)| = o(n)$ by Cesàro convergence. On the interior all heights are at least $c_z n$, so the absolute contribution is at most $n \sup_{h \geq c_z n, t} |g_h(t)| = o(n)$. It follows that

$$\liminf_{n \rightarrow \infty} \inf_t \frac{1}{n} \log[\kappa^{-n} n^{-pn} Z_n(\lambda(t))] \geq \Phi_p(z) - O_z(d).$$

The limits are taken in order: first $n \rightarrow \infty$ at fixed polygon and refinement, then $d \downarrow 0$, then improve the polygon. Combining with (B.5) for $B = \mathcal{K}$ proves the uniform variational limit

$$\sup_{t \in [0, 1]} \left| \frac{1}{n} \log[\kappa^{-n} n^{-pn} Z_n(\lambda(t))] - \Phi_p(y_*) \right| \longrightarrow 0. \quad (\text{B.11})$$

Using (B.5) on the closed complement of an η_0 -ball and the gap (B.8) yields the needed conclusion.

Proposition B.2 (Uniform fixed-neighborhood concentration). *For every fixed $p > 0$, family $g_h(t)$ satisfying the displayed uniform hypotheses, and fixed $\eta_0 > 0$, there are $c_{\eta_0} > 0$ and n_{η_0} such that*

$$\sup_{0 \leq t \leq 1} \mathbb{P}_{n,t}(\|y_n - y_*\|_\infty \geq \eta_0) \leq e^{-c_{\eta_0} n} \quad (n \geq n_{\eta_0}). \quad (\text{B.12})$$

Only fixed-neighborhood concentration and the displayed closed-set upper estimate have been proved here. No rate for shrinking neighborhoods, fluctuation theorem, or full lower large-deviation principle on arbitrary open sets is asserted.

B.4 Evaluation of the free energy constant

Set $v = y'_*$ and use $A_* = \mathcal{E}'(v)$ from the calibration proof. Since $1 - v^2 = (y_*/H_p)^p$,

$$\mathcal{E}(v) + \frac{p}{2} \log y_* = \log 2 + \frac{p}{2} \log H_p + v A_*.$$

Also $A'_* = p/(2y_*)$, and $y_* A_*$ tends to zero at both endpoints. Integrating by parts,

$$\int_0^2 y'_* A_* \, ds = - \int_0^2 y_* A'_* \, ds = -p.$$

Consequently $\Phi_p(y_*) = \log 4 + p \log H_p - p = \log d_p - p$. The uniform variational limit therefore becomes exactly (35.5), including the factor κ^n . This completes the proof of every assertion of Lemma 35.1.

Part IV

Diagonal power weighted Dyck paths: all orders asymptotics and integer safe inversion for OEIS A291333

Bundle Report 204, archive `Report204.zip`, delivered as `Report204.tex` (16 pages). Delivered title block “Diagonal power weighted Dyck paths / All orders asymptotics and integer safe inversion for OEIS A291333 / Report 204 / 4 October 2026”. Printed as delivered, with `[write]` remarks and notes; Report 204’s Section k is Section $k + 43$ here, with its statements and equations. Its labels carry `pwd:dg::`; its two unlabelled sections were labelled; its citation key `AD` is `avdoshkin`. This Part is self-contained.

Abstract of this Part (as delivered). Let $Z_n(p)$ be the sum, over Dyck paths of semilength n , of the product of the p th powers of the descent heights. We give an exact staircase-partition representation and prove a full expansion for $Z_n(\tau n + \sigma)/(n!)^{\tau n + \sigma}$, uniform for τ in a fixed compact positive interval and bounded σ . The expansion holds in an exponentially weighted ℓ^1 norm at every fixed order. Its coefficients are Boltzmann partition moments and can be computed by finite mixed-moment or row-box algorithms. A separate scalar estimate is uniform in linearly growing truncation order and proves an exponentially accurate specified truncation. We also obtain the first total-variation correction, analytic coefficient-cutoff bounds, and an all-fixed-order Lambert W inverse with an integer-safe two-ceiling enclosure. The diagonal $Z_n(n)$ is OEIS A291333. Its leading equivalent, with constant $\prod_{j \geq 1} (1 - e^{-j})^{-1}$, was recorded by Václav Kotěšovec in the OEIS in 2017 and updated in 2018; it is re-proved here, not claimed as new. Exact checks and high-precision diagnostics accompany the proofs in a reproducible source archive.

44 Model and scope

A Dyck path in $\mathcal{D}_n$ starts at height zero, has n upsteps and n downsteps, never goes below zero, and ends at zero. If h_i is the height immediately before its i th downstep, set

$$Z_n(p) = \sum_{P \in \mathcal{D}_n} \prod_{i=1}^n h_i(P)^p, \quad p \in \mathbb{R}. \quad (44.1)$$

All heights in this product are positive integers, so real powers cause no ambiguity. Empty products give $Z_0(p) = 1$. The formal continued fraction is

$$\sum_{n \geq 0} Z_n(p) x^n = \frac{1}{1 - \frac{1^p x}{1 - \frac{2^p x}{1 - \frac{3^p x}{1 - \dots}}}}. \quad (44.2)$$

To see this, let F_h count excursions above height h . Decomposition into a sequence of arches gives $F_h = (1 - (h+1)^p x F_{h+1})^{-1}$. Each coefficient uses only finitely many heights. Thus (44.2) is a formal identity, with no analytic convergence assertion.

The array A290569 has entries $Z_n(k)$, and its main diagonal A291333 is

$$a_n = Z_n(n). \quad (44.3)$$

Both identifications are recorded in the OEIS [5, 6]. For example, the five paths of semilength three have descent-height products 6, 4, 2, 2, 1, giving $a_3 = 6^3 + 4^3 + 2^3 + 2^3 + 1 = 297$.

The regime studied here is $p = \tau n + \sigma$. The fixed-column regime $p = k$ does not supply uniform estimates in this regime. Throughout, unless stated otherwise,

$$0 < a \leq \tau \leq b < \infty, \quad |\sigma| \leq M, \quad 0 \leq M < \infty. \quad (44.4)$$

The constants a, b, M are fixed. An expansion to every fixed order means a Poincaré expansion: for each chosen order there is a corresponding remainder bound. Section 48 separately proves a scalar bound uniform in a growing order. It does not establish convergence, divergence, Borel summability, an optimal truncation rule, a true exponential scale, or a full transseries. No assertion here covers $\tau \rightarrow 0$ or $\tau \rightarrow \infty$.

Prior leading equivalent. Write

$$Q(\tau) = \prod_{j=1}^{\infty} (1 - e^{-\tau j})^{-1} = \sum_{\lambda \in \mathcal{P}} e^{-\tau |\lambda|}, \quad (44.5)$$

where $\mathcal{P}$ includes the empty partition. The OEIS attributes $a_n \sim Q(1)(n!)^n$ to Kotěšovec, 26 August 2017, updated 21 July 2018 [5]. The constant also occurs in a different tree/partition sequence, A317103 [7]. Our purpose is a proof with uniform corrections, quantitative partition-law refinements, explicit computation and inversion. This article makes no global priority claim.

Remark 44.1 ([write] The OEIS entries, quoted, 7 October 2026). OEIS A291333, read on 7 October 2026 in the internal format, is at revision #12 (21 July 2018), author Ilya Gutkovskiy (22 August 2017), offset 0; its name is “ $a(n) = [x^n] \frac{1}{1-x/(1-2^n x/(1-3^n x/(1-4^n x/(1-5^n x/(1-6^n x/(1-7^n x/(1-8^n x/(1-9^n x/(1-10^n x/(1-11^n x/(1-12^n x/(1-13^n x/(1-14^n x/(1-15^n x/(1-16^n x/(1-17^n x/(1-18^n x/(1-19^n x/(1-20^n x/(1-21^n x/(1-22^n x/(1-23^n x/(1-24^n x/(1-25^n x/(1-26^n x/(1-27^n x/(1-28^n x/(1-29^n x/(1-30^n x/(1-31^n x/(1-32^n x/(1-33^n x/(1-34^n x/(1-35^n x/(1-36^n x/(1-37^n x/(1-38^n x/(1-39^n x/(1-40^n x/(1-41^n x/(1-42^n x/(1-43^n x/(1-44^n x/(1-45^n x/(1-46^n x/(1-47^n x/(1-48^n x/(1-49^n x/(1-50^n x/(1-51^n x/(1-52^n x/(1-53^n x/(1-54^n x/(1-55^n x/(1-56^n x/(1-57^n x/(1-58^n x/(1-59^n x/(1-60^n x/(1-61^n x/(1-62^n x/(1-63^n x/(1-64^n x/(1-65^n x/(1-66^n x/(1-67^n x/(1-68^n x/(1-69^n x/(1-70^n x/(1-71^n x/(1-72^n x/(1-73^n x/(1-74^n x/(1-75^n x/(1-76^n x/(1-77^n x/(1-78^n x/(1-79^n x/(1-80^n x/(1-81^n x/(1-82^n x/(1-83^n 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- ...))))), a continued fraction.”, its formula lines are “ $a(n) = A290569(n,n)$.” and Kotěšovec’s “ $a(n) \sim c * (n!)^n \sim c * 2^{(n/2)} * \pi^{(n/2)} * n^{(n*(2*n+1)/2)} / \exp(n^{2-1/12})$, where $c = 1/QPochhammer(\exp(-1)) = 1.9824409074128737036856824655613120156828827\dots$ ” (26 August 2017, updated 21 July 2018), the case $\tau = 1$, $\sigma = 0$ of Theorem 46.1 at order zero. Its b-file, by Robert Israel, covers $0 \leq n \leq 30$; the source compared the first eleven displayed values. The write computed $Z_n(n)$ for $n \leq 30$ with its own code: all 31 b-file terms agree. A290569 is quoted in Part I (Remark 1.7). The source could open neither A291333 directly nor A291332; the write did not read A291332 or A317103 either. Nothing was submitted to the OEIS.

45 An exact partition identity

Let $\delta_n = (n-1, n-2, \dots, 1, 0)$ be the staircase. Partitions are weakly decreasing nonnegative row lengths with trailing zeros suppressed. Row and column indices start at one.

Proposition 45.1. *There is a bijection between $\mathcal{D}_n$ and the partitions $\lambda \subseteq \delta_n$ such that*

$$R_n(\tau, \sigma) := \frac{Z_n(\tau n + \sigma)}{(n!)^{\tau n + \sigma}} = \sum_{\lambda \subseteq \delta_n} r_n(\lambda)^{\tau n + \sigma}, \quad (45.1)$$

$$r_n(\lambda) = \prod_{i=1}^n \left(1 - \frac{\lambda_i}{n-i+1}\right) = \prod_{(i,j) \in \lambda} \frac{n-i-j+1}{n-i-j+2}. \quad (45.2)$$

Every factor is positive. Every partition with $|\lambda| < n$ lies in δ_n .

Proof. Let u_i be the number of upsteps before downstep i and put $\lambda_i = n - u_i$. The numbers u_i are nondecreasing, $i \leq u_i \leq n$, and $u_n = n$. These are precisely the conditions that $\lambda \subseteq \delta_n$. Conversely, before each successive downstep insert enough upsteps to reach $u_i = n - \lambda_i$. The conditions above ensure a unique Dyck path. Its i th descent height is $u_i - i + 1 = n - i + 1 - \lambda_i$. Division by $n!$ gives the first product in (45.2); telescoping each row gives the second.

If a partition has size m , then $\lambda_i \leq m - i + 1$ whenever row i is present, since the preceding $i-1$ rows are nonempty. For $m < n$ this is at most $n-i$, proving the last claim. $\square$

Define the statistics

$$D_k(\lambda) = \frac{1}{k} \sum_i ((i-1+\lambda_i)^k - (i-1)^k) \quad (45.3)$$

$$= \frac{1}{k} \sum_{(i,j) \in \lambda} ((i+j-1)^k - (i+j-2)^k), \quad k \geq 1. \quad (45.4)$$

Telescoping proves the second formula. It is symmetric in row and column indices, so every D_k is invariant under partition conjugation. In particular,

$$D_1(\lambda) = m := |\lambda|, \quad 0 \leq D_k(\lambda) \leq m^k. \quad (45.5)$$

Indeed, a cell (i, j) forces at least $i+j-1$ cells in its row and column together. Thus $i+j-1 \leq m$, and the mean value theorem bounds its divided difference in (45.4) by m^{k-1} .

Set $\rho(\lambda) = \max_{i: \lambda_i > 0} (i-1+\lambda_i)$, with $\rho(\emptyset) = 0$. For $|z|\rho(\lambda) < 1$ define

$$g_\lambda(z; \tau, \sigma) = \exp \left(-\tau \sum_{k \geq 1} D_k z^{k-1} - \sigma \sum_{k \geq 1} D_k z^k \right). \quad (45.6)$$

These series converge in the stated disk by (45.3). Since $\rho(\lambda) \leq n-1$ in the staircase, the convergent expansion of each row logarithm gives

$$g_\lambda(1/n; \tau, \sigma) = r_n(\lambda)^{\tau n + \sigma} \quad (\lambda \subseteq \delta_n). \quad (45.7)$$

This remains valid for staircase shapes with $m > n$. The smaller disk $|z|m < 1$ used for estimates below is a sufficient estimation domain, not the full domain of the identity.

Put $A_r = \tau D_{r+1} + \sigma D_r$ and define $B_r = B_r(\lambda; \tau, \sigma)$ by

$$g_\lambda(z) = e^{-\tau m} \sum_{r \geq 0} B_r z^r, \quad B_0 = 1, \quad r B_r = - \sum_{j=1}^r j A_j B_{r-j}. \quad (45.8)$$

The recurrence follows by differentiating the formal exponential $\exp(-\sum_{j \geq 1} A_j z^j)$. For later use,

$$\begin{aligned} B_1 &= -A_1, & B_2 &= \frac{1}{2} A_1^2 - A_2, \\ B_3 &= -A_3 + A_1 A_2 - \frac{1}{6} A_1^3, & B_4 &= -A_4 + A_1 A_3 + \frac{1}{2} A_2^2 - \frac{1}{2} A_1^2 A_2 + \frac{1}{24} A_1^4. \end{aligned} \quad (45.9)$$

46 Uniform expansion in weighted summation norm

The identity (44.5) follows by expanding independent geometric series. It is finite for every $\tau > 0$: for example, $-\log(1 - e^{-\tau j}) \leq e^{-\tau j}/(1 - e^{-\tau})$, a summable majorant.

Theorem 46.1. *Assume (44.4). Fix an integer $K \geq 0$ and $0 \leq \eta < a/2$. Define*

$$c_r(\tau, \sigma) = \frac{1}{Q(\tau)} \sum_{\lambda \in \mathcal{P}} e^{-\tau|\lambda|} B_r(\lambda; \tau, \sigma). \quad (46.1)$$

These sums are absolutely convergent, $c_0 = 1$, and uniformly in τ, σ ,

$$\sum_{\lambda \in \mathcal{P}} e^{\eta|\lambda|} \left| \mathbf{1}_{\lambda \subseteq \delta_n} g_\lambda(1/n) - e^{-\tau|\lambda|} \sum_{r=0}^K B_r(\lambda) n^{-r} \right| = O(n^{-K-1}). \quad (46.2)$$

The first term is defined to be zero outside the staircase. Consequently,

$$R_n(\tau, \sigma) = Q(\tau) \left(\sum_{r=0}^K c_r(\tau, \sigma) n^{-r} + O(n^{-K-1}) \right). \quad (46.3)$$

The constants depend only on a, b, M, K and, in (46.2), the fixed η .

Proof. Choose $0 < \delta < 1/2$ with $b\delta/(1-\delta) \leq a/4$, and let

$$C = \exp\left(\frac{M\delta}{1-\delta}\right). \quad (46.4)$$

For $m \geq 1$ and $|z| \leq \delta/m$, (45.5) gives

$$\begin{aligned} \left| \tau \sum_{k \geq 2} D_k z^{k-1} \right| &\leq \frac{bm\delta}{1-\delta} \leq \frac{am}{4}, \\ \left| \sigma \sum_{k \geq 1} D_k z^k \right| &\leq \frac{M\delta}{1-\delta}. \end{aligned}$$

Hence $|g_\lambda(z)| \leq Ce^{-3am/4}$. Cauchy's estimate gives the uniform bound

$$|e^{-\tau m} B_r(\lambda)| \leq Ce^{-3am/4} (m/\delta)^r. \quad (46.5)$$

For $m \leq \delta n/2$, summing the coefficient majorants after order K gives

$$\left| g_\lambda(1/n) - e^{-\tau m} \sum_{r=0}^K B_r n^{-r} \right| \leq 2Ce^{-3am/4} \left(\frac{m}{\delta n} \right)^{K+1}. \quad (46.6)$$

All these partitions fit the staircase by Proposition 45.1. After multiplication by $e^{\eta m}$, the sum of (46.6) is $O(n^{-K-1})$, since $3a/4 - \eta > 0$ and every fixed polynomial moment of the partition series converges.

Here are explicit tail justifications needing no asymptotic formula for the partition numbers $p(m)$. For $c > 0$,

$$\sum_{m>N} p(m)e^{-cm} \leq e^{-cN/2} Q(c/2). \quad (46.7)$$

Polynomial factors can be absorbed into a smaller exponential. For $n \geq 2M/a$, the exponent $\tau n + \sigma$ is positive. From the row product, $-\log(1-x) \geq x$ and $n-i+1 \leq n$ imply

$$0 \leq \mathbf{1}_{\lambda \subseteq \delta_n} g_\lambda(1/n) \leq e^{-(\tau+\sigma/n)m} \leq e^{-am/2}. \quad (46.8)$$

The actual-weight tail $m > \delta n/2$ is exponentially small in the weighted norm because $a/2 - \eta > 0$. The finitely many coefficient tails are exponentially small by (46.5), since $3a/4 - \eta > 0$. This proves (46.2); the empty partition contributes exactly 1 and has no positive-order remainder. The same coefficient estimate proves absolute convergence in (46.1). Summing (46.2) at $\eta = 0$ proves (46.3); $1 \leq Q(\tau) \leq Q(a)$ is uniform. $\square$

46.1 Observables and normalized laws

Give a partition the probability

$$P_{n,\tau,\sigma}(\lambda) = \frac{\mathbf{1}_{\lambda \subseteq \delta_n} g_\lambda(1/n)}{R_n(\tau, \sigma)}, \quad \nu_\tau(\lambda) = \frac{e^{-\tau|\lambda|}}{Q(\tau)}. \quad (46.9)$$

This is the weighted path law transported by the bijection. If an observable f satisfies $|f(\lambda)| \leq C_f e^{\eta|\lambda|}$ with a fixed $\eta < a/2$, then (46.2) can be multiplied and summed against f . All fixed polynomials in $m, D_2, D_3, \dots$ satisfy such a bound with any fixed positive η .

For a fully explicit normalized expansion, define $T_0 = 1$ and

$$T_r(\lambda) = B_r(\lambda) - \sum_{j=1}^r c_j T_{r-j}(\lambda). \quad (46.10)$$

Finite series division in (46.2), using (46.3) and $Q \geq 1$, yields

$$P_{n,\tau,\sigma} = \nu_\tau \sum_{r=0}^K T_r n^{-r} + O_{\ell_\eta^1}(n^{-K-1}). \quad (46.11)$$

Here $\|v\|_{\ell_\eta^1} = \sum_\lambda e^{\eta|\lambda|} |v(\lambda)|$. The coefficient sequences have finite weighted norm by the preceding bounds, so all products in this finite division are legitimate. This proves total-variation convergence and convergence of every fixed polynomial moment.

Under ν_τ , the multiplicities M_j of parts of size j are independent and

$$\mathbb{P}(M_j = s) = (1 - q^j)q^{js}, \quad q = e^{-\tau}, \quad s = 0, 1, \dots \quad (46.12)$$

Indeed the product of these probabilities is $Q(\tau)^{-1}q^{\sum jM_j}$. The sum of $\mathbb{P}(M_j > 0)$ is finite, so the product law is supported on partitions. Formula (46.11) quantifies the approximation to this product law.

47 Coefficients and coefficient algorithms

Let

$$L_s(q) = \sum_{j \geq 1} \frac{j^s q^j}{1 - q^j}, \quad q = e^{-\tau}. \quad (47.1)$$

All expectations in this section are under ν_τ . If $n(\lambda) = \sum_i (i-1)\lambda_i$, then

$$D_2 = n(\lambda) + n(\lambda') + \frac{m}{2}, \quad \mathbb{E}D_2 = L_2 - \frac{1}{2}L_1. \quad (47.2)$$

For the expectation identity, $n(\lambda') = \frac{1}{2} \sum_i \lambda_i(\lambda_i - 1)$ and conjugation preserves ν_τ . Also $\mathbb{E} \sum_i \lambda_i^2 = L_2$ and $\mathbb{E}m = L_1$, by (46.12).

It follows that

$$c_1(\tau, \sigma) = -\tau L_2 + (\tau/2 - \sigma)L_1, \quad (47.3)$$

$$c_2(\tau, \sigma) = \frac{1}{2}\mathbb{E}(\tau D_2 + \sigma m)^2 - \tau \mathbb{E}D_3 - \sigma \mathbb{E}D_2. \quad (47.4)$$

Define logarithmic coefficients by

$$\log \left(1 + \sum_{r \geq 1} c_r z^r \right) = \sum_{r \geq 1} \ell_r z^r, \quad \ell_r = c_r - \frac{1}{r} \sum_{j=1}^{r-1} j \ell_j c_{r-j}. \quad (47.5)$$

In particular $\ell_1 = c_1$ and

$$\ell_2 = \frac{1}{2} \text{Var}(\tau D_2 + \sigma m) - \tau \mathbb{E}D_3 - \sigma \mathbb{E}D_2. \quad (47.6)$$

Taking the logarithm of the positive scalar expansion is justified uniformly: its leading normalized term is 1, its correction is uniformly $O(n^{-1})$, and the Taylor remainder for $\log(1+w)$ is uniform for $|w| \leq 1/2$.

For the diagonal we abbreviate $c_r = c_r(1, 0)$ and $\ell_r = \ell_r(1, 0)$. Then

$$a_n = (n!)^n Q(1) \left(1 + \frac{c_1}{n} + \frac{c_2}{n^2} + \dots + \frac{c_K}{n^K} + O(n^{-K-1}) \right), \quad (47.7)$$

$$c_1 = - \sum_{j \geq 1} \frac{j^2 - j/2}{e^j - 1}, \quad c_2 = \frac{1}{2} \mathbb{E}D_2^2 - \mathbb{E}D_3, \quad (47.8)$$

$$c_3 = -\mathbb{E}D_4 + \mathbb{E}(D_2 D_3) - \frac{1}{6} \mathbb{E}D_2^3.$$

These are fixed, n -independent constants, not finite- n fits.

47.1 Finite mixed moments of multiplicities

Fix a largest permitted part J . Process $j = J, J-1, \dots, 1$. Suppose S rows have already been inserted and M new rows of length j are to be appended. Then

$$S_{\text{new}} = S + M, \quad (D_k)_{\text{new}} = D_k + \frac{1}{k} \sum_{s=0}^{M-1} ((S+s+j)^k - (S+s)^k). \quad (47.9)$$

For fixed j the summand is a polynomial of total degree at most $k-1$ in S, s ; summation makes the added term a polynomial of degree at most k in S, M . Give S, M weight one and D_k weight k . Therefore the vector of mixed moments of weighted degree at most $2K$ is closed under (47.9). Only the finitely many D_k needed through D_{K+1} enter the coefficients.

The newly drawn M is independent of earlier rows. Its raw moments are explicit:

$$\mathbb{E}M^r = \sum_{\ell=0}^r \left\{ \begin{matrix} r \\ \ell \end{matrix} \right\} \ell! \left(\frac{q^j}{1-q^j} \right)^\ell. \quad (47.10)$$

This follows from $M^r = \sum_{\ell} \left\{ \begin{matrix} r \\ \ell \end{matrix} \right\} (M)_{\ell}$ and the geometric generating function; the zeroth moment is one. Expand each updated monomial, take the moments in M by (47.10), and use the stored old moments in S, D_k . This is a finite exact algebraic algorithm at every chosen J, K , with rational functions of the q^j as inputs.

Every term of B_r has weighted degree at most $2r$: a factor A_j has weight at most $j+1 \leq 2j$, and its formal order is j . Thus the same closed vector evaluates all $c_r^{(J)}$ for $r \leq K$. These are expectations conditional on the largest part being at most J ; their normalization is

$$Q_J(\tau) = \prod_{j=1}^J (1 - e^{-\tau j})^{-1}. \quad (47.11)$$

The moment bounds in Theorem 46.1 justify convergence as $J \rightarrow \infty$. Section 50 supplies a diagonal cutoff bound.

47.2 A separate row-box algorithm

An independent way to compute the coefficients is to sum shapes contained in an $L \times L$ square. Work with polynomials in z modulo z^{K+1} . Define

$$D_{k,i}(j) = \frac{(i-1+j)^k - (i-1)^k}{k},$$

$$w_i(j) = e^{-\tau j} \exp \left[- \sum_{r=1}^K (\tau D_{r+1,i}(j) + \sigma D_{r,i}(j)) z^r \right] \pmod{z^{K+1}}. \quad (47.12)$$

Use the boundaries $F(i, 0) = 1$ and $F(L+1, j) = 1$, and compute

$$F(i, j) = F(i, j-1) + w_i(j) F(i+1, j), \quad i = L, L-1, \dots, 1, \quad j = 1, \dots, L. \quad (47.13)$$

The first branch restricts row i to length below j ; the second fixes its length to j and caps the next row by j . The boundaries count the all-zero trailing rows once. Induction proves

$$[z^r] F(1, L) = \sum_{\lambda \subseteq L \times L} e^{-\tau|\lambda|} B_r(\lambda). \quad (47.14)$$

After division by $Q(\tau)$ this is an unnormalized restriction of the full-law coefficient. It is not the conditional multiplicity moment $c_r^{(J)}$ and must not be divided by Q_J by mistake. Naive truncated polynomial multiplication costs $O(L^2 K^2)$ arithmetic operations. The two algorithms give genuinely different finite truncations and an effective cross-check.

48 A uniform scalar Gevrey remainder

The next theorem keeps track of the order dependence rather than extrapolating a fixed-order big- O statement.

Theorem 48.1. *Under (44.4), choose δ, C as in (46.4) and set*

$$B = \frac{4}{a\delta}, \quad A = 2CQ(a/2), \quad A' = (1 + 2C)Q(a/4). \quad (48.1)$$

For every $r \geq 0$,

$$|c_r(\tau, \sigma)| \leq CQ(a/2)B^r r!. \quad (48.2)$$

For positive integers $n \geq 2M/a$ and every integer $0 \leq K \leq n/(2B)$,

$$\left| R_n - Q(\tau) \sum_{r=0}^K c_r n^{-r} \right| \leq AB^{K+1}(K+1)!n^{-K-1} + A'e^{-a\delta n/8}. \quad (48.3)$$

All constants are independent of K, n, τ, σ within these ranges.

Proof. The elementary inequality

$$m^r \leq r!(4/a)^r e^{am/4} \quad (48.4)$$

comes from retaining one term of the exponential series. Combine it with (46.5), sum over nonempty partitions, and use $Q(\tau) \geq 1$ to get (48.2). The empty partition is handled directly, with $c_0 = 1$ and $C, Q \geq 1$.

For $m \leq \delta n/2$, apply (48.4) with $r = K + 1$ to (46.6). Summation gives the first term of (48.3) with the displayed A .

For $m > \delta n/2$, the actual weights are bounded by $e^{-am/2}$, hence their sum is at most $Q(a/4)e^{-a\delta n/8}$. For each coefficient use (46.5) and (48.4) to obtain

$$\begin{aligned} \sum_{m > \delta n/2} \sum_{|\lambda|=m} |e^{-\tau m} B_r| n^{-r} &\leq Cr!(B/n)^r \sum_{m > \delta n/2} p(m) e^{-am/2} \\ &\leq CQ(a/4) e^{-a\delta n/8} r! (B/n)^r. \end{aligned}$$

The ratio of successive terms in the finite sum from $r = 0$ to K is $(r+1)B/n \leq KB/n \leq 1/2$. Its sum is at most two, including $K = 0$. Adding the actual-weight tail gives A' . $\square$

Corollary 48.2. *For all sufficiently large n , choose*

$$K + 1 = \left\lfloor \frac{n}{4B} \right\rfloor. \quad (48.5)$$

Then the error on the left of (48.3) is at most

$$A4^{-\lfloor n/(4B) \rfloor} + A'e^{-a\delta n/8}. \quad (48.6)$$

Thus this specified linear truncation is exponentially accurate, uniformly on the compact parameter set.

Proof. Write $s = K + 1 \geq 1$. The choice is in the theorem's range, and $s! \leq s^s$ implies $(B/n)^s s! \leq (Bs/n)^s \leq 4^{-s}$. $\square$

Remark 48.3. The coefficient upper bound has Gevrey order one. It is an upper bound and need not describe the true coefficient growth. The scalar bound (48.3) is uniform in K ; the weighted- ℓ^1 theorem was stated at fixed K . No uniform-in- K weighted- ℓ^1 result is inferred here. The explicit exponential rate in (48.6) is a proved rate from conservative majorants, not an optimality statement or an identification of separate exponential sectors.

49 Total variation and the first correction

For probabilities on partitions use $d_{\text{TV}}(P, \nu) = \frac{1}{2} \sum_{\lambda} |P(\lambda) - \nu(\lambda)|$.

Proposition 49.1. *Uniformly under (44.4),*

$$d_{\text{TV}}(P_{n,\tau,\sigma}, \nu_{\tau}) = \frac{1}{2n} \mathbb{E}_{\nu_{\tau}} |B_1 - c_1| + O(n^{-2}). \quad (49.1)$$

On the diagonal, put $q = e^{-1}$ and $\mu = \mathbb{E}_{\nu_1} D_2 = -c_1$. Then

$$\frac{1}{2} < \mu < 2, \quad d_{\text{TV}}(P_{n,1,0}, \nu_1) = \frac{\mu(1+q) - q/2}{Q(1)n} + O(n^{-2}). \quad (49.2)$$

Proof. At order one (46.11) says $P_n - \nu_{\tau} = n^{-1} \nu_{\tau}(B_1 - c_1) + O_{\ell^1}(n^{-2})$. The inequality $||x+y| - |x|| \leq |y|$ proves (49.1), including the factor $1/2$.

For the diagonal, the first five terms of the series for e give $e > 8/3$, while comparison of the remaining factorial tail with a geometric series gives $e < 3$. Thus $1/3 < q < 3/8$. The first two terms of the positive series for μ at $q = 1/3$ give

$$\mu > \frac{q}{2(1-q)} + \frac{3q^2}{1-q^2} \Big|_{q=1/3} = \frac{5}{8} > \frac{1}{2}.$$

For the upper bound, separate $j = 1$ and replace $1 - q^j$ by $1 - q^2$ for $j \geq 2$:

$$\mu \leq \frac{q}{2(1-q)} + \frac{q(1+q)/(1-q)^3 - q/[2(1-q)^2] - q/2}{1-q^2} < \frac{27237}{13750} < 2. \quad (49.3)$$

The penultimate value is obtained at $q = 3/8$. Monotonicity holds because before summation the majorant is a sum of positive increasing terms.

Now $D_2(\emptyset) = 0$, $D_2((1)) = 1/2$, and $D_2 \geq 2$ on every other diagram: a diagram of size at least two contains (2) or (1, 1), each with $D_2 = 2$, and every added cell has positive contribution. Since $\mathbb{E}(\mu - D_2) = 0$, half its absolute expectation is its positive-part expectation, supported just on $\emptyset$ and (1). Their Boltzmann probabilities are $1/Q$ and q/Q , so the result is $[\mu(1+q) - q/2]/Q$. $\square$

50 Analytic bounds for coefficient cutoffs

In this section only, $\tau = 1, \sigma = 0$. Let $c_r^{(J)}$ be the multiplicity coefficient with parts at most J , as in Section 47.1. Define

$$\beta_r = [z^r] \exp\left(\frac{z}{1-z}\right). \quad (50.1)$$

In particular $\beta_1 = 1$, $\beta_2 = 3/2$, $\beta_3 = 13/6$ and $\beta_4 = 73/24$. We use β_r here to keep these majorants distinct from the inverse coefficients b_r below.

Proposition 50.1. *For $r \geq 1$, $0 < \theta < 1$, and $J+1 \geq 2r/(1-\theta)$, set*

$$U_r(J, \theta) = \beta_r \exp\left(\frac{\pi^2}{6\theta}\right) (J+1)^{2r} e^{-(1-\theta)(J+1)}, \quad (50.2)$$

$$V_r(J) = \beta_r e^{\pi^2/3} (4r/e)^{2r} \frac{e^{-(J+1)}}{1 - e^{-1}}. \quad (50.3)$$

Then

$$|c_r - c_r^{(J)}| \leq U_r(J, \theta) + V_r(J). \quad (50.4)$$

For $r = 0$, $c_0 = c_0^{(J)} = 1$, so its error is exactly zero.

Proof. For a nonempty partition of size m , $D_{j+1} \leq m^{j+1} \leq m^{2j}$. Taking absolute coefficients of the exponential yields

$$|B_r(\lambda)| \leq \beta_r m^{2r}. \quad (50.5)$$

The Euler product obeys

$$\log Q(\theta) = \sum_{k \geq 1} \frac{1}{k(e^{\theta k} - 1)} \leq \frac{1}{\theta} \sum_{k \geq 1} k^{-2} = \frac{\pi^2}{6\theta}. \quad (50.6)$$

If a part exceeds J , then $m \geq J + 1$. The function $m^{2r} e^{-(1-\theta)m}$ decreases for $m \geq 2r/(1-\theta)$. Multiply its maximum on this tail by $\sum_m p(m) e^{-\theta m} = Q(\theta)$ and use (50.6); the unnormalized missing coefficient sum is at most U_r .

Let E_J be the event that every part is at most J . The finite multiplicity law is $\nu_1(\cdot \mid E_J)$ and $\mathbb{P}(E_J) = Q_J(1)/Q(1)$. Consequently,

$$c_r - c_r^{(J)} = E[B_r \mathbf{1}_{E_J^c}] - \mathbb{P}(E_J^c) c_r^{(J)}. \quad (50.7)$$

The first term is bounded by U_r , using $Q(1) \geq 1$. By independence, or just a union bound,

$$\mathbb{P}(E_J^c) \leq \sum_{j > J} \mathbb{P}(M_j > 0) = \frac{e^{-(J+1)}}{1 - e^{-1}}. \quad (50.8)$$

For $r \geq 1$, maximizing $m^{2r} e^{-m/2}$ over $m \geq 0$ gives $(4r/e)^{2r}$. Divide the restricted sum by $Q_J \geq 1$, split $e^{-m} = e^{-m/2} e^{-m/2}$, and apply (50.5):

$$|c_r^{(J)}| \leq \beta_r Q(1/2) (4r/e)^{2r} \leq \beta_r e^{\pi^2/3} (4r/e)^{2r}. \quad (50.9)$$

Equations (50.7)–(50.9) prove (50.4). The constant coefficient is handled separately as stated. $\square$

The row-box method admits a closely related bound. A diagram outside $L \times L$ has size at least $L + 1$, so for $r \geq 1$ and the same condition with L replacing J ,

$$\left| c_r - \frac{[z^r]F(1, L)}{Q(1)} \right| \leq U_r(L, \theta). \quad (50.10)$$

There is no V_r term because no conditional renormalization is performed. For $r = 0$, the error is at most $e^{\pi^2/(6\theta)} e^{-(1-\theta)(L+1)}$. These estimates control the exact finite-cutoff error, not floating-point roundoff.

For $J = 200$ and $\theta = 0.09$, ordinary high-precision evaluation of $U_r + V_r$ gives approximately

$$1.28 \cdot 10^{-67}, \quad 7.76 \cdot 10^{-63}, \quad 4.53 \cdot 10^{-58}, \quad 2.57 \cdot 10^{-53} \quad (r = 1, 2, 3, 4).$$

The formulas are rigorous analytic bounds. These printed evaluations use ordinary high-precision arithmetic and are not directed-rounding enclosures; neither they nor the computed coefficient decimals certify every displayed digit.

51 Smooth inversion and integer thresholds

Define, for $Y > 1$,

$$N(Y) = \min\{n \geq 1 : a_n \geq Y\}, \quad L = \log Y. \quad (51.1)$$

The use of $Y > 1$ avoids the initial equality $a_0 = a_1 = 1$.

Lemma 51.1. *The sequence a_n is strictly increasing for $n \geq 1$ and unbounded.*

Proof. Wrap a size- n Dyck path in an outer upstep and downstep. Each old descent height h becomes $h + 1$; its factor at exponent $n + 1$ is $(h + 1)^{n+1} > h^n$. The final descent has height one. This is an injection, so summing proves strict increase. The single path consisting of all upsteps followed by all downsteps has weight $(n!)^n$, proving unboundedness. $\square$

For fixed $K \geq 0$ define the smooth approximation

$$H_K(x) = x \log \Gamma(x + 1) + \log Q(1) + \sum_{r=1}^K \ell_r x^{-r}. \quad (51.2)$$

Theorem 46.1 and (47.5) imply

$$\log a_n = H_K(n) + O(n^{-K-1}). \quad (51.3)$$

Let ψ be the logarithmic derivative of Γ . Convexity of $\log \Gamma$ and $\Gamma(x + 1) = x\Gamma(x)$ imply $\log x \leq \psi(x + 1) \leq \log(x + 1)$. Stirling's formula [32] therefore gives directly

$$H'_K(x) = 2x \log x - x + O(\log x) > 0 \quad (51.4)$$

for sufficiently large x . This argument differentiates the exact Gamma expression, not an uncontrolled remainder. Denote by $x_K(L)$ the unique sufficiently large solution of $H_K(x) = L$.

Theorem 51.2. *For each fixed $K \geq 0$ there is a constant $C_K > 0$ such that for all sufficiently large Y , writing $x = x_K(\log Y)$,*

$$\left\lceil x - C_K x^{-K-2} / \log x \right\rceil \leq N(Y) \leq \left\lceil x + C_K x^{-K-2} / \log x \right\rceil. \quad (51.5)$$

The constants and lower threshold on Y are existential asymptotic constants, not effective numerical certificates.

Proof. On $[x - 2, x + 2]$, (51.4) gives $H'_K \geq cx \log x$, while the integer residual in (51.3) is bounded by Cx^{-K-1} . Choose a fixed C_K strictly larger than C/c after enlarging the constants uniformly on this interval, and put $\varepsilon = C_K x^{-K-2} / \log x$. Consider precisely the integers

$$m_- = \lceil x - \varepsilon \rceil - 1, \quad m_+ = \lceil x + \varepsilon \rceil.$$

For large x they lie in $[x - 2, x + 2]$, with $m_- < x - \varepsilon$ and $m_+ \geq x + \varepsilon$. The derivative lower bound exceeds the possible integer residual at each, so $\log a_{m_-} < L < \log a_{m_+}$. Lemma 51.1 gives $m_- + 1 \leq N(Y) \leq m_+$, which is (51.5), including integer endpoints. $\square$

The two ceilings may differ near an integer, even when the radius tends to zero. Thus the theorem does not license the unconditional formula $N(Y) = \lceil x_K(\log Y) \rceil$. If both ceilings coincide, their common value is the threshold. At a finite input, resolving an ambiguous rounding decision requires a certified residual bound or direct exact comparison with neighboring a_n .

51.1 The Lambert scale and the first three corrections

Let W_0 be the real principal Lambert function, defined by $W_0(z)e^{W_0(z)} = z$ for $z > 0$ [33]. Set

$$t = \sqrt{\frac{2L}{W_0(2L/e^2)}}, \quad u = \log t, \quad H = 2u - 1, \quad A = \frac{u + \log(2\pi)}{2}, \quad (51.6)$$

so that $t^2(\log t - 1) = L$ exactly. Also put $d = 1/12 + \log Q(1)$. Multiplying Stirling's expansion by n gives

$$\log a_n = n^2(\log n - 1) + \frac{n}{2} \log(2\pi n) + d + \frac{\ell_1}{n} + \frac{\ell_2 - 1/360}{n^2} + O(n^{-3}). \quad (51.7)$$

The constant $1/12$ and the absence of a Stirling n^{-1} term after multiplication are important for inversion.

For each fixed $K \geq 1$,

$$x_K(L) = t + \alpha_0(u) + \frac{\alpha_1(u)}{t} + \frac{\alpha_2(u)}{t^2} + O(t^{-3}), \quad (51.8)$$

where

$$\alpha_0 = -A/H, \quad (51.9)$$

$$\alpha_1 = -\frac{(H+2)\alpha_0^2/2 + (A+1/2)\alpha_0 + d}{H}, \quad (51.10)$$

$$\alpha_2 = -\frac{(H+2)\alpha_0\alpha_1 + \alpha_0^3/3 + (A+1/2)\alpha_1 + \alpha_0^2/4 + \ell_1}{H}. \quad (51.11)$$

The rational function $\alpha_0(u)$ tends to $-1/4$. Replacing it by that limit loses a correction of order $1/\log t$, larger than every inverse power of t . The logarithmic dependence must therefore be retained in a power-scale inverse.

51.2 An arbitrary fixed-order recurrence

Let $\mathcal{B}_j$ denote Bernoulli numbers, to distinguish them from $B_r(\lambda)$. Define

$$b_r = \ell_r + s_r, \quad s_r = 0 \text{ (} r \text{ odd)}, \quad s_{2k-2} = \frac{\mathcal{B}_{2k}}{2k(2k-1)} \quad (k \geq 2). \quad (51.12)$$

Thus $b_1 = \ell_1$ and $b_2 = \ell_2 - 1/360$. Hold u fixed and use a new formal variable ε . Define

$$\begin{aligned} \Psi(\varepsilon, v; u) = & \varepsilon^{-1} \left\{ (1 + \varepsilon v)^2 [u - 1 + \log(1 + \varepsilon v)] - (u - 1) \right\} \\ & + \frac{1}{2}(1 + \varepsilon v)[u + \log(2\pi) + \log(1 + \varepsilon v)] \\ & + d\varepsilon + \sum_{r \geq 1} b_r \varepsilon^{r+1} (1 + \varepsilon v)^{-r}. \end{aligned} \quad (51.13)$$

The apparent pole is removable and the constant term is $Hv + A$. Starting with (51.9), put

$$\alpha_j(u) = -\frac{1}{H} [\varepsilon^j] \Psi \left(\varepsilon, \sum_{i=0}^{j-1} \alpha_i(u) \varepsilon^i; u \right), \quad j \geq 1. \quad (51.14)$$

Only $b_1, \dots, b_{j-1}$ enter. The required list is empty when $j = 1$; the initial coefficient $j = 0$ requires no b_r . The coefficient of a newly added α_j at order ε^j is exactly H ; every other use of it has higher order. This proves that the recurrence is triangular. Expanding its first terms gives (51.10) and (51.11).

Theorem 51.3. *Fix integers $J \geq 0$ and $K \geq \max(0, J - 1)$. Then*

$$x_K(L) = t + \sum_{j=0}^J \alpha_j(\log t)t^{-j} + O(t^{-J-1}), \quad L \rightarrow \infty. \quad (51.15)$$

Every α_j is a rational function of u , bounded as $u \rightarrow \infty$.

Proof. Induct on j in (51.14). Each numerator is a polynomial in finitely many previous α_i and b_r , with at most affine dependence on u . If the preceding coefficients are bounded, the numerator is $O(u)$; division by $H = 2u - 1$ preserves boundedness. Rationality is preserved as well. After choosing a sufficiently large u_0 , all these fixed rational functions and all their fixed derivatives are bounded on $u \geq u_0$; the only introduced denominators are powers of H .

Write $\tilde{x}_J = t + \sum_{j=0}^J \alpha_j(u)t^{-j}$. The leading expansion of $H_K(t + v) - L$, uniformly for bounded v , is $t(Hv + A) + O(u)$. Taking two fixed sufficiently large displacements of opposite sign proves $x_K = t + O(1)$ by eventual monotonicity. The same bound holds for $\tilde{x}_J$.

Use Stirling's formula to an order leaving $O(x^{-J})$ in $H_K(x)$, retaining $b_1, \dots, b_{J-1}$. The list is empty for $J = 0, 1$; at $J = 0$, the constant d may still be retained with smaller error. Omitted logarithmic correction terms are $O(x^{-J})$ because $K \geq J - 1$ when $J \geq 1$. Ordinary Taylor remainders in $\log(1 + \varepsilon v)$ and $(1 + \varepsilon v)^{-r}$ are uniform for bounded v and $\varepsilon = 1/t$ small. Their prefactors grow at most as $O(u)$. The recurrence cancels coefficients through ε^J in (51.13), which is the residual divided by t , hence

$$H_K(\tilde{x}_J) - L = O(ut^{-J}).$$

On the intervening bounded interval about t , (51.4) gives $H'_K(x) \geq ctu$. The mean value theorem proves the error $O(t^{-J-1})$ in (51.15). $\square$

This fixed-order expansion approximates a defined smooth inverse. It introduces no unproved interpolation of the discrete sequence. To recover an integer threshold one must use Theorem 51.2 or exact comparisons; a floating-point root followed by a ceiling is not an integer certificate.

Remark 51.4 ([write] The inverse and the transseries volume, 7 October 2026). Statement by statement, against the volume named in the Guide (its symbols carry the subscript “vol”).

- (a) *Instance.* By Lemma 51.1, a_n increases strictly for $n \geq 1$, so $N(Y)$ is the integer staircase of `p0:def:three-inverses` with $n_1 = 1$ and `p0:thm:staircase(1)` applies.
- (b) *Analogue.* Theorem 51.2 is, as stated and proved, an analogue of `p0:thm:staircase(2)`: a two-ceiling bracket proved directly at the integers from the smooth model H_K .
- (c) *Instance.* The scale t of (51.6), with $t^2(\log t - 1) = L$, is `p0:prop:factorial-core` after the substitution $X_{\text{vol}} = t^2$: $X(\frac{1}{2} \log X - 1) = L$ has $\kappa_{\text{vol}} = \frac{1}{2}$, $d_{\text{vol}} = -1$, and the proposition's $X = L/(\kappa_{\text{vol}}v)$ with $v = W_0(2Le^{-2})$ is $2L/W_0(2L/e^2) = t^2$. This is the double-factorial normalization of `p0:rem:factorial-core` up to the scale of L .
- (d) *Formal instance, first coefficient checked.* With $x = t(1 + E)$ the core $x^2(\log x - 1)$ gives $t^2[HE + h(E)]$ with $H = 2u - 1$, $u = \log t$, and $h(E) = (u - 1)E^2 + (1 + E)^2 \log(1 + E) - E$, so the recurrence (51.14) is the triangular recurrence of `p0:thm:core-reversion` written for the additive displacement $v = tE$, with $\Lambda_{\text{vol}} = H$ and coefficients in a ring containing u ; its first coefficient $\alpha_0 = -A/H$ of (51.9) is the volume's e_1 . The write did not compare α_1, α_2 . Theorem 51.3 has its own residual-over-slope proof, whose last step is `p0:thm:backward-error`.

No novelty is claimed for any of these inversions.

52 Computational verification and diagnostics

The proofs above are analytic and combinatorial. The accompanying programs test identities and normalization, compute finite-cutoff coefficients, and evaluate asymptotic diagnostics. They are not premises for the theorems.

52.1 Exact checks

The exact verification suite uses integers, rational numbers and symbolic polynomial operations. It compares three constructions of $Z_n(p)$: direct path enumeration, a height-transfer dynamic program and a finite formal continued fraction. It checks the staircase bijection and row/cell products path by path, the conjugation identities for D_k , their size bounds, coefficient polynomials, geometric moments, finite row-box enumeration, and the triangular inverse recurrence. It also compares the first eleven diagonal values with the OEIS entry. The largest initial terms are stored in the machine-readable output rather than typeset as a long digit table.

All checks use explicit conditional failures. They remain active under `python -O`; the suite does not rely on removable Python `assert` statements. The canonical result files deliberately omit the interpreter optimization flag so successful normal and optimized runs can be byte-compared. The README specifies the complete build and archive manifest.

52.2 High-precision diagnostics

The following values are ordinary high-precision numerical diagnostics, rounded for readability:

Quantity	Approximate value
$Q(1)$	1.98244090741287370369
c_1	-1.72818020667799661965
c_2	4.37507557565279903972
c_3	-10.48810716563746719452
c_4	24.67555987770631379244
ℓ_2	2.88177216227599748368
$[\mu(1 + e^{-1}) - e^{-1}/2]/Q(1)$	1.09965570555819126120

These coefficients come from the algorithms in Section 47, not from fitting finite- n data. Finite-part and finite-square computations have different normalization and are compared accordingly. Analytic truncation errors and numerical roundoff are distinct: ordinary `mpmath` arithmetic has not been replaced here by directed interval arithmetic.

For a normalized forward check let

$$E_K(n) = n^{K+1} \left(\frac{R_n(1, 0)}{Q(1)} - \sum_{r=0}^K c_r n^{-r} \right).$$

The theorem predicts $E_K(n) \rightarrow c_{K+1}$. A transfer computation without enormous factorial-sized intermediates gives:

n	$E_0(n)$	$E_1(n)$	$E_2(n)$	$E_3(n)$
10	-1.375520	3.526598	-8.484773	20.033338
20	-1.532882	3.905960	-9.382304	22.116058
40	-1.624994	4.127452	-9.904941	23.326636
80	-1.675084	4.247721	-10.188327	23.982379
160	-1.701240	4.310475	-10.336082	24.324094

For each downstep numbered $d + 1$, the transfer divides its height factor by $(n - d)^p$. The product of the denominators is $(n!)^p$ for every path, so this computes R_n directly. This normalization is also tested against exact small cases.

For inverse diagnostics choose $L = \log a_n$ and compute the large root of $H_4(x) = L$. These are real-root errors, not threshold errors:

n	$x_4(\log a_n) - n$	$(x_4 - n)n^6 \log n$	$nd_{\text{TV}}(P_{n,1,0}, \nu_1)$
10	$-2.95433 \cdot 10^{-6}$	-6.802602	0.988176
20	$-3.75043 \cdot 10^{-8}$	-7.190586	1.042846
40	$-4.85625 \cdot 10^{-10}$	-7.337623	1.070987
80	$-6.41083 \cdot 10^{-12}$	-7.364266	1.085257
160	$-8.61809 \cdot 10^{-14}$	-7.338070	1.092440

The TV column approaches the value $1.0996557 \dots$ from Proposition 49.1. Its finite- n evaluation checks that only the empty and singleton partitions have positive probability difference in these cases. To justify this shortcut, adding a cell (i, j) multiplies the likelihood ratio P_n/ν_1 by $e(1 - 1/(n - i - j + 2))^n < 1$ inside the staircase. Every diagram of size at least two contains (2) or $(1, 1)$, which have the same likelihood ratio. Thus checking the singleton and this size-two maximum covers all larger diagrams; excluded diagrams have ratio zero. The general asymptotic theorem does not require that extra finite- n observation. The smooth-root residual scale tests the indexing in (51.3). None of these numbers supplies the unknown effective C_K in Theorem 51.2.

General-parameter diagnostics also include $(\tau, \sigma) = (0.5, -0.7), (1.3, 0.4), (2, 1)$, with finite boxes and their sizes stated in the result file. At smaller τ a fixed box cuts off more mass; its output is not relabeled an exact infinite coefficient. This is why the error analysis is stated separately from numerical agreement.

52.3 Reproduction contract

The public archive contains this TeX source and PDF, a build script, pinned Python dependencies, exact and numerical source code, canonical JSON results and a SHA-256 manifest. From its extracted root, `python build.py` runs the checks, regenerates results, compiles the PDF and recreates the archive. Running `python -O build.py` performs the same complete work with optimization enabled. Consult `README.md` for the concrete toolchain and file list. Fixed PDF metadata, sorted archive members and fixed ZIP metadata make outputs deterministic in the recorded environment. Cross-platform byte identity across different TeX engines or dependency versions is not promised.

The archive contains the public mathematics and reproducible computational evidence. Its results distinguish exact checks from numerical tolerance checks, and no successful tolerance test is represented as an interval certificate. No network access is required by the build once the stated dependencies are installed.

53 Sources and limits of the source screen

The source check was bounded and read-only. The OEIS-hosted indexed A291333 record retrieved on 4 October 2026 supplies the definition, diagonal identification and Kotěšovec attribution. Direct opening of that entry returned a cache miss, so the indexed record is not represented as a fresh uncached revision check. A site-wide footer date is not an entry revision date. The opened A290569 entry confirms the fixed-column continued fraction and the main diagonal; the opened A317103 entry confirms the same product constant in a different model. Attempts to retrieve the neighboring A291332 entry did not yield a usable definition, so no claim about its detailed overlap is made.

Avdoshkin and Dymarsky’s weighted-Dyck discussion [12], Appendix C, is relevant prior context for macroscopic weighted-path methods. It studies fixed power-law parameters and selected modifications. It is not used to transfer a fixed-power asymptotic to $p = \tau n + \sigma$; the proofs here instead use the exact partition identity and small-diagram concentration.

A bounded screen of a public report repository [34] included identifier searches, cached path listings and a limited local text corpus. Selected nearby weighted-Dyck reports were distinguished by their peak-weight or fixed-exponent models. Neither a negative identifier result nor this limited overlap screen constitutes exhaustive literature or repository clearance. Accordingly, this report re-proves the already recorded leading equivalent and presents the further theorems with explicit hypotheses, without asserting a first proof worldwide or absence of earlier work.

54 Further questions

The proofs leave several concrete problems open.

1. *Approach to the small- τ boundary.* When τ decreases with n , typical partition size grows and the present compact-parameter bounds deteriorate. Identifying the crossover to a macroscopic path regime requires estimates beyond Theorem 46.1.
2. *True high-order growth.* The bound $|c_r| \leq CB^r r!$ does not show that factorial growth occurs. Determining late coefficients could identify sharper truncation scales. No conclusion about divergence or Borel summability follows from the current upper bound.
3. *Effective threshold certificates.* Explicit finite- n logarithmic remainders combined with validated Gamma and Lambert computations could turn Theorem 51.2 into a numerical certification algorithm with a computable threshold.
4. *Validated coefficient arithmetic.* The analytic cutoff estimates already isolate truncation error. Directed-rounding implementations of the finite moment recurrences could add rigorous arithmetic enclosures and certify selected digits.
5. *Higher distributional terms.* The polynomials T_r give all fixed-order signed-law corrections. Higher TV terms involve sign boundaries of these corrections; the diagonal first-order split is unusually simple because $1/2 < \mu < 2$.

[write] **7 October 2026.** Under Vladimir's standing rule of 4 October 2026, the write found no claim of this Part that is unproved as stated and none that is wrong. It recomputed $Q(1) = 1.98244090741287370368\dots$, $c_1 = -1.72818020667799661965\dots$, the total-variation constant $1.09965570555819126119\dots$ (the tables print the roundings $\dots 370369$ and $\dots 126120$, declared rounded) and $E_0(10)$, $E_0(20)$, $E_0(40)$ of the forward table: all agree.

[write] **The merged bibliography, 7 October 2026.** The four delivered bibliographies are merged, and every detail of the delivered entries is kept: an entry cited by more than one Part is printed once, with each Part's own note on what it inspected; bracketed Part numbers mark entries of one Part only. Renamed keys: Part II's `oeis` is `oeisA338634`; Part IV's `AD` is `avdoshkin`. Part I's `oeisclassical` is printed as delivered although Part I does not cite it.

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- [3] OEIS Foundation Inc., [A000698](#), [A167872](#), [A321963](#), [A000182](#), [A002105](#), [A326328](#), [A395382](#), [A395383](#), and [A104133](#). Displayed source terms used for exact checks; signs, scales and offsets are recorded in the companion data. [Part I]
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